Regularizing Forecast Combination with Random Subsets*

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Abstract

How much should be estimated when combining many forecasts? Equal weighting is robust to estimation error but ignores heterogeneity in forecast quality, while estimated optimal weights exploit this heterogeneity at the cost of sampling uncertainty. We demonstrate that combining forecasts using random subsets creates a regularization path between equal weighting and fully estimated weights. By analyzing the trade-off between estimation and approximation, we find that, due to the shared sample used by subset regressions, the optimal subset size grows at the rate $T^{1/4}$ in contrast to the $T^{1/2}$ rate for individually evaluated subsets. We propose a feasible finite-sample criterion that selects the data-supported degree of regularization. Using the real-time U.S. Survey of Professional Forecasters (SPF) data, we show that the criterion selects an interior subset size close to the ex-post loss-minimizing choice. Across 17 variables, the feasible selector generally achieves forecast accuracy close to this infeasible benchmark. It preserves the robust survey consensus while avoiding the instability of fully estimated weights.

Keywords: Forecast combination, random subsets, regularization.

JEL: C53, C52, C32.

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1 Introduction

Forecast combination involves a basic regularization trade-off. Using equal weights helps to avoid weight estimation errors. However, it does not take into account possible persistent differences in forecast accuracy. On the other hand, estimated optimal weights can capture these differences, but they often lead to large sampling errors, especially with limited data. This paper investigates how much weight estimation the available sample can support.

This question is especially important for survey forecasts. In particular, Bowles et al. (2007) note that there is variation in forecast accuracy across the ECB Survey of Professional Forecasters respondents. Yet, the short sample size – especially in comparison to the number of forecasters – often prevents decision-makers from estimating these differences. Therefore, a useful combination rule should be able to determine how much weight estimation is supported by the available information.

This paper establishes that the random subset method (RSM) reframes the choice of how much to estimate as a one-dimensional regularization problem indexed by the subset size k. The latter controls the dimension of each local weight-estimation problem. Similar to Diebold and Pauly (1990), in population, RSM produces equal weights when $k = 1$, meaning a single forecast is drawn per subset; when k equals the total number of forecasts, it recovers the fully estimated optimal weights. Intermediate values of k form a regularization path determining how much heterogeneity in forecast quality is estimated.

We characterize the leading excess risk of the shared-sample infinite-bagged[1] RSM estimator, which can be presented as follows:

$$L_{T,m}^{\mathrm{bag},\infty}(k) - Q^{\mathrm{opt}} \asymp \underbrace{\frac{\mathcal{D}}{mk^2}}_{\text{approximation cost}} + \underbrace{\frac{bk^2}{mT}}_{\text{estimation cost}}, \qquad k^* \asymp \left(\frac{\mathcal{D}T}{b}\right)^{1/4},$$

where m is the number of forecasts, T is the sample size, b is the common-error variance, and $\mathcal{D}$ is the precision heterogeneity constant.

The two terms have a direct regularization interpretation. The approximation cost term decreases with k: the larger the subset size, the closer the RSM weights are to the fully estimated optimal weights. Simultaneously, the estimation cost term increases with

[1]Bagging usually randomly selects observations rather than predictors. In the computer science literature, this is usually labeled "feature bagging" (Munson and Caruana, 2009). Throughout, 'bagging' refers to "feature bagging".

k since more coefficients need to be estimated from the same panel of observations. Correspondingly, the subset size k^* that minimizes the asymptotic total risk can be explicitly derived. Hence, k^* measures the amount of local estimation complexity supported by the sample. Additionally, we propose a feasible rule to estimate k^* in the finite-sample setup that we call the *RSM Selector*, which allows us to choose in real time how much heterogeneity to estimate.

Moreover, the form of the optimal subset size, $k^* \asymp \left(\frac{\mathcal{D}T}{b}\right)^{1/4}$, shows that the complexity of the forecast combination increases rather slowly with the length of the historical sample. Even with substantially more data, one should therefore remain conservative about estimating differences among forecasters. For example, holding the common-error variance b and the precision heterogeneity parameter $\mathcal{D}$ fixed in the leading formula, doubling the optimal subset size k^* requires sixteen times as much historical data. That is why a large forecast pool can still support only a small local estimation dimension if the sample size is short.

These results make the relation to the forecast-pool analysis of Elliott and Liao (2026) straightforward: small population gains from optimal weights can be overwhelmed by the cost of estimating them. This trade-off, in our case, is characterized by k^*: the smaller the value of k^*, the less heterogeneity the available sample can support.

It is crucial to distinguish between RSM and the standard regression framework. The RSM estimator averages many subset regressions, but those regressions reuse the same observations. This means that their estimation errors remain correlated across different draws. We show that the shared-sample estimation penalty in our forecast-combination setting is of order $k^2/(mT)$, as presented above. By contrast, if a single subset regression is estimated, its loss has a more standard k/T penalty. That leads to a different square-root rule of choosing k^* for the expected loss of a single regression.

We conduct Monte Carlo simulations to demonstrate the finite-sample regularization performance of RSM. The obtained results are consistent with the developed asymptotic theory. When forecast precisions are nearly homogeneous, little is gained by estimating weights. In this case, the proposed *RSM Selector* remains close to equal weighting by selecting small subset sizes. On the other hand, if precision heterogeneity is substantial, the *RSM Selector* chooses larger k, as the benefits of moving away from equal weights are larger. Additional exercises show that the derived shared-sample risk criterion tracks

the simulated infinite-bag risk. The theory-predicted increase in the optimal subset size as the training sample grows is also observed.

Empirically, the paper evaluates the feasible *RSM Selector* using real-time forecasts of quarterly U.S. GDP-price inflation one and four quarters ahead. The *RSM Selector* chooses an interior degree of regularization close to the ex-post loss-minimizing fixed subset size. Empirical evidence shows that the median eligible pool contains around 30 forecasters at $h = 1$ and $h = 4$, whereas the *RSM Selector* chooses median subset sizes of only 3 and 2, respectively. The ex-post best fixed size is 2 at $h = 1$ and 4 at $h = 4$. The realized loss curve is U-shaped with a nearly flat region across small values of k. However, loss increases substantially as k approaches the full panel size of m. The results are robust to different panel balancing rules, motivated by Conflitti et al. (2015), and time periods included in the sample.

In terms of forecast precision, RSM compares favorably with the survey consensus while outperforming combinations with fully estimated or constrained weights. We also evaluate the performance of the *RSM Selector* across 17 SPF variables. As before, it selects small subsets across all the variables. In most cases, the resulting forecast accuracy is close to that of the infeasible ex-post best fixed subset size. Moreover, when $\widehat{k} = 2$, RSM reduces the root mean squared forecast error (RMSFE) relative to equal weighting by 0.65 percent in the pooled evaluation, which is significant at the 10% level. These patterns suggest that regularization via RSM preserves the survey consensus while avoiding the instability of fully estimated optimal weights. Hence, the *RSM Selector* identifies how much weight estimation the data support, and the resulting forecasts confirm that this choice is informative.

Related literature. The paper is closely related to the forecast-combination puzzle: equal weighting can outperform estimated optimal weights when finite-sample estimation error offsets their population advantage (Timmermann, 2006; Smith and Wallis, 2009; Claeskens et al., 2016). In particular, Elliott and Liao (2026) show that under forecast-error structure assumptions motivated by the US SPF, the population gains from estimating optimal weights can indeed be small. They conclude that the best approach may be to select a specific subset and average the corresponding forecasts.[2] However, they rely on a single predetermined subset of forecasts chosen by its previous forecast per-

[2]A similar conclusion has been drawn, among others, in Diebold et al. (2025); Cui and Hu (2025).

formance. Diebold and Shin (2019) use partially-egalitarian LASSO for determining the subset of forecasters whose forecasts are then averaged. Instead, we explicitly minimize the asymptotic risk of the infinitely-bagged RSM estimator to derive the optimal subset size k^*.

Many existing combination methods shrink weights toward equality via penalties (Diebold and Pauly, 1990; Stock and Watson, 2004). On the other hand, RSM uses the subset dimension itself to supply the regularization path, allowing the shared training sample to determine the optimal rate.

Complete subset regression and RSM have been predominantly applied to subsets of predictors (Elliott et al., 2013, 2015; Boot and Nibbering, 2019). Instead, we apply this method for forecast combination. We draw subsets of forecasts themselves and estimate combination weights within each subset. Elliott et al. (2013) show that in the orthogonal-predictor case, the complete subset variance implies the analogous $\frac{k^2}{mT}$ order. We prove that this penalty order persists in a forecast combination framework subject to sum-to-one constraints and two-component errors. Additionally, we derive the approximation error cost, which allows us to define k^* and its asymptotic $T^{1/4}$ rate and propose the feasible *RSM Selector*. Unlike previous studies, which choose the subset size by past forecasting performance, we derive k^* and show how to estimate it in real time.

Our empirical design follows the literature on combining survey forecasts and evaluating them in real time. US and EU SPF evidence emphasizes both the resilience of the survey mean and the stabilizing role of constraints or shrinkage (Genre et al., 2013; Conflitti et al., 2015). Our use of origin-specific data vintages and first-release evaluation follows the principle that neither weight estimation nor forecast ranking should use revised information that was unavailable at the forecast origin (Croushore, 2011).

Outline. This paper proceeds as follows. Sections 2 and 3 develop the theory and the *RSM Selector*. Section 4 reports the main Monte Carlo evidence; Section 5 evaluates the *RSM Selector* in real-time US SPF data. Finally, Section 6 concludes. The appendices contain additional theoretical results, proofs, auxiliary simulations and empirical results.

2 Theoretical Model

We restate the main theoretical finding. The leading shared-sample risk trade-off derived below is as follows:

$$L_{T,m}^{\text{bag},\infty}(k) - Q^{\text{opt}} \asymp \underbrace{\frac{\mathcal{D}}{mk^2}}_{\text{approximation cost}} + \underbrace{\frac{bk^2}{mT}}_{\text{estimation cost}}, \qquad k^* \asymp \left(\frac{\mathcal{D}T}{b}\right)^{1/4}, \tag{1}$$

where m is the number of forecasts, T is the sample size, b is the common-error variance, and $\mathcal{D}$ is the precision heterogeneity constant.

This means larger subsets move the bagged population weights toward the oracle, but they also require the shared training sample to estimate more within-subset coefficients. Figure 1 gives the resulting regularization path before the formal derivation.[3]

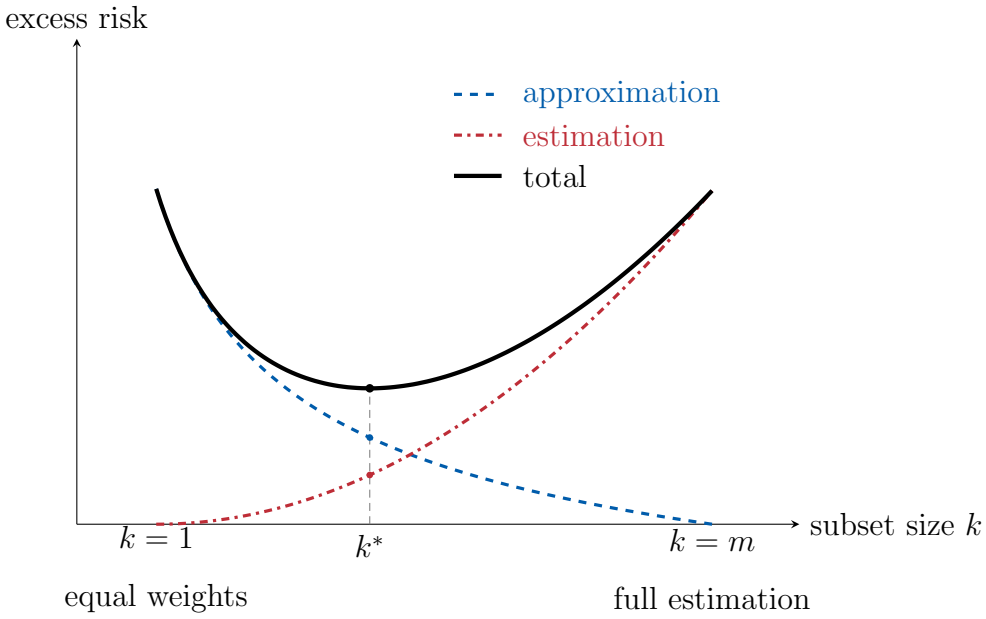


Figure 1: Subset size as a regularization path

More formally, suppose there are m unbiased individual forecasts $f_t = (f_{1,t}, \ldots, f_{m,t})'$ of a scalar target y_t. Let ι_r denote the r-dimensional vector of ones. The realized forecast-error vector is defined as:

$$e_t = f_t - y_t \iota_m. \tag{2}$$

It is worth mentioning that expectations below, denoted $\mathbb{E}$, are taken over the unconditional distribution of realized forecast errors across origins. Later, we introduce

[3]The curves are schematic. Proposition 10 derives their leading orders.

expectations over random subset selection, which are denoted separately by $\mathbb{E}_S$. We fix the forecast horizon h and suppress its subscript for simplicity.

Assumption 1 (Forecast errors)**.** $\mathbb{E}(e_t) = 0$ *and* $\mathrm{Var}(e_t) = \Omega_{e,m} \succ 0$.

Assumption 2 (Two-component forecast-error structure)**.** *The forecast errors satisfy:*

$$e_t = h_t \iota_m + \xi_t, \tag{3}$$

where $\mathbb{E}(h_t) = 0$, $\mathrm{Var}(h_t) = b > 0$, $\mathbb{E}(\xi_t) = 0$, $\mathrm{Var}(\xi_t) = D$, *and* $\mathrm{Cov}(h_t, \xi_t) = 0$. *Moreover,*

$$D = \mathrm{diag}(\tau_1^{-1}, \ldots, \tau_m^{-1}), \qquad 0 < \tau_L \leq \tau_i \leq \tau_U < \infty$$

uniformly in i *and* m. *Denote*

$$\bar{\tau}_m = \frac{1}{m}\sum_{i=1}^{m} \tau_i, \qquad v_{\tau,m} = \frac{1}{m}\sum_{i=1}^{m} (\tau_i - \bar{\tau}_m)^2,$$

and assume $\bar{\tau}_m \to \bar{\tau} > 0$ *and* $v_{\tau,m} \to v_\tau > 0$. *Finally, assume that the precision-profile constant*

$$\mathcal{D}_m = \frac{1}{\bar{\tau}_m^6} \frac{1}{m} \sum_{i=1}^{m} \tau_i \{v_{\tau,m} - \bar{\tau}_m(\tau_i - \bar{\tau}_m)\}^2 = \frac{\mu_1\mu_3 - \mu_2^2}{\mu_1^5}, \tag{4}$$

where $\mu_j = \frac{1}{m}\sum_{i=1}^{m} \tau_i^j$ *converges to a finite constant* $\mathcal{D} > 0$.

Assumption 3 (Gaussian shared sample)**.** *The blocks* h_t *and* ξ_t *are independent centered Gaussian variables, and* $(h_t, \xi_t')'$ *is i.i.d. across the* T *training observations. The evaluation error is an independent draw from the same distribution. Moreover,* $T^{1/4} = o(m)$ *and* $m = o(T)$.

Assumption 1 is a standard assumption in the forecast combination literature. It implies that the forecasts are unconditionally unbiased with some positive forecast-error covariance matrix. Assumption 2 is a reduced-form specification for the forecast-error vector. The common component h_t may combine an unforecastable target innovation with forecast-construction disturbances shared across forecasters. In particular,

$$\Omega_{e,m} = \mathrm{Var}(e_t) = b\iota_m\iota_m' + D. \tag{5}$$

Hence, h_t captures shocks shared by all forecasters, while ξ_t collects forecaster-specific errors. This common-loading structure is related to the forecast-error decompositions

used by Davies and Lahiri (1995, 1999); Lahiri and Sheng (2010).[4] The boundedness and convergence conditions in Assumption 2 keep the precision distribution nondegenerate as m grows. The stronger Gaussian condition in Assumption 3 is imposed only for the exact feasible-loss calculations and the shared-sample expansion, in the spirit of Kan and Zhou (2007).

For any adding-up weight vector w satisfying $\iota_m' w = 1$, quadratic forecast risk is as follows:[5]

$$Q(w) = \mathbb{E}(w' e_t e_t' w) = w' \Omega_{e,m} w. \tag{6}$$

The equal-weight and population-optimal combinations have the following closed forms:

$$w^{\text{ave}} = m^{-1} \iota_m, \qquad Q^{\text{ave}} = m^{-2} \iota_m' \Omega_{e,m} \iota_m, \tag{7}$$

$$w^{\text{opt}} = \frac{\Omega_{e,m}^{-1} \iota_m}{\iota_m' \Omega_{e,m}^{-1} \iota_m}, \qquad Q^{\text{opt}} = (\iota_m' \Omega_{e,m}^{-1} \iota_m)^{-1}. \tag{8}$$

When $w^{\text{ave}} \neq w^{\text{opt}}$, equal weighting has strictly higher population risk: $Q^{\text{ave}} > Q^{\text{opt}}$. Estimating the optimal weights, however, introduces sampling error that can erase this population gain. This tension is a leading explanation of the forecast-combination puzzle.

2.1 Random Subset Method for Forecast Combination

Before formally introducing RSM, we formulate the subset sampling rule.

Assumption 4 (Subset sampling)**.** *Each subset S is drawn uniformly from the $\binom{m}{k}$ subsets of size k, without replacement within a subset and independently of the data. The subsets $S_1, \ldots, S_G$ are independent draws.*[6]

[4] Unlike these more general approaches, we assume that a forecast-specific bias component is 0, since the forecasts are assumed to be unbiased. Additionally, we assume that matrix D is diagonal. Notice that this specification is equivalent to a single factor structure with equal loadings. General factor models with heterogeneous loadings and residual cross-sectional dependence are also analyzed; see, for instance, Lee and Seregina (2020).

[5] The analysis uses symmetric quadratic loss. When w is estimated, the expression is conditional on the training sample and applies to an independent evaluation observation. Asymmetric loss is outside the scope of the paper; see Elliott and Timmermann (2004). Moreover, Aruoba et al. (2012) show that equal weighting is optimal under the minimax loss function.

[6] In other words, a given forecast cannot be selected twice in the same subset. However, since the subsets are taken independently, we allow the case when $S_g = S_{g'}$ for $g \neq g'$.

Let P_S be the $m \times k$ selection matrix that embeds a subset weight vector in $\mathbb{R}^m$ by placing its entries in their original coordinates and filling the remaining coordinates with zeros. Notice that the embedding matrix P_S satisfies the following properties: $P_S' P_S = I_k$ and $P_S' \iota_m = \iota_k$. Let $\Omega_{e,k,S}$ denote the $k \times k$ principal submatrix of $\Omega_{e,m}$ indexed by S:

$$\Omega_{e,k,S} = P_S' \Omega_{e,m} P_S = b\iota_k \iota_k' + D_S, \qquad D_S = \operatorname{diag}(\tau_i^{-1} : i \in S). \tag{9}$$

Within subset S, the population-optimal weights and their embedding can be written as follows:

$$w_S = \frac{\Omega_{e,k,S}^{-1} \iota_k}{\iota_k' \Omega_{e,k,S}^{-1} \iota_k}, \qquad v_S = P_S w_S, \qquad \iota_m' v_S = 1. \tag{10}$$

Three population criteria must be distinguished. First, the average-subset risk evaluates one randomly selected subset and then averages its loss:

$$Q_k^{\text{sub}} = \mathbb{E}_S[v_S' \Omega_{e,m} v_S] = \mathbb{E}_S[Q(S)], \qquad Q(S) = (\iota_k' \Omega_{e,k,S}^{-1} \iota_k)^{-1}. \tag{11}$$

Second, finite bagging averages G independently sampled embedded weight vectors before evaluating loss:

$$Q_k^{\text{bag},G} = \mathbb{E}_{S_1,\ldots,S_G}(\bar{v}_{k,G}' \Omega_{e,m} \bar{v}_{k,G}), \qquad \bar{v}_{k,G} = G^{-1} \sum_{g=1}^{G} v_{S_g}. \tag{12}$$

Finally, complete, or infinite, bagging evaluates the limit as $G \to \infty$:

$$Q_k^{\text{bag},\infty} = \bar{v}_k' \Omega_{e,m} \bar{v}_k, \qquad \bar{v}_k = \mathbb{E}_S(v_S). \tag{13}$$

The last object is equivalent to averaging over all size-k subsets, as in complete subset regression (Elliott et al., 2013). Moreover, due to the convexity of the population loss function in v_S, we can show, by applying Jensen's inequality, similar to Boot and Nibbering (2019), that $Q_k^{\text{bag},\infty} \leq Q_k^{\text{sub}}$. Strict convexity of $v' \Omega_{e,m} v$ makes equality possible only when v_S is constant almost surely, which under uniform fixed-size sampling occurs when $k = m$.

Proposition 1 (Finite-bagging bridge)**.** *For every* $1 \leq G < \infty$,

$$Q_k^{\text{bag},G} = Q_k^{\text{bag},\infty} + \frac{1}{G}\left(Q_k^{\text{sub}} - Q_k^{\text{bag},\infty}\right). \tag{14}$$

Consequently, $Q_k^{\text{bag},\infty} \leq Q_k^{\text{bag},G} \leq Q_k^{\text{sub}}$.

The proof is in Appendix A.4. It is known that averaging multiple regression models via bagging is guaranteed never to increase the expected mean squared error under a squared-loss function (Breiman, 1996). In our case, the identity shows that variation due to subset sampling vanishes as $G \to \infty$.

Notice the following special cases:

- when $G = 1$, $Q_k^{\text{bag},1} = Q_k^{\text{sub}}$;
- as $G \to \infty$, $Q_k^{\text{bag},G} \to Q_k^{\text{bag},\infty}$.

The implementable finite-bag estimator requires a rate condition on the number of subset draws.

Assumption 5 (Number of subset draws G)**.** *For results transferring the infinite-bag risk expansion and optimal-subset rule to the implementable finite-bag estimator, let* $G = G_{T,m} \to \infty$ *and assume*

$$\frac{mT^{1/4}}{G} \to 0.$$

We can analyze how the RSM weights change depending on the subset size k in some general cases. Our analysis closely follows Elliott et al. (2013).

Proposition 2 (RSM as a unifier)**.** $\bar{v}_1 = w^{\text{ave}}$ *and* $\bar{v}_m = w^{\text{opt}}$.

The endpoint identities show that RSM contains both standard benchmarks.

3 The Approximation-Estimation Trade-off

Define the subset and full-panel sums of precisions τ_i by:

$$A_S = \sum_{i \in S} \tau_i, \qquad A_m = \sum_{i=1}^{m} \tau_i = m\bar{\tau}_m. \tag{15}$$

Proposition 3 (Subset loss under two components)**.** *For every subset* S,

$$w_{i,S} = \frac{\tau_i}{A_S} \quad (i \in S), \qquad Q(S) = v_S' \Omega_{e,m} v_S = b + \frac{1}{A_S}. \tag{16}$$

For the full panel and equal weighting, we have:

$$w_i^{\text{opt}} = \frac{\tau_i}{A_m}, \qquad Q^{\text{opt}} = b + \frac{1}{m\bar{\tau}_m}, \tag{17}$$

$$w_i^{\text{ave}} = 1/m, \qquad Q^{\text{ave}} = b + \frac{1}{m^2} \sum_{i=1}^{m} \tau_i^{-1}. \tag{18}$$

Appendix A.1 derives these identities. The subset loss $Q(S)$ is the sum of the irreducible common component b and the inverse of the total precision in the subset. Along any nested sequence, adding a forecast strictly lowers population risk because every τ_i is positive. The minimum is attained at the full panel, i.e., $k = m$. Also, notice that the optimal subset weight, unlike the subset loss, does not depend on b: it is only a function of the forecast precision vector τ.

To illustrate the relationship between the two losses, Q^{opt} and Q^{ave}, let $B = \bar{\tau}_m^{-1}$ and $H = (m^{-1}\sum_{i=1}^m \tau_i^{-1})\bar{\tau}_m$, where $\bar{\tau}_m$ is the mean precision and $m^{-1}\sum_{i=1}^m \tau_i^{-1}$ is the mean idiosyncratic variance.

Then, B is the reciprocal of average precision. It illustrates the scale of idiosyncratic noise. Intuitively, it depicts how noisy a typical forecast is without taking into account heterogeneity. More precise forecasts lead to a larger $\bar{\tau}_m$ value. This, in turn, decreases the value of B and allows for achieving lower loss.

The quantity H is the product of mean variance $m^{-1}\sum_{i=1}^m \tau_i^{-1}$ and mean precision $\bar{\tau}_m$. It measures the dispersion of idiosyncratic forecast variances and equals the ratio of their arithmetic mean to their harmonic mean. Thus, $H \geq 1$, with equality if and only if all τ_i are equal, the case in which equal weights are optimal.

The endpoint risks can therefore be written as follows:

$$Q^{\text{opt}} = b + \frac{B}{m}, \qquad Q^{\text{ave}} = b + \frac{BH}{m}, \qquad Q^{\text{ave}} - Q^{\text{opt}} = \frac{B(H-1)}{m} \geq 0.$$

Thus, equal weighting cannot dominate the unrestricted optimum in the population. Persistent precision heterogeneity implies $H > 1$ and creates room for an interior RSM combination to improve on equal weighting. Yet, the population risk gap is $B(H-1)/m$ and therefore shrinks with the pool size when B and H remain bounded. This is consistent with Elliott and Liao (2026): as the number of forecasts grows, the population risks of equal and optimal weighting become close even when their weights differ.

Proposition 4 (Average subset loss)**.** *Under Assumptions 2 and 4, for every k, the exact average-subset risk is as follows:*

$$Q_k^{\text{sub}} = b + \mathbb{E}_S(A_S^{-1}). \tag{19}$$

As $m \geq k \to \infty$,

$$Q_k^{\text{sub}} = b + \frac{1}{k\bar{\tau}_m} + \frac{m-k}{k^2(m-1)}\frac{v_{\tau,m}}{\bar{\tau}_m^3} + O(k^{-5/2}), \tag{20}$$

and hence

$$Q_k^{\text{sub}} = b + \frac{1}{k\bar{\tau}_m} + O(k^{-2}). \tag{21}$$

The proof is in Appendix A.5.

For infinite bagging, let us introduce the following notation:

$$g_i(k) = k\mathbb{E}_S(A_S^{-1} \mid i \in S), \qquad z_k = \frac{1}{k} - \frac{1}{m}. \tag{22}$$

Proposition 5 (Bagged loss representation)**.** *The infinite-bagged weights and their excess risk satisfy*

$$\bar{v}_{i,k} = \frac{\tau_i}{m} g_i(k), \qquad \sum_{i=1}^{m} \tau_i g_i(k) = m, \tag{23}$$

$$Q_k^{\text{bag},\infty} = Q^{\text{opt}} + R_k^{\text{bag}}, \qquad R_k^{\text{bag}} = \sum_{i=1}^{m} \frac{(\bar{v}_{i,k} - w_i^{\text{opt}})^2}{\tau_i} = \frac{1}{m^2} \sum_{i=1}^{m} \tau_i \left(g_i(k) - \frac{1}{\bar{\tau}_m} \right)^2 \geq 0. \tag{24}$$

The proof can be found in Appendix A.6. Most importantly, R_k^{bag} measures the population cost of using infinitely-bagged RSM relative to the optimal weighting. The same endpoints can be seen here. When $k = 1$, R_1^{bag} depicts the difference between the equal weighting and optimal weighting losses. This amount is expressed as a variance-weighted squared difference between the corresponding weights. On the other hand, when $k = m$, $R_m^{bag} = 0$.

For further results, we need to study the asymptotic behavior of R_k^{bag}.

Proposition 6 (Asymptotic behavior of R_k^{bag})**.** *Under Assumptions 2 and 4, as $k \to \infty$ with $k/m \to 0$,*

$$R_k^{bag} = \frac{\mathcal{D}_m}{m} z_k^2 \{1 + o(1)\}. \tag{25}$$

The proof can be found in Appendix B.

3.1 Estimation Error Inflation

The population risks above treat weights as known. We now fit the unrestricted optimal weights on the common training sample and evaluate them on an independent error draw. Let $L_{T,m}^{\text{sub}}(k)$ denote the expected test loss when one fitted size-k subset is evaluated, and the result is averaged over S. Let $L_{T,m}^{\text{bag},\infty}(k)$ denote the expected test loss of the complete-subset estimator, which averages all fitted subset-weight vectors before evaluation.

Proposition 7 (Exact average subset inflation)**.** *Under Assumptions 2 and 3, if* $T > k$*, then*

$$L_{T,m}^{\text{sub}}(k) = Q_k^{\text{sub}}\, \mathcal{I}_{\text{sub}}(T,k), \qquad \mathcal{I}_{\text{sub}}(T,k) = \frac{T-1}{T-k}. \tag{26}$$

Thus, $\mathcal{I}_{\text{sub}}(T,k) = 1 + (k-1)/(T-k)$*.*

For each fixed S, the adding-up restriction leaves $k-1$ free Gaussian regression coefficients. Proposition 15 proves that the expected independent-test loss is $Q(S)(T-1)/(T-k)$. This result is in line with the existing literature; see Okhrin and Schmid (2006); Rosset and Tibshirani (2020).

For infinite bagging, all subset regressions reuse the same observations. Their estimation errors are correlated, so Equation (26) cannot be applied after averaging the weight vectors. This is why we derive the RSM-specific inflation factor as follows:

Proposition 8 (Shared-sample infinite-bag inflation)**.** *Let* $k = k_{T,m}$ *satisfy* $k \to \infty$*,* $k/m \to 0$*, and* $k < T$*. Under Assumptions 1–4,*

$$L_{T,m}^{\text{bag},\infty}(k) = Q_k^{\text{bag},\infty} + \frac{\Gamma_{m,k}}{T} + o\left(\frac{k^2}{mT}\right), \tag{27}$$

where the complete first-order coefficient $\Gamma_{m,k}$ *is given in* (57)*. Equivalently, the infinite-bag inflation factor can be written as:*

$$\mathcal{I}_\infty(T,m,k) = 1 + \frac{\Gamma_{m,k}}{TQ_k^{\text{bag},\infty}} + o\left(\frac{k^2}{mT}\right) = 1 + \frac{k^2}{mT}\{1 + o(1)\}. \tag{28}$$

Moreover,

$$\Gamma_{m,k} = b\frac{k^2}{m}\{1 + o(1)\}. \tag{29}$$

The proof can be found in Appendix B. By combining the results in Proposition 8 and Equation (25), we obtain the RSM approximation-estimation trade-off as stated in Equation (1).

3.2 Optimal subset size

With both inflation factors established, we can solve the two distinct risk–complexity problems. The first result optimizes the leading feasible average subset criterion.

Proposition 9 (Square-root rule for the average-subset criterion)**.** *Under Assumptions 2–4, combining Equation* (21) *and* (26) *gives the following leading continuous criterion:*

$$\widetilde{L}_{T,m}^{\text{sub}}(k) = \left(b + \frac{1}{k\bar{\tau}_m}\right)\frac{T-1}{T-k}, \qquad 0 < k < T. \tag{30}$$

It has the unique minimizer

$$\widetilde{k}^{\text{sub}}_{T,m} = \frac{-1 + \sqrt{1 + b\bar{\tau}_m T}}{b\bar{\tau}_m}. \tag{31}$$

If, additionally, $T^{1/2} = o(m)$*, then:*

$$\widetilde{k}^{\text{sub}}_{T,m} \sim \sqrt{\frac{T}{b\bar{\tau}}}. \tag{32}$$

Proposition 10 (Optimal infinitely bagged subset size)**.** *Let*

$$k^*_{T,m} \in \arg\min_{1 \le k \le \min\{m, T-1\}} L^{\text{bag},\infty}_{T,m}(k).$$

Under Assumptions 1–4,

$$\frac{k^*_{T,m}}{T^{1/4}} \longrightarrow \left(\frac{\mathcal{D}}{b}\right)^{1/4}, \qquad k^*_{T,m} \sim \left(\frac{\mathcal{D}T}{b}\right)^{1/4}. \tag{33}$$

The proof is in Appendix B.8. The two leading terms are:

$$R^{\text{bag}}_k \sim \frac{\mathcal{D}}{mk^2}, \qquad \frac{\Gamma_{m,k}}{T} \sim \frac{bk^2}{mT}.$$

Balancing the two terms gives the fourth-root rate. Notice that the common $1/m$ factor cancels, so the total number of forecasts, m, affects feasibility and boundary behavior (in particular, via $\mathcal{D}$). However, it does not directly enter the formula for selecting $k^*_{T,m}$.

Moreover, the finite-bagged RSM has the same leading optimizer when the number of subset draws grows sufficiently quickly, as stated in Assumption 5. Similarly to Equation (14), we can write the feasible loss bridge as follows:

$$L^{\text{bag},G}_{T,m}(k) = L^{\text{bag},\infty}_{T,m}(k) + \frac{1}{G}\left\{L^{\text{sub}}_{T,m}(k) - L^{\text{bag},\infty}_{T,m}(k)\right\}. \tag{34}$$

The proof can be found in Appendix B.10.

Remark 1 (Fixed forecast pool)**.** *If* m *is fixed while* $T \to \infty$*, estimation becomes asymptotically costless and* $k = m$ *is an asymptotically optimal boundary choice.*

The asymptotic rule has a feasible, boundary-corrected empirical theory-guided implementation. Let $\widehat{C} = T^{-1}\sum_{t=1}^{T} e_t e_t'$ be the training second-moment matrix. Then, it is possible to estimate the common-error variance $\widehat{b}$ by the mean off-diagonal element of $\widehat{C}$, and construct $\widehat{\mathcal{D}}_m$ from the implied idiosyncratic variances as in Equation (4). We select:

$$\widehat{k} = \arg\min_{1 \le k \le \min\{m, T-2\}} \left\{\widehat{\mathcal{D}}_m \left(\frac{1 - k/m}{k}\right)^2 + \widehat{b}\frac{k^2}{T}\right\}. \tag{35}$$

The factor $1 - k/m$ restores the exact zero approximation cost at $k = m$. Proposition 19 proves that $\widehat{k}/k^*_{T,m} \to_p 1$, and the fixed-size expected-risk curve evaluated at $\widehat{k}$ has vanishing leading regret relative to $k^*_{T,m}$. However, at $k = 1$ the exact weight-estimation penalty is zero, whereas the leading penalty term in Equation (35) assigns a positive penalty $(\widehat{b}/T)$. We call the rule in Equation (35) the *RSM Selector*: it chooses k and hence the degree of regularization using only training-sample second moments. Appendix D.3 states the numerical safeguards used in the empirical implementation. The *RSM Selector* can be described as follows:

RSM Selector

1. Construct the $T \times m$ matrix of training forecast errors and compute $\widehat{C} = T^{-1} \sum_{t=1}^{T} e_t e_t'$.
2. Estimate $\widehat{b}$ by the mean off-diagonal elements of $\widehat{C}$. Namely, $\widehat{b} = \frac{1}{m(m-1)} \sum_{i \neq j} \widehat{C}_{ij}$.
3. Compute $\widehat{\tau}_i = \frac{1}{\widehat{C}_{ii} - \widehat{b}}$ and $\widehat{\mathcal{D}}_m = \frac{1}{\widehat{\tau}_m^6} \frac{1}{m} \sum_{i=1}^{m} \widehat{\tau}_i \{\widehat{v}_{\tau,m} - \widehat{\tau}_m(\widehat{\tau}_i - \widehat{\tau}_m)\}^2$.
4. Evaluate the criterion in Equation (35) on the admissible integer grid and choose its smallest minimizer $\widehat{k}$.
5. At $\widehat{k}$, estimate subset-optimal weights and average the resulting subset forecasts. Whenever computationally feasible, use exact enumeration of the available subsets.

This procedure is used in the empirical exercise. Later, we also use the direct plug-in version of the asymptotic fourth-root rule, rounding Equation (33) to the nearest admissible integer: $\widehat{k} = \left(\widehat{\mathcal{D}}_m T/\widehat{b}\right)^{1/4}$. We compare its selected subset sizes and forecast performance with those produced by the boundary-corrected *RSM Selector*. A simple numerical example of both estimation strategies is presented in Appendix B.11.

4 Monte Carlo Simulations

We conduct several Monte Carlo exercises, each corresponding to a distinct theoretical object. The first evaluates the feasible finite-G forecasting procedure and asks whether RSM improves out-of-sample MSPE relative to standard combination benchmarks. The

second isolates the shared-sample estimation penalty and tests both the coefficient $\Gamma_{m,k}/T$ and its implications for the optimal subset size. Additional results can be found in Appendix C.

4.1 Simulation design

4.1.1 Finite-G forecasting performance

For each replication, the target, y_t, and the ith candidate forecast, $f_{i,t}$, are generated as follows:

$$y_t = \mu_t + \varepsilon_t, \tag{36}$$

$$f_{i,t} = \mu_t + c_t + \xi_{i,t}, \qquad i = 1, \dots, m, \tag{37}$$

where the predictable component follows

$$\mu_t = 0.6\mu_{t-1} + \nu_t.$$

The initial value is drawn from its stationary distribution,

$$\mu_1 \sim N\left(0, \frac{1}{1-0.6^2}\right).$$

The innovations are mutually independent over t and across components, with

$$\nu_t \sim N(0,1), \qquad \varepsilon_t \sim N(0,s), \qquad c_t \sim N(0,q), \qquad \xi_{i,t} \sim N(0,\sigma_i^2).$$

We set $s = q = 1$. The term c_t is common to all candidate forecasts, whereas $\xi_{i,t}$ generates heterogeneity in forecast precision. Because all combination weights are constrained to sum to one, the common predictable component μ_t does not affect the population comparison between alternative combination weights. In the forecast-error representation of Assumption 2, $e_t = (c_t - \varepsilon_t)\iota_m + \xi_t$, so the common error is $h_t = c_t - \varepsilon_t$ and has variance $b = q + s = 2$ in this experiment.

We characterize each precision profile by B and H, as discussed in Section 2. To generate a precision profile, for each H, we draw a fixed set of centered Gaussian values z_i and set:

$$\tau_i \propto \exp(\omega z_i).$$

The parameter ω is chosen numerically so that the finite pool attains the specified value of H exactly. The resulting precisions are then rescaled so that their sample mean

equals $1/B$. For a given value of H, the same normalized precision profile is used across different values of B, and the resulting precision vector is held fixed across all Monte Carlo replications.

The main experiment fixes the following values:

$$m = 50, \qquad T = 240, \qquad T_{\text{test}} = 100, \qquad B \in \{3, 8\}, \qquad H \in \{1.05, 6\}.$$

These values of H are chosen to reflect different amounts of heterogeneity across forecasters. As shown in Section 2, $H \geq 1$, with equality holding only under homogeneity. Thus, the case when $H = 1.05$ represents an approximately homogeneous forecast pool. Meanwhile, $H = 6$ represents substantial heterogeneity where RSM can potentially be more useful.

For every subset size $k = 1, \ldots, m$, we take $G = 800$ independent draws from the collection of all size-k subsets. For each subset S_g, the combination weights are estimated by constrained least squares:

$$\widehat{w}_{S_g} = \arg\min_{w} \sum_{t=1}^{T} \left(y_t - \sum_{i \in S_g} w_i f_{i,t} \right)^2 \quad \text{subject to} \quad \sum_{i \in S_g} w_i = 1.$$

The finite-bagged RSM forecast is obtained by averaging the resulting subset forecasts as follows:

$$\widehat{y}_t^{\text{RSM}}(k) = \frac{1}{G} \sum_{g=1}^{G} \sum_{i \in S_g} \widehat{w}_{i,S_g} f_{i,t}.$$

We compare the forecast combination performance of the bagged RSM with three benchmarks: equal weighting, fully estimated optimal weights, and the infeasible oracle that uses the population-optimal weights. The fully-estimated and RSM weights are estimated on the T training observations. Later, those weights remain fixed for evaluation.

The forecast combination performance is evaluated using the out-of-sample mean squared prediction error

$$\widehat{\text{MSPE}} = \frac{1}{T_{\text{test}}} \sum_{t=T+1}^{T+T_{\text{test}}} (y_t - \widehat{y}_t)^2 .$$

We repeat this exercise 500 times. There could be a potential problem with using RSM directly to choose the subset size after evaluation due to Monte Carlo noise. It may happen that, given the chosen subset size k, it outperforms other values purely by chance. To control for this "optimistic" bias, we split the replications into two groups: even and

odd. The latter are used to select the candidate k. The evaluation is then performed on the former replications, keeping k fixed.

4.1.2 Shared-sample infinite-bag risk

The second Monte Carlo experiment aims to verify the shared-sample results (both the derived risk trade-off and the optimal subset selection rule), derived in Section 3.1 and in Section 3.2, respectively.

Unlike in the previous setup, we consider several τ profiles. The first one, labeled the balanced two-point profile, is constructed in such a way that half of the forecasts are assigned $\tau_i = 0.01$ and the other half are assigned $\tau_i = 0.09$. Consequently, for this collection of τ_i, $\bar{\tau}_m = 0.05$, $H_m = 2.778$, and $\mathcal{D}_m = 4.608$.

Besides, we define a deterministic smooth profile of τ. Specifically, for each $i \in \{1, \dots, m\}$, the values are defined as follows:

$$z_{i,m} = 2\frac{i - \frac{1}{2}}{m} - 1, \qquad \tau_{i,m} = 0.05\,\frac{\exp(\omega_m z_{i,m})}{m^{-1}\sum_{j=1}^{m}\exp(\omega_m z_{j,m})},$$

where ω_m is numerically optimized so that $\mathcal{D}_m$ equals 4.608. Both profiles are constructed such that they share the same mean ($\bar{\tau}_m$) and the same precision-profile constant ($\mathcal{D}_m$), even though the distributions of precisions are substantially different.

In each replication, we simulate the $T \times m$ forecast-error matrix directly:

$$e_t = h_t \iota_m + \xi_t, \qquad h_t \sim N(0, b), \qquad \xi_{i,t} \sim N(0, \tau_i^{-1}), \qquad b = 0.2.$$

The common and idiosyncratic innovations are mutually independent and independent across time. All subset regressions within a replication use this same training sample.

The balanced two-point experiment considers $T \in \{256, 512, 1024, 2048, 4096\}$ and sets m equal to the nearest even integer to $5\sqrt{T}$, giving $m \in \{80, 114, 160, 226, 320\}$. This sequence satisfies $T^{1/4} = o(m)$ and $m = o(T)$, as required by Assumption 3. The five designs use, respectively, $\{100, 100, 100, 120, 150\}$ replications. The smooth robustness design uses $\{80, 100, 120\}$ replications at $T \in \{256, 1024, 4096\}$. For every candidate k, we make $G = 800$ independent uniform subset draws.

For a weight vector v, risk is evaluated analytically as follows:

$$L(v) = v' C_m v = b + \sum_{i=1}^{m} \frac{v_i^2}{\tau_i}, \qquad \Omega_{e,m} = b\iota_m \iota_m' + \operatorname{diag}\left(\frac{1}{\tau_1}, \dots, \frac{1}{\tau_m}\right)$$

The finite-G feasible and population risks are separately converted to their expected infinite-bag counterparts using the bridge in Equation (34):

$$\widehat{L}_\infty = \frac{G\widehat{L}_G - \widehat{L}_1}{G-1},$$

where $\widehat{L}_G$ is the risk of the average of the G subset weight vectors and $\widehat{L}_1$ is their average individual risk. For each (T, m, k) cell, the empirical shared-sample penalty is the Monte Carlo average of

$$\widehat{L}_\infty^{\text{feasible}}(k) - \widehat{L}_\infty^{\text{population}}(k).$$

4.2 Simulation Results

4.2.1 Forecasting performance

Table 1 reports the evaluation-sample results. When $H = 1.05$, bagged RSM and equal weighting have essentially identical MSPE. When $H = 6$, RSM reduces MSPE relative to equal weighting by 11.75 percent for $B = 3$ and 25.38 percent for $B = 8$. The fully estimated combination performs substantially worse than RSM, illustrating the sampling cost of estimating all $m = 50$ weights. The selected k in the nearly homogeneous designs should not be interpreted structurally because the corresponding loss curves are essentially flat.

B	H	Selected k	RSM	Equal weights	Gain vs. EW (%)	Oracle	Fully estimated
3	1.05	4	2.050	2.050	-0.03 (0.06)	2.047	2.569
8	1.05	4	2.151	2.152	0.05 (0.08)	2.144	2.694
3	6	8	2.073	2.349	11.75 (0.34)	2.037	2.570
8	6	11	2.201	2.950	25.38 (0.42)	2.134	2.696

Notes: The table evaluates the selected k using the 250 even-numbered replications. Gain against equal weighting is $100(\overline{L}_{\text{EW}} - \overline{L}_{\text{RSM}})/\overline{L}_{\text{EW}}$. Its paired delta-method Monte Carlo standard error, in percentage points, is shown in parentheses.

Table 1: Monte Carlo Performance and Benchmark Comparisons

The RSM results mentioned in Table 1 are based on the ex-post loss-minimizing choice of k across an independent set of training replications (the odd-numbered draws), which is then evaluated out-of-sample on the reported even-numbered replications. However, this

B	H	Method	Median k	MSPE	Gain vs. EW(%)
3	1.05	Fourth-root plug-in	2	2.049	0.03
3	1.05	RSM Selector	2	2.049	0.03
8	1.05	Fourth-root plug-in	3	2.149	0.13
8	1.05	RSM Selector	3	2.149	0.14
3	6	Fourth-root plug-in	12	2.078	11.54
3	6	RSM Selector	11	2.076	11.63
8	6	Fourth-root plug-in	17	2.209	25.13
8	6	RSM Selector	15	2.205	25.26

Notes: Both rules estimate their tuning quantities on each training sample and are evaluated on the 250 even-numbered replications. The direct plug-in rounds $(\widehat{D}_m T/\widehat{b})^{1/4}$ to the nearest admissible subset size.

Table 2: RSM selector and direct fourth-root plug-in performance

choice is infeasible in practice. That is why it is crucial to validate the *RSM Selector* and the direct fourth-root plug-in rule. We use the same simulated data for both approaches. The corresponding results can be found in Table 2.

The two RSM rules produce similar forecast combination results across all designs. As for the selected subset size, in the near-homogeneous case, $H = 1.05$, both approaches agree on a small subset size, with the same median k. When heterogeneity is present, i.e., $H = 6$, some differences can be observed: although it does not affect the overall performance, the fourth-root rule leads to a slightly larger k. The boundary correction makes the RSM Selector slightly more conservative. Most importantly, both feasible rules produce essentially the same gains relative to the equal weighting when compared to the ex-post loss-minimizing k, shown in Table 1.

4.2.2 Shared-sample mechanism

Figure 2 compares the estimated shared-sample penalty with its theoretical counterpart $\Gamma_{m,k}/T$ for every triplet (T, m, k), as defined in Section 4.1.2.

Almost all the points lie on the 45-degree line, indicating the close match between the estimated and theoretical penalties.

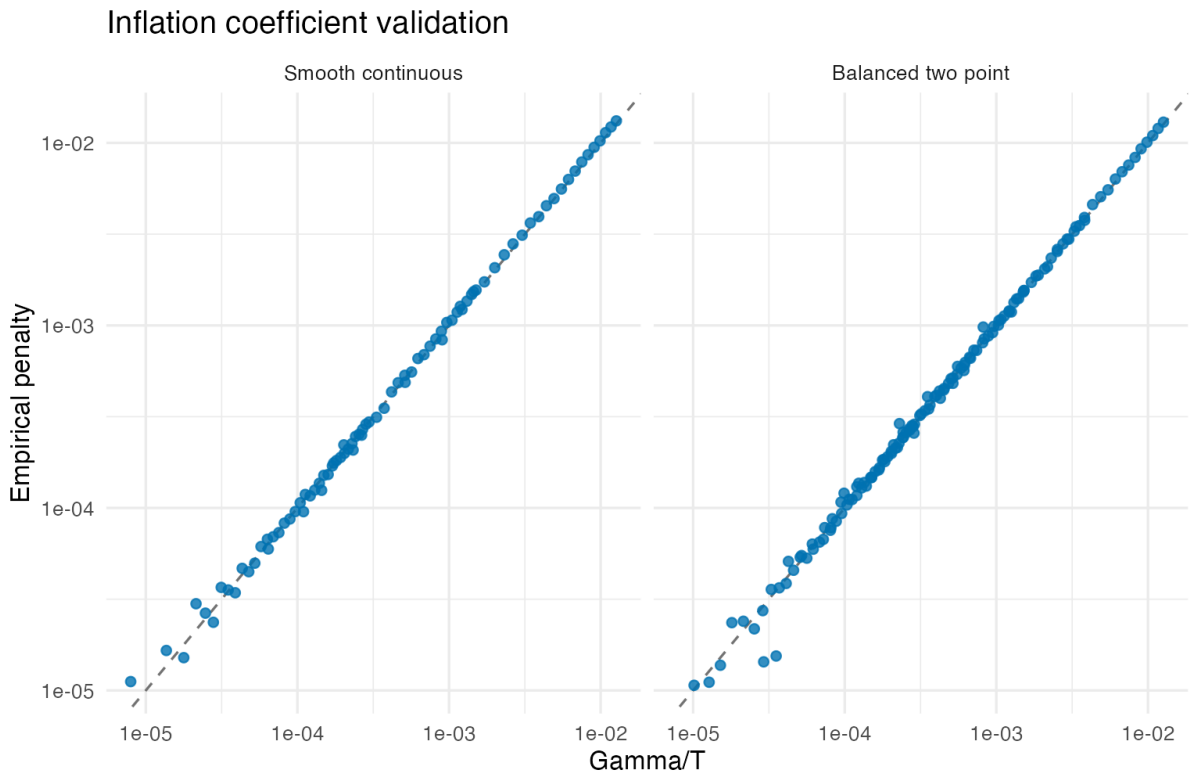


Figure 2: Validation of the shared-sample inflation coefficient. Each point represents one (T, m, k) cell. The horizontal axis is the theoretical penalty $\Gamma_{m,k}/T$, and the vertical axis is the empirical penalty. The dashed line is the 45-degree line. Both axes are logarithmic.

We also study how the optimal subset size k^* changes as the training sample T grows. We compare three distinct ways of choosing k. First, the empirical k^* is computed by directly minimizing a control-variate estimate of the simulated infinite-bagged risk. The other two procedures follow the theoretical results: the minimization of the complete criterion $Q_k^{\text{bag},\infty} + \frac{\Gamma_{m,k}}{T}$ and the leading asymptotic fourth-root rule. Figure 3 compares the subset size differences across the selection rules.

The complete criterion tracks the empirical optimum closely under both precision profiles. Besides, the leading fourth-root rule also captures the gradual increase in the optimal subset size as the training sample grows, although it does display larger finite-sample discrepancies.

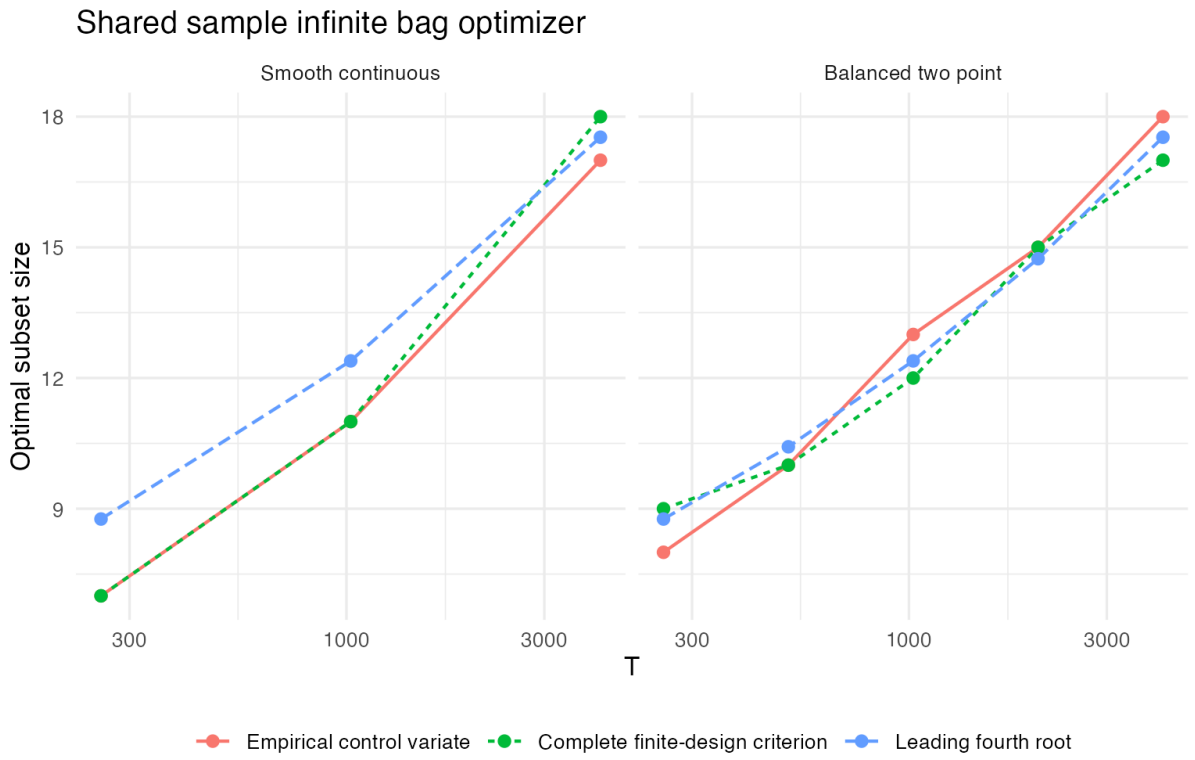


Figure 3: Optimal subset size for shared-sample infinite bagging under the balanced two-point baseline and the smooth continuous precision profile.

To summarize, the finite-G RSM generates forecasting gains compared to equal weighting when forecast precision is sufficiently heterogeneous. Even when the forecasts are nearly homogeneous, the RSM Selector and its fourth-root counterpart select median values of k close to equal weighting. Moreover, the feasible rules produce similar results to the infeasible ex-post loss-minimizing k. The derived finite-design shared-sample coefficient accurately explains the feasible estimation penalty. Finally, we illustrate the theory-predicted growth of the subset size k with the growing sample size.

5 Empirical Application

5.1 Data

For the empirical exercise, we use individual forecasts of annualized quarter-on-quarter US GDP price-index inflation from the Survey of Professional Forecasters (SPF) at horizons $h = 1$ and $h = 4$. For a survey submitted in quarter t, the h-quarter-ahead target is

constructed from two adjacent level forecasts in the same vintage as follows:[7]

$$100\left[\left(\frac{PGDP_{h+2}}{PGDP_{h+1}}\right)^4 - 1\right].$$

For example, the $h = 4$ target is quarterly inflation in quarter $t + 4$; it is not cumulative four-quarter or year-over-year inflation. Historical forecast errors at each origin use only the GDP price-index vintage available in the middle of that quarter, and combinations are scored against the first official release. This timing convention follows the real-time-data principle that a forecasting procedure should be estimated on the information set actually available at the origin (Croushore and Stark, 2001; Stark, 2010; Croushore, 2011).

The US SPF is known to be unbalanced. To balance the panel, we use a similar procedure to Conflitti et al. (2015).[8] At each origin, there are two requirements that a forecaster needs to fulfill to be included in the sample. Namely, a forecaster must submit a current forecast (it is never imputed) and must have responded in at least 12 of the last 48 quarters whose outcomes are observed. Then, missing historical values are filled using the mean of that forecaster's earlier observed forecasts in the window, whereas a leading gap (the first missing forecasts) is backfilled with the first observed own forecast. This panel-construction procedure allows for entering and exiting of forecasters without using future participation (Capistrán and Timmermann, 2009; Bürgi, 2023).

The common evaluation sample runs from Q2 2004 through Q2 2025. We choose this sample as the Philadelphia Fed took over the survey in Q2 1990, and the official SPF documentation cautions that participant identifiers in the earlier surveys are less reliable. Given the balancing rules, Q2 2004 is the first target available at both horizons. The Full sample contains 85 targets per horizon. To assess performance outside unusually volatile episodes, we introduce a Calm sample, in which the global financial crisis (Q3 2008–Q4 2009) and the broad pandemic disruption (Q1 2020–Q2 2021) are removed. That leaves 73 targets.

Additionally, we repeat the same experiment for 17 SPF variables (including GDP price inflation).[9] We preserve the same design as before for each variable while fixing

[7] Under standard SPF data conventions, the index j in PGDP_j corresponds to the forecast for calendar quarter $t + j - 2$. For example, $PGDP_1$ denotes the $t + 1 - 2 = t - 1$ period.

[8] The difference between balancing rules is discussed in Appendix D.4.

[9] Outcomes for the T-bill rate and 10-year Treasury yield are from FRED-MD dataset (McCracken and Ng, 2016).

$h = 1$. This exercise allows us to analyze whether the *RSM Selector* generally captures the right amount of weight estimation the data can support. Later, we pool the forecast losses across variables. In particular, pooling helps to test whether there are gains from departing from equal weighting at origins where the *RSM Selector* chooses $\widehat{k} \geq 2$.

The real-time SPF exercise is known to be a deliberately demanding setup for any forecast combination method. First, as established in the literature, the simple survey mean is difficult to beat because the underlying cross-sectional differences in forecaster precision are modest, leaving very little population gain to be extracted (Conflitti et al., 2015; Elliott and Liao, 2026). Also, the real-time balancing rule restricts the available training data to a short rolling window of $T = 48$ quarters. Therefore, the current setup acts as a rigorous test of the *RSM Selector's* ability to avoid overfitting.

5.2 RSM Regularization Performance

Table 3 summarizes the selected subset sizes of both feasible and infeasible estimators. At every origin, the *RSM Selector* minimizes the boundary-corrected plug-in criterion using the training data, as discussed in the box in Section 3.2. The median selected subset size is three at $h = 1$ and two at $h = 4$ for both samples. Every selected combination is computed by exact enumeration, so its forecast contains no subset-draw simulation error.

h	Sample	Quarters	Median selected k	Selected range	Best fixed k
1	Full	85	3	1–5	2
1	Calm	73	3	1–5	2
4	Full	85	2	1–3	4
4	Calm	73	2	1–3	4

Table 3: RSM Selector and Ex-Post Fixed-k Benchmark

Figure 4 reports the Full and Calm fixed-k loss curves. As shown in Table 3, the minimum is obtained when $k = 2$ at $h = 1$ and $k = 4$ at $h = 4$ for both samples.

The curves are U-shaped over the subset sizes feasible at every origin. It is worth mentioning, however, that fixed $k = 1$ produces nearly the same results as the ex-post fixed k. In other words, the RMSFE curves are almost flat around small values of k. This

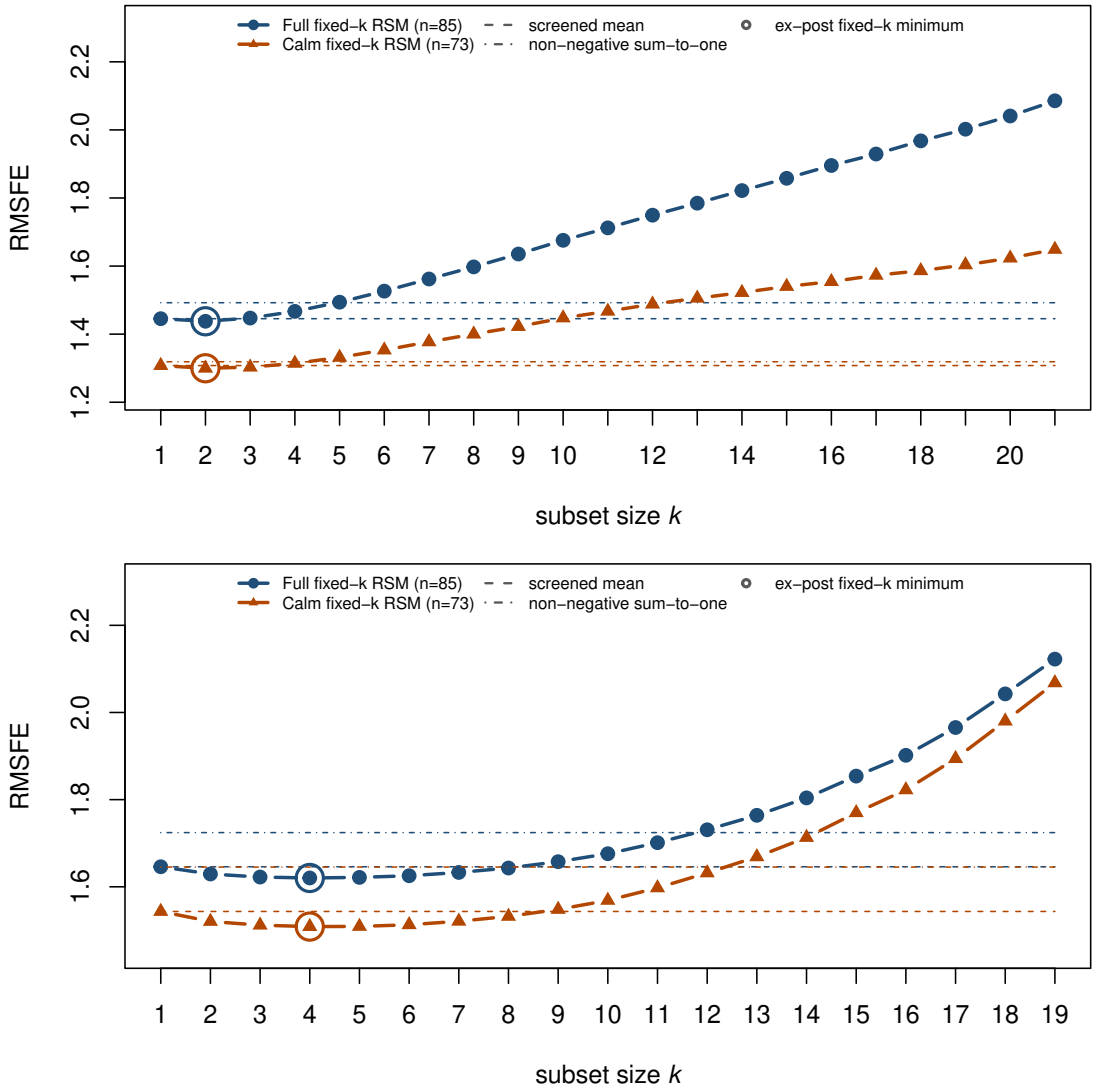


Figure 4: Real-Time Fixed-k RSM Loss Curves

Notes: The upper panel shows $h = 1$ and the lower panel $h = 4$.

illustrates that the amount of heterogeneity in the SPF sample is rather modest. This conclusion is later reiterated while evaluating forecasting performance in more detail.

h	Sample	Selector RMSFE	Fixed-k RMSFE	Gain vs fixed-k(%)
1	Full	1.430	1.438	+0.52
1	Calm	1.286	1.300	+1.03
4	Full	1.646	1.620	-1.56
4	Calm	1.539	1.509	-1.99

Table 4: RSM Selector versus the Infeasible Best Fixed Subset Size

In Table 4, the infeasible forecast combination performance is compared to the *RSM Selector*. In each scenario, the latter performs similarly to its infeasible counterpart in terms of RMSFE. Moreover, at $h = 1$, the best fixed-k produces slightly worse results. Overall, every difference from the infeasible ex-post benchmark is within two percent, consistent with the ability of the *RSM Selector* to capture the right amount of heterogeneity to estimate in the real-time setup.

Figure 5 depicts how the subset size k selected by the *RSM Selector* (blue line) and the best recursive k over the previous 20 origins (orange line) evolve over time. As stated in Table 3, the chosen size ranges from 1 to 5 when $h = 1$. Meanwhile, for the $h = 4$ case, the selected k is between 1 and 3.

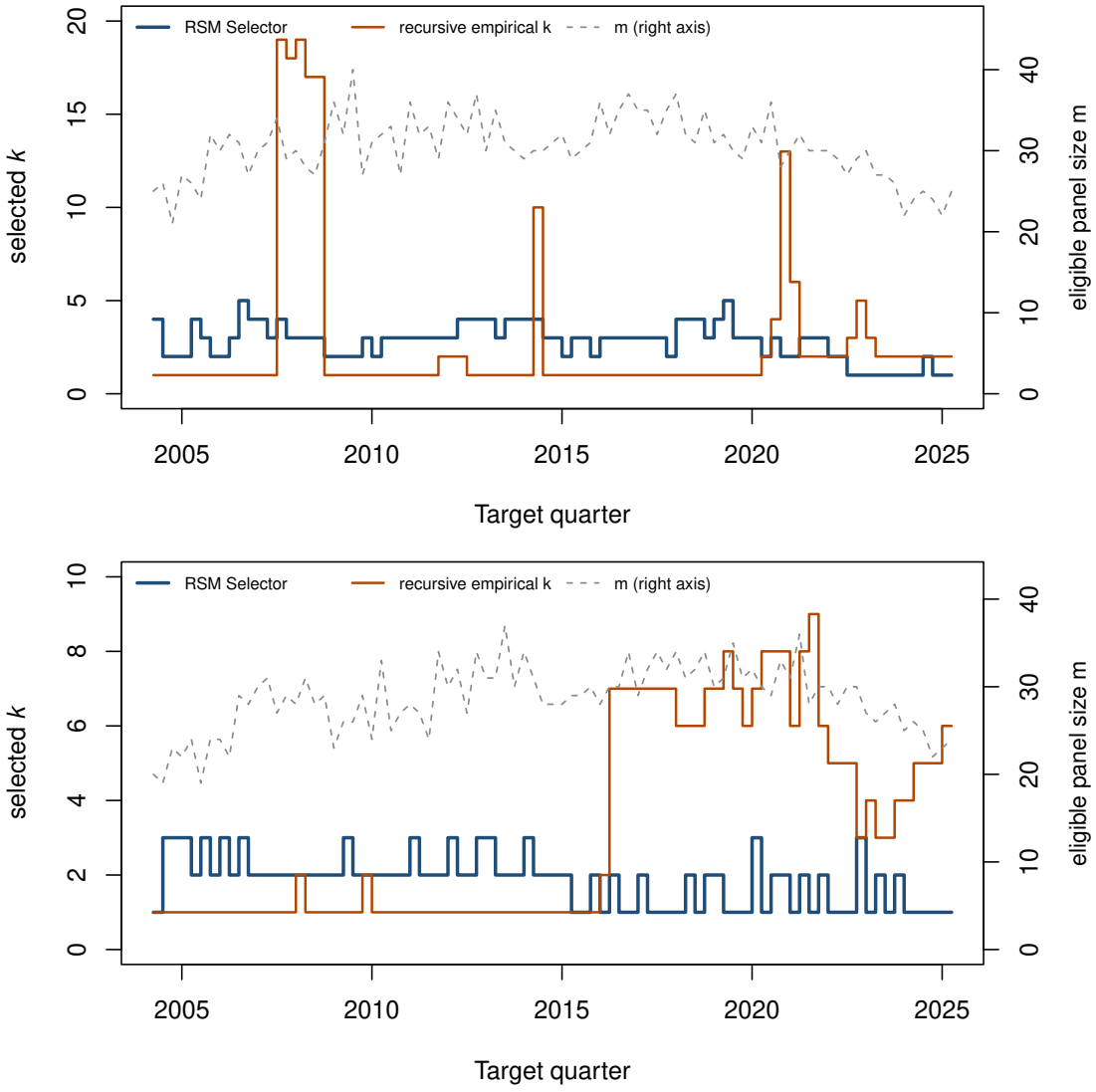


Figure 5: RSM Selector and Recursive Empirical Subset Size

Notes: The upper panel shows $h = 1$ and the lower panel $h = 4$.

Even though most of the time the best empirical subset size is 1 for both horizons, the recursive strategy may spontaneously produce rather huge values of k especially during unstable times (such as the financial crisis and the COVID-19 recession). Additionally, Figure 6 shows both the estimated heterogeneity constant $\widehat{\mathcal{D}}_m$ and the common-error variance $\widehat{b}$. For $h = 1$, before 2022, the *RSM Selector* chooses a median subset size of 3 since $\widehat{\mathcal{D}}_m$ dominates. However, after the 2022 inflation spike, when all the forecasters severely underestimated the true magnitude, $\widehat{\mathcal{D}}_m$ falls sharply whereas $\widehat{b}$ increases. As a result, the *RSM Selector* chooses equal weighting at most origins afterwards.

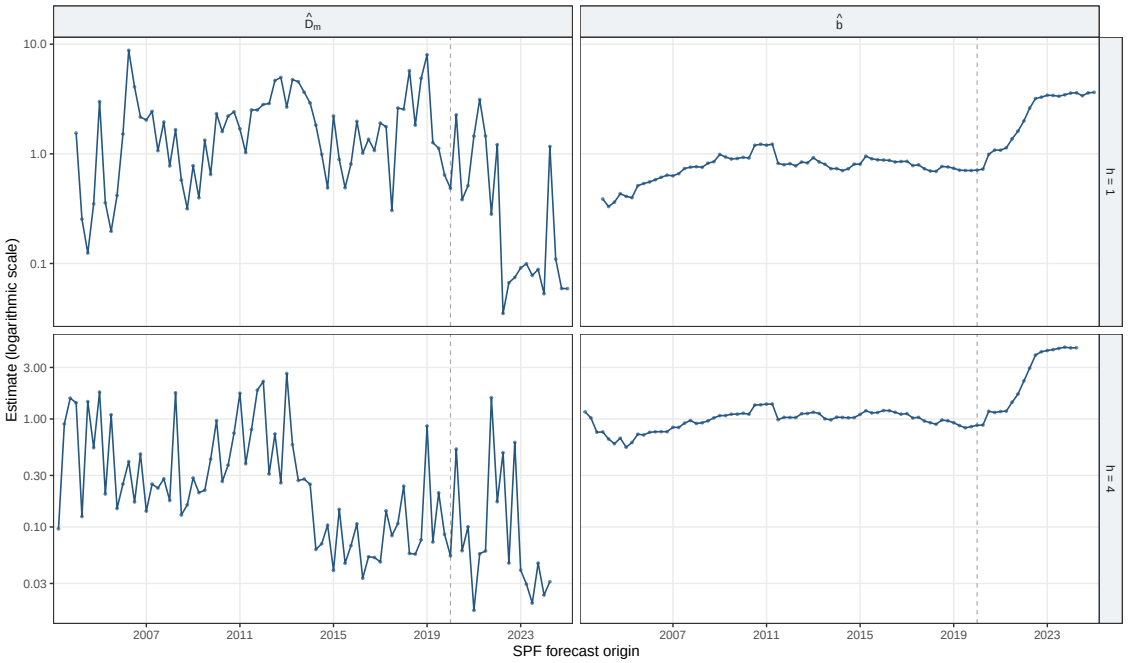


Figure 6: Estimated Heterogeneity Constant $\widehat{\mathcal{D}}_m$ and Common-Error Variance $\widehat{b}$

5.3 Forecast evaluation and regularization benchmarks

Table 5 compares several forecast combination and regularization methods. Besides the *RSM Selector*, we also estimate both survey means (mean of all available forecasters and mean only of those retained in the sample), non-negative sum-to-one weights (Conflitti et al., 2015), partially-egalitarian LASSO (peLASSO) (Diebold and Shin, 2019), and unrestricted sum-to-one weights. In Appendix D.1, the description of each model and performance of some other benchmarks can be found.

At $h = 1$, RSM has the lowest RMSFE among the methods considered.[10] Its Full

[10]HAC(4) correction is used to adjust for serial correlation and overlapping observations (Croushore,

	Full	Calm
Panel A: $h = 1$		
All-respondent mean	1.012 (.234)	1.018 (.237)
Equal weights, screened panel	1.011 (.239)	1.017 (.241)
RSM Selector	**1.000** (–)	**1.000** (–)
peLASSO, rolling-validation λ	1.026^{*} (.081)	1.022 (.176)
Non-negative sum-to-one weights	1.043^{**} (.031)	1.025 (.140)
Unrestricted sum-to-one weights	1.902^{***} (< .001)	1.727^{***} (< .001)
Panel B: $h = 4$		
All-respondent mean	1.011^{**} (.021)	1.011^{**} (.015)
Equal weights, screened panel	1.000 (.473)	1.003 (.206)
RSM Selector	**1.000** (–)	**1.000** (–)
peLASSO, rolling-validation λ	1.060^{*} (.086)	1.079^{*} (.082)
Non-negative sum-to-one weights	1.048^{**} (.010)	1.069^{***} (.004)
Unrestricted sum-to-one weights	2.265^{***} (.004)	2.439^{***} (.009)

Entries are each method's RMSFE divided by the RSM RMSFE. Parentheses contain p-values from a one-sided Diebold–Mariano with HAC(4) correction. *, **, and *** denote significance at the 10, 5, and 1 percent levels.

Table 5: Real-Time SPF GDP Price-Inflation Forecast Evaluation

and Calm gains over screened equality are 1.05 and 1.63 percent. The corresponding gains over the all-respondent mean are 1.17 and 1.76 percent. Yet, the improvement upon both equal-weighting benchmarks is not significant. Relative to non-negative sum-to-one weights, RSM lowers RMSFE by 4.16 and 2.46 percent; the Full-sample difference is significant at 5 percent. Since non-negative weights are usually sparse, this result is consistent with evidence that a well-designed model averaging technique can outperform sparse selection in forecasting settings (Kotchoni et al., 2019). Also, RSM lowers RMSFE relative to peLASSO in both samples, with a significant gain in the Full sample at the 10% level.

RSM and screened equal weights produce similar results at the longer forecasting horizon. Relative to the all-respondent mean, however, RSM lowers RMSFE by 1.07 and 1.12 percent. The gains are significant at the 5% level. As before, RSM also improves on non-negative sum-to-one weights by 4.57 and 6.47 percent.

Figure 7 illustrates the corresponding forecaster-level regularization for the common Q1 2022 target. RSM spreads the combination across the eligible panel while allowing moderate deviations from equality (at $h = 4$, the *RSM Selector* chooses $k = 1$ for this target, so RSM coincides with equal weights), whereas non-negative sum-to-one weighting concentrates its mass on fewer forecasters.

Also, Tables 11–14 show that the selected sizes and forecasting performance are stable when the GDP price deflator is replaced by CPI inflation and when the boundary-corrected criterion is replaced by the direct fourth-root approximation.

5.4 Pooled SPF Results

We repeat the same exercise independently for 17 SPF variables. Table 6 reports the selected subset sizes.[11]

2025). The results are robust if other lags are selected.

[11] Per-variable forecast evaluation can be found in Appendix D.5.

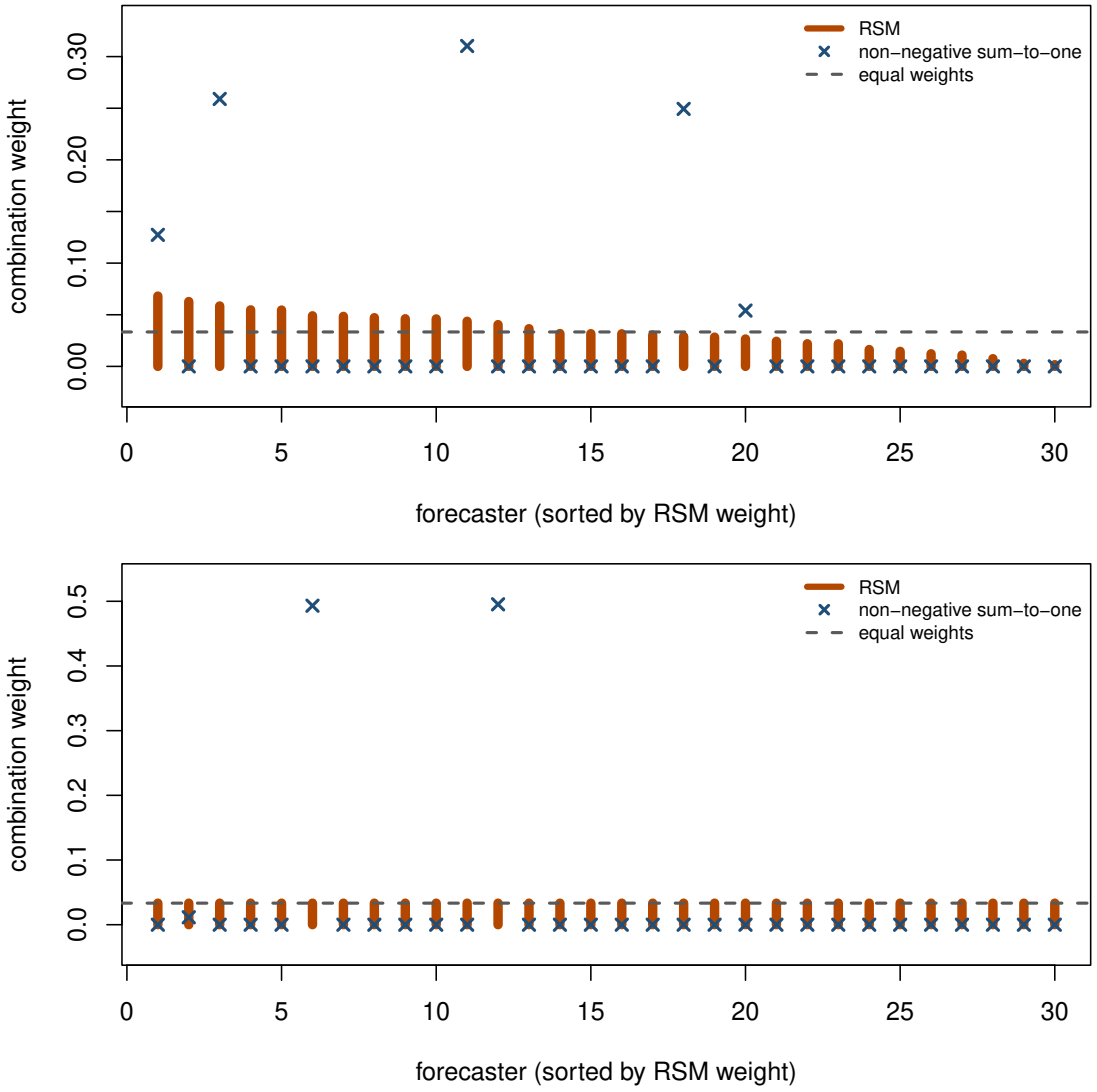


Figure 7: Real-time combination-weight profiles for Q1 2022

Notes: The upper panel shows $h = 1$ and the lower panel $h = 4$.

			RSM Selector			Ex-post best fixed k	
Variable	Origins	m	Median $\widehat{k}$	Range	$\widehat{k}=1$ (%)	k	Rel. RMSFE
Payroll employment growth	37	26	1	1–3	51	3	0.999
Unemployment rate	85	31	2	1–6	8	5	0.995
Housing starts	85	29	3	1–5	2	7	0.993
Industrial production growth	85	28	1	1–3	64	1	1.002
Nominal GDP growth	85	31	2	1–4	31	1	0.997
Real consumption growth	85	31	2	1–4	34	2	1.002
Real GDP growth	85	32	1	1–2	76	1	1.001
Real nonresidential investment growth	85	30	2	1–3	41	1	0.990
Real residential investment growth	85	29	1	1–3	65	2	0.993
Real state & local government growth	85	29	2	1–3	49	2	0.988
Core CPI inflation	24	28	2	1–4	46	9	0.933
Core PCE inflation	24	28	2	1–4	50	2	0.987
CPI inflation	85	31	1	1–3	51	1	1.002
PCE inflation	24	28	1	1–2	58	2	0.996
GDP price inflation	85	30	3	1–5	13	2	1.005
10-year Treasury yield	83	23	1	1–2	87	10	0.905
3-month T-bill rate	83	29	3	1–6	6	7	0.982

Notes:

Table 6: Selected and Ex-Post Best Subset Sizes across SPF Variables

The *RSM Selector*, across all variables, selects small subset sizes $\widehat{k}$: the median selected size is between 1 and 3 for every variable. In most cases, the ex-post best subset size is also of a small order. For 12 of 17 variables it is at most 3. Those findings are in line with the GDP price inflation case. More specifically, they confirm that the exploitable heterogeneity that a panel can support is small across most of the SPF variables. Typically, the infeasible ex-post best fixed k improves on the *RSM Selector* by at most 2%. This finding empirically supports the use of the *RSM Selector* in real time.

There are 5 variables (Unemployment rate, Housing starts, 3-month T-bill rate, Core CPI inflation, and 10-year Treasury yield) whose ex-post best fixed subset size exceeds 3. Those deviations from the *RSM Selector* selection can be explained in several ways. First, the Unemployment rate jumped to almost 13% in Q2 2020 and remained high through Q4 2020. With those quarters excluded from the evaluation, the best fixed k falls to 2, which matches the median selection of the *RSM Selector*. So, the difference between the subset sizes is caused predominantly by those COVID-related quarters. The

3-month T-bill rate variable is also affected by only a handful of quarters (Q4 2008 and Q3 2022 - Q4 2022).

Second, the RMSFE curve can essentially be flat over the whole region of small and moderate k. In particular, this is the case for the Housing starts variable. Again, the forecasting gains of using the best fixed subset size are negligible and are entirely caused by three quarters (Q3 2007 – Q4 2007, Q4 2020). Thus, in this case, the choice of k does not lead to economically meaningful differences. Instead, it illustrates that the many candidate subset sizes produce virtually the same forecasts.

Third, there are variables for which the number of evaluation origins is limited, making the comparison less reliable. This happens because those variables entered the SPF later. Core CPI inflation is one of those variables – it has only 24 evaluation origins. Moreover, all the forecast combination gain over the *RSM Selector* comes from three specific quarters (Q1 2022 – Q3 2022). Notice that for the Core PCE inflation, which has the same number of evaluation origins, the median selected subset size is the same as the ex-post fixed one.

The only SPF variable that does not fall into one of the special cases mentioned above is the 10-year Treasury yield. There, the best fixed k produces substantial gains in comparison to the *RSM Selector*, whose median size equals 1, meaning that the selection is too conservative. One potential explanation is that the 48-quarter estimation window cannot detect the persistent difference in accuracy across forecasters since the common-error variance b dominates $\mathcal{D}_m$, forcing $\widehat{k}$ to be small.

Before pooling, we address the problem of scaling. The SPF variables are measured using various units; thus directly comparing forecast errors is unreliable. Similar to Hyndman and Koehler (2006), we scale the errors by their full-sample equal-weight mean squared error (MSE).[12] Essentially, scaling makes each error unit-free, allowing for pooling different variables fairly. Table 7 presents the forecast error comparison when variables are scaled and pooled.

Across the SPF variables, when the *RSM Selector* selects $\widehat{k} = 2$, RSM reduces the RMSFE by 0.65 percent compared to equal weighting. This gain is statistically significant at the 10% level. These results support the real-time choice of k: when the *RSM Selector*

[12] Hyndman and Koehler (2006) scale the corresponding errors by the in-sample Mean Absolute Error (MAE). We use MSE instead for two reasons. First, the DM test uses squared-error differences and dividing by the MSE seems more natural. Second, the theory for the *RSM Selector* is based on the squared-loss function.

$\widehat{k}$	Gain over Equal Weighting
$\widehat{k} = 1$ (502 origins)	0
$\widehat{k} = 2$ (426 origins)	0.65* (.095)
$\widehat{k} \geq 2$ (708 origins)	0.63 (.107)

Notes: Entries are percentage reductions in RMSFE relative to equal weighting, pooled across variables. Each variable's squared-error differential is divided by its equal-weight MSE. The scaled errors are averaged across variables at each date. Parentheses contain p-values from a one-sided Diebold–Mariano test with HAC(4) correction. *, **, and *** denote significance at the 10, 5, and 1 percent levels.

Table 7: Pooled Forecast Combination Performance

selects $\widehat{k} \geq 2$, RSM reduces RMSFE relative to equal weighting.

6 Conclusion

This paper treats forecast combination as a regularization problem: how much heterogeneity in forecast quality can a given time-series sample support? By analyzing the trade-off between the approximation gains of exploiting forecaster heterogeneity and the estimation costs of recovering it, we demonstrate that RSM provides a regularization path indexed by the subset size k. The optimal RSM subset size grows at rate $T^{1/4}$. We propose the *RSM Selector*, which chooses the degree of regularization in the real-time setup from training data. Additionally, we validate RSM and the *RSM Selector* via Monte Carlo simulations.

In the real-time SPF application, using GDP price inflation and CPI, the *RSM Selector* chooses small subset sizes close to the ex-post optimal fixed size, and realized loss is U-shaped over the common feasible k range. We then extend the analysis to 17 SPF variables to validate the empirical findings across a broader set of variables. In most cases, the *RSM Selector* chooses small subsets with little loss in forecast accuracy relative to the infeasible best fixed subset size. When it selects $\widehat{k} = 2$, RSM delivers modest but statistically significant gains over equal weighting.

These findings help to explain the resilience of the survey mean. Forecasters can differ in their forecast accuracy, but the short samples, as well as their irregular participation in the forecasting rounds, prevent decision-makers from reliably estimating the

existing heterogeneity. By selecting $\widehat{k}$, the *RSM Selector* detects if the panel supports any weight estimation ($\widehat{k} \geq 2$) or none ($\widehat{k} = 1$). Thus, RSM retains its flexibility to exploit cross-sectional differences if the sample size is long enough and forecasters sufficiently heterogeneous, yet shrinks toward the equal-weight benchmark when estimation error threatens to overwhelm gains.

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Appendices

A Population and Bagging Results

A.1 Proof of Proposition 3

Proof. Apply the Sherman–Morrison–Woodbury formula[13] (Golub and Van Loan, 2013) to $\Omega_{e,k,S}$ to obtain:

$$\Omega_{e,k,S}^{-1} = D_S^{-1} - \frac{bD_S^{-1}\iota_k\iota_k' D_S^{-1}}{1+b\iota_k' D_S^{-1}\iota_k} = D_S^{-1} - \frac{bD_S^{-1}\iota_k\iota_k' D_S^{-1}}{1+bA_S}$$

Multiply by ι_k:

$$\Omega_{e,k,S}^{-1}\iota_k = D_S^{-1}\iota_k - \frac{bD_S^{-1}\iota_k\iota_k' D_S^{-1}\iota_k}{1+bA_S} = D_S^{-1}\iota_k\left(1-\frac{bA_S}{1+bA_S}\right) = \frac{D_S^{-1}\iota_k}{1+bA_S}$$

Premultiply by ι_k' to obtain:

$$\iota_k'\Omega_{e,k,S}^{-1}\iota_k = \frac{\iota_k' D_S^{-1}\iota_k}{1+bA_S} = \frac{A_S}{1+bA_S}.$$

Now, we solve the constrained minimization problem as in Bates and Granger (1969):

$$w_S = \frac{\Omega_{e,k,S}^{-1}\iota_k}{\iota_k'\Omega_{e,k,S}^{-1}\iota_k} = \frac{D_S^{-1}\iota_k/(1+bA_S)}{A_S/(1+bA_S)} = \frac{D_S^{-1}\iota_k}{A_S} \implies w_{i,S} = \tau_i/A_S$$

$$\min_{\iota_k' w=1} w'\Omega_{e,k,S}w = \frac{1}{\iota_k'\Omega_{e,k,S}^{-1}\iota_k} = \frac{1+bA_S}{A_S} = b + \frac{1}{A_S}.$$

The embedding $v_S = P_S w_S$ satisfies

$$v_S'\Omega_{e,m}v_S = w_S'P_S'\Omega_{e,m}P_S w_S = w_S'\Omega_{e,k,S}w_S = b + \frac{1}{A_S},$$

using $\Omega_{e,k,S} = P_S'\Omega_{e,m}P_S$ from (9).

For the full-panel case, stated in Equation (17), $S = \{1,\dots,m\}$, $P_S = I_m$, $A_S = A_m = m\bar{\tau}_m$. Hence:

$$w_i^{\text{opt}} = \frac{\tau_i}{A_m} \qquad Q^{\text{opt}} = b + \frac{1}{A_m} = b + \frac{1}{m\bar{\tau}_m}.$$

Finally, Equation (18) is obtained by evaluation of the quadratic form after plugging $w^{\text{ave}} = \frac{1}{m}\iota_m$:

$$Q^{\text{ave}} = m^{-2}\iota_m'\Omega_{e,m}\iota_m = m^{-2}\iota_m'(b\iota_m\iota_m' + D)\iota_m = m^{-2}\left(bm^2 + \sum_{i=1}^m \tau_i^{-1}\right) = b + \frac{1}{m^2}\sum_{i=1}^m \tau_i^{-1}.$$

□

[13] In this case, the formula is valid as $D_S = \text{diag}(\tau_i^{-1} | i \in S) \succ 0$, thus invertible, and $1 + bA_S > 0$.

A.2 Moments of the subset precision sum

Lemma 1 (Moments of A_S)**.** *Under Assumptions 2 and 4:*

(1) $k\tau_L \le A_S \le k\tau_U$.

(2) For $m \ge 2$,

$$\mathbb{E}_S(A_S) = k\bar{\tau}_m, \qquad \mathrm{Var}_S(A_S) = \frac{k(m-k)}{m-1} v_{\tau,m}. \tag{38}$$

(3) For $m \ge 3$, $2 \le k \le m$, *and every* i,

$$\mathbb{E}_S(A_S \mid i \in S) = k\bar{\tau}_m + \frac{m-k}{m-1}(\tau_i - \bar{\tau}_m), \tag{39}$$

$$\mathrm{Var}_S(A_S \mid i \in S) = \frac{(k-1)(m-k)}{m-2}\left\{\frac{m}{m-1} v_{\tau,m} - \frac{m}{(m-1)^2}(\tau_i - \bar{\tau}_m)^2\right\}. \tag{40}$$

(4) The constant $C = (4^{7/4}+4)(\tau_U - \tau_L)^3 < \infty$, *which depends only on* $\tau_U - \tau_L$, *satisfies, uniformly in* i, m, k,

$$\mathbb{E}_S|A_S - k\bar{\tau}_m|^3 \le Ck^{3/2}, \tag{41}$$

$$\mathbb{E}_S(|A_S - k\bar{\tau}_m|^3 \mid i \in S) \le Ck^{3/2}. \tag{42}$$

Proof. Statement (1) holds because by definition $A_S = \sum_{i \in S} \tau_i$. By Assumption 2, each τ_i satisfies $\tau_L \le \tau_i \le \tau_U$. Thus, $k\tau_L \le A_S \le k\tau_U$.

Statement (2) contains two unconditional moments of A_S. Assumption 4 imposes that $\Pr(i \in S) = \frac{k}{m}$ for every i. So, the unconditional expected value $\mathbb{E}_S(A_S)$ is as follows:

$$\mathbb{E}_S(A_S) = \sum_{i=1}^{m} \tau_i \Pr(i \in S) = \frac{k}{m}\sum_{i=1}^{m} \tau_i = \frac{k}{m} \cdot m\bar{\tau}_m = k\bar{\tau}_m.$$

For the unconditional variance $\mathrm{Var}_S(A_S)$, we use the results from Cochran (1977), which gives:

$$\mathrm{Var}_S(A_S) = \frac{k(m-k)}{m-1}\frac{1}{m}\sum_{i=1}^{m}(\tau_i - \bar{\tau}_m)^2 = \frac{k(m-k)}{m-1} v_{\tau,m}.$$

Statement (3) deals with the conditional moments. Conditional on inclusion $i \in S$, the remaining $k-1$ elements are sampled uniformly from the other $m-1$. The corresponding mean precision has the following form:

$$\bar{\tau}_{-i} = \frac{1}{m-1}\sum_{j \ne i} \tau_j = \frac{m\bar{\tau}_m - \tau_i}{m-1} = \bar{\tau}_m - \frac{\tau_i - \bar{\tau}_m}{m-1}.$$

Hence:

$$\mathbb{E}_S(A_S \mid i \in S) = \tau_i + (k-1)\bar{\tau}_{-i} = \tau_i + (k-1)\left(\bar{\tau}_m - \frac{\tau_i - \bar{\tau}_m}{m-1}\right) = k\bar{\tau}_m + (\tau_i - \bar{\tau}_m)\left(\frac{m-k}{m-1}\right).$$

To calculate the conditional variance $\mathrm{Var}_S(A_S \mid i \in S)$, first derive the variance of the $m-1$ precisions around their mean:

$$\begin{aligned}
v_{\tau,-i} &= \frac{1}{m-1}\sum_{j\neq i}(\tau_j - \bar{\tau}_{-i})^2 = \frac{1}{m-1}\sum_{j\neq i}\left(\tau_j - \bar{\tau}_m + \frac{\tau_i - \bar{\tau}_m}{m-1}\right)^2 \\
&= \frac{1}{m-1}\left\{\sum_{j\neq i}(\tau_j - \bar{\tau}_m)^2 + \frac{2(\tau_i - \bar{\tau}_m)}{m-1}\sum_{j\neq i}(\tau_j - \bar{\tau}_m) + \frac{(\tau_i - \bar{\tau}_m)^2}{m-1}\right\} \\
&= \frac{1}{m-1}\left\{m v_{\tau,m} - (\tau_i - \bar{\tau}_m)^2 - \frac{2(\tau_i - \bar{\tau}_m)^2}{m-1} + \frac{(\tau_i - \bar{\tau}_m)^2}{m-1}\right\} \\
&= \frac{1}{m-1}\left\{m v_{\tau,m} - (\tau_i - \bar{\tau}_m)^2 - \frac{(\tau_i - \bar{\tau}_m)^2}{m-1}\right\} \\
&= \frac{m}{m-1}v_{\tau,m} - \frac{m}{(m-1)^2}(\tau_i - \bar{\tau}_m)^2.
\end{aligned}$$

The same formula mentioned in Cochran (1977) can be applied to drawing $k-1$ elements without replacement from the remaining $m-1$:

$$\begin{aligned}
\mathrm{Var}_S(A_S \mid i \in S) &= \frac{(k-1)\{(m-1)-(k-1)\}}{(m-1)-1} v_{\tau,-i} \\
&= \frac{(k-1)(m-k)}{(m-2)}\left\{\frac{m v_{\tau,m}}{m-1} - \frac{m(\tau_i - \bar{\tau}_m)^2}{(m-1)^2}\right\}.
\end{aligned}$$

Statement (4) bounds both conditional and unconditional centered third moments. For the latter, by Lyapunov's inequality we obtain:

$$\mathbb{E}_S\left|A_S - k\bar{\tau}_m\right|^3 \leq \left\{\mathbb{E}_S\left(A_S - k\bar{\tau}_m\right)^4\right\}^{3/4}. \tag{43}$$

To express the upper-bound differently, we use of Hoeffding's theorem (Hoeffding, 1963; Bardenet and Maillard, 2015). It states that for any convex function, the expected value of a sum of random variables sampled without replacement is less than or equal to the expected value of the same function applied to independent random variables sampled with replacement. In our case, $A_S - k\bar{\tau}_m = \sum_{i\in S}(\tau_i - \bar{\tau}_m)$ is the sum of randomly selected k elements without replacement from the set of $\{\tau_1 - \bar{\tau}_m, \ldots, \tau_m - \bar{\tau}_m\}$. The fourth-moment formula for the sum with replacement has the following form:

$$\frac{k}{m}\sum_{i=1}^{m}(\tau_i - \bar{\tau}_m)^4 + 3k(k-1)v_{\tau,m}^2$$

Hence,

$$\mathbb{E}_S\left|A_S - k\bar{\tau}_m\right|^3 \le \left\{\mathbb{E}_S\left(A_S - k\bar{\tau}_m\right)^4\right\}^{3/4} \le \left\{\frac{k}{m}\sum_{i=1}^{m}(\tau_i - \bar{\tau}_m)^4 + 3k(k-1)v_{\tau,m}^2\right\}^{3/4}. \quad (44)$$

Using the fact that precisions are bounded,

$$|\tau_i - \bar{\tau}_m| \le \tau_U - \tau_L,$$

we obtain:

$$\begin{aligned}\mathbb{E}_S\left|A_S - k\bar{\tau}_m\right|^3 &\le \left\{\left[k + 3k(k-1)\right](\tau_U - \tau_L)^4\right\}^{3/4} \\ &= \left\{(3k^2 - 2k)(\tau_U - \tau_L)^4\right\}^{3/4} \\ &\le \left\{4k^2(\tau_U - \tau_L)^4\right\}^{3/4} \\ &= 4^{3/4}(\tau_U - \tau_L)^3 k^{3/2}. \end{aligned} \quad (45)$$

For the conditional bound, first consider $m \ge 2$. Conditional on $i \in S$, the other $k-1$ precisions are sampled without replacement from the remaining $m-1$ precisions. After centering at their finite-population mean, their range is, as in the previous case, at most $\tau_U - \tau_L$. The same Hoeffding–Lyapunov argument gives:

$$\mathbb{E}_S\left(\left|A_S - \mathbb{E}_S(A_S \mid i \in S)\right|^3 \,\middle|\, i \in S\right) \le 4^{3/4}(\tau_U - \tau_L)^3(k-1)^{3/2} \le 4^{3/4}(\tau_U - \tau_L)^3 k^{3/2}.$$

The conditional-mean identity in statement (3) implies that:

$$\left|\mathbb{E}_S(A_S \mid i \in S) - k\bar{\tau}_m\right| = \frac{m-k}{m-1}\left|\tau_i - \bar{\tau}_m\right| \le \tau_U - \tau_L.$$

Next, notice that

$$A_S - k\bar{\tau}_m = A_S - \mathbb{E}_S(A_S \mid i \in S) + \mathbb{E}_S(A_S \mid i \in S) - k\bar{\tau}_m.$$

By using the triangular inequality $|a+b|^3 \le 4\left(|a|^3 + |b|^3\right)$, we obtain:

$$\left|A_S - k\bar{\tau}_m\right|^3 \le 4\left|A_S - \mathbb{E}_S(A_S \mid i \in S)\right|^3 + 4\left|\mathbb{E}_S(A_S \mid i \in S) - k\bar{\tau}_m\right|^3. \quad (46)$$

Finally, take a conditional expectation:

$$\mathbb{E}_S\left(\left|A_S - k\bar{\tau}_m\right|^3 \,\middle|\, i \in S\right) \le 4^{7/4}(\tau_U - \tau_L)^3 k^{3/2} + 4(\tau_U - \tau_L)^3 \le \left(4^{7/4} + 4\right)(\tau_U - \tau_L)^3 k^{3/2},$$

where the last inequality uses $k^{3/2} \ge 1$.

If $m = k = 1$, then $A_S = \bar{\tau}_m$, so both third moments are zero. Thus, with

$$C = \left(4^{7/4} + 4\right)(\tau_U - \tau_L)^3,$$

both bounds hold uniformly in i, m, and k. □

A.3 Reciprocal identities

Lemma 2 (Exact reciprocal expansions)**.** *For $a > 0$, $\mu > 0$,*

$$\frac{1}{a} = \frac{1}{\mu} - \frac{a-\mu}{\mu^2} + \frac{(a-\mu)^2}{a\mu^2}, \tag{47}$$

$$\frac{1}{a} = \frac{1}{\mu} - \frac{a-\mu}{\mu^2} + \frac{(a-\mu)^2}{\mu^3} - \frac{(a-\mu)^3}{a\mu^3}. \tag{48}$$

Proof. Putting the right-hand side of (47) over the common denominator $a\mu^2$ gives numerator

$$a\mu - a(a-\mu) + (a-\mu)^2 = a\mu - \mu(a-\mu) = \mu^2,$$

and hence the fraction equals $1/a$. Similarly, for (48), the common denominator is $a\mu^3$, and the numerator is μ^3.

□

A.4 Finite-bag population bridge

Proof of Proposition 1. Expanding the quadratic form in Equation (12) gives:

$$Q_k^{\text{bag},G} = \frac{1}{G^2} \sum_{g=1}^{G} \sum_{\ell=1}^{G} \mathbb{E}(v'_{S_g} \Omega_{e,m} v_{S_\ell}).$$

There are G diagonal terms, each equal to Q_k^{sub}. For $g \neq \ell$, independence of the two subset draws and $\mathbb{E}(v_{S_g}) = \mathbb{E}(v_{S_\ell}) = \bar{v}_k$ imply that:

$$\mathbb{E}(v'_{S_g} \Omega_{e,m} v_{S_\ell}) = (\mathbb{E}v_{S_g})' \Omega_{e,m} (\mathbb{E}v_{S_\ell}) = \bar{v}'_k \Omega_{e,m} \bar{v}_k = Q_k^{\text{bag},\infty}.$$

There are $G(G-1)$ ordered off-diagonal pairs. Consequently,

$$\begin{aligned} Q_k^{\text{bag},G} &= \frac{G}{G^2} Q_k^{\text{sub}} + \frac{G(G-1)}{G^2} Q_k^{\text{bag},\infty} \\ &= Q_k^{\text{bag},\infty} + \frac{1}{G}(Q_k^{\text{sub}} - Q_k^{\text{bag},\infty}). \end{aligned}$$

Moreover, $Q_k^{\text{sub}} - Q_k^{\text{bag},\infty} = \mathbb{E}_S[(v_S - \bar{v}_k)' \Omega_{e,m} (v_S - \bar{v}_k)] \geq 0$ because $\Omega_{e,m}$ is positive definite. Hence:

$$Q_k^{\text{bag},\infty} \leq Q_k^{\text{bag},G} \leq Q_k^{\text{sub}}.$$

□

A.5 Average-subset expansion

Proof of Proposition 4. Equation (16) gives the exact identity $Q_k^{\text{sub}} = b + \mathbb{E}_S(A_S^{-1})$. Use (48) with $a = A_S$ and $\mu = k\bar{\tau}_m$, then take $\mathbb{E}_S$:

$$\mathbb{E}_S(A_S^{-1}) = \frac{1}{k\bar{\tau}_m} + \frac{\text{Var}_S(A_S)}{k^3\bar{\tau}_m^3} - \frac{1}{k^3\bar{\tau}_m^3}\mathbb{E}_S\left\{\frac{(A_S - k\bar{\tau}_m)^3}{A_S}\right\}.$$

The linear term vanishes by (38). Substituting the variance from the same equation yields:

$$\frac{\text{Var}_S(A_S)}{k^3\bar{\tau}_m^3} = \frac{m-k}{k^2(m-1)}\frac{v_{\tau,m}}{\bar{\tau}_m^3}.$$

For the remainder, Lemma 1(1) and (41) give:

$$\left|\mathbb{E}_S\left\{\frac{(A_S - k\bar{\tau}_m)^3}{A_S}\right\}\right| \leq \frac{Ck^{3/2}}{k\tau_L} = O(k^{1/2}).$$

After division by $k^3\bar{\tau}_m^3$, this is $O(k^{-5/2})$ uniformly because $\bar{\tau}_m \geq \tau_L$. Substitution into the preceding expansion therefore gives

$$Q_k^{\text{sub}} = b + \frac{1}{k\bar{\tau}_m} + \frac{m-k}{k^2(m-1)}\frac{v_{\tau,m}}{\bar{\tau}_m^3} + O(k^{-5/2}).$$

Since the middle term is $O(k^{-2})$ under the uniform precision bounds, $Q_k^{\text{sub}} = b + 1/(k\bar{\tau}_m) + O(k^{-2})$. □

A.6 Infinite-bagged risk representation

Proof of Proposition 5. Put $\delta = \bar{v}_k - w^{\text{opt}}$. Both vectors add to one, hence $\iota_m'\delta = 0$. The first-order condition for the full-panel optimum implies $\Omega_{e,m}w^{\text{opt}} = Q^{\text{opt}}\iota_m$, so

$$(w^{\text{opt}})'\Omega_{e,m}\delta = Q^{\text{opt}}\iota_m'\delta = 0.$$

Expanding around w^{opt} and using $\iota_m'\delta = 0$,

$$\begin{aligned} Q_k^{\text{bag},\infty} - Q^{\text{opt}} &= 2(w^{\text{opt}})'\Omega_{e,m}\delta + \delta'\Omega_{e,m}\delta \\ &= \delta'(b\iota_m\iota_m' + D)\delta \\ &= b(\iota_m'\delta)^2 + \delta' D\delta = \delta' D\delta = \sum_{i=1}^{m}\frac{(\bar{v}_{i,k} - w_i^{\text{opt}})^2}{\tau_i} = R_k^{bag}. \end{aligned}$$

Next, Equation (16) and the componentwise form of the embedding $v_S = P_S w_S$ established in the proof of Proposition 3 give $v_{S,i} = \mathbf{1}\{i \in S\}\tau_i/A_S$. Since $\Pr(i \in S) = k/m$, we have:

$$\bar{v}_{i,k} = \mathbb{E}_S(v_{S,i}) = \frac{k}{m}\tau_i\mathbb{E}_S(A_S^{-1} \mid i \in S) = \frac{\tau_i}{m}g_i(k).$$

Using $w_i^{\text{opt}} = \tau_i/(m\bar{\tau}_m)$ in the preceding squared distance gives

$$R_k^{bag} = \sum_{i=1}^{m} \frac{(\bar{v}_{i,k} - w_i^{\text{opt}})^2}{\tau_i} = \sum_{i=1}^{m} \frac{1}{\tau_i} \left\{ \frac{\tau_i}{m} \left(g_i(k) - \frac{1}{\bar{\tau}_m} \right) \right\}^2 = \frac{1}{m^2} \sum_{i=1}^{m} \tau_i \left(g_i(k) - \frac{1}{\bar{\tau}_m} \right)^2,$$

Finally,

$$\sum_{i=1}^{m} \tau_i g_i(k) = m \sum_{i=1}^{m} \bar{v}_{i,k} = m,$$

because every v_S, and hence $\bar{v}_k$, adds to one. □

A.6.1 Endpoint identities

Proof of Proposition 2. Both identities follow from the representation $\bar{v}_{i,k} = \tau_i g_i(k)/m$ in Equation (23).

Take $k = 1$. Conditional on $i \in S$ the subset of size one is necessarily $S = \{i\}$, so $A_S = \tau_i$ with probability one and

$$g_i(1) = 1 \cdot \mathbb{E}_S(A_S^{-1} \mid i \in S) = \frac{1}{\tau_i}, \qquad \text{hence} \qquad \bar{v}_{i,1} = \frac{\tau_i}{m} \cdot \frac{1}{\tau_i} = \frac{1}{m}$$

for every i. Therefore $\bar{v}_1 = m^{-1}\iota_m = w^{\text{ave}}$ by (7).

Take $k = m$. There is exactly one subset of size m, namely $S = \{1, \ldots, m\}$, so under Assumption 4 it is drawn with probability one. Then $A_S = A_m = m\bar{\tau}_m$, so:

$$g_i(m) = m \cdot \frac{1}{m\bar{\tau}_m} = \frac{1}{\bar{\tau}_m}, \qquad \text{hence} \qquad \bar{v}_{i,m} = \frac{\tau_i}{m} \cdot \frac{1}{\bar{\tau}_m} = \frac{\tau_i}{m\bar{\tau}_m} = w_i^{\text{opt}},$$

□

Global and conditional weights. The population global weight already obtained in Equation (23) can be written as

$$\bar{v}_{i,k} = \mathbb{E}_S(v_{S,i}) = \Pr(i \in S)\, \mathbb{E}_S(w_{i,S} \mid i \in S) = \frac{k}{m}\, \mathbb{E}_S(w_{i,S} \mid i \in S).$$

Consequently,

$$\bar{v}_{i,k} > \frac{1}{m} \quad \Longleftrightarrow \quad \mathbb{E}_S(w_{i,S} \mid i \in S) > \frac{1}{k}.$$

Thus a forecaster receives more than the equal global weight exactly when its mean weight, conditional on inclusion, exceeds the equal weight within a size-k subset.

B Estimation-Error Inflation for Shared-Sample Infinite Bagging

B.1 Local form of the population approximation error

Proposition 11 (Local form of R_k^{bag})**.** *Under Assumptions 2 and 4, as $k \to \infty$ with $k/m \to 0$,*

$$g_i(k) - \frac{1}{\bar{\tau}_m} = z_k \left\{ \frac{v_{\tau,m} - \bar{\tau}_m(\tau_i - \bar{\tau}_m)}{\bar{\tau}_m^3} + o(1) \right\} \qquad \text{uniformly in } i, \tag{49}$$

and consequently

$$R_k^{bag} = \frac{\mathcal{D}_m}{m} z_k^2 \{1 + o(1)\}, \qquad \mathcal{D}_m \geq \frac{\tau_L}{\bar{\tau}_m^6} \left(v_{\tau,m}^2 + \bar{\tau}_m^2 v_{\tau,m} \right), \tag{50}$$

and the lower bound is strictly positive for all sufficiently large m because $v_{\tau,m} \to v_\tau > 0$ under Assumption 2.

Proof. Apply the exact reciprocal identity (48) with $a = A_S$ and $\mu = k\bar{\tau}_m$, multiply both sides by k, and then take $\mathbb{E}_S(\cdot \mid i \in S)$. Since $g_i(k) = k\mathbb{E}_S(A_S^{-1} \mid i \in S)$, this gives:

$$g_i(k) - \frac{1}{\bar{\tau}_m} = -\frac{\mathbb{E}_S(A_S - k\bar{\tau}_m \mid i \in S)}{k\bar{\tau}_m^2} + \frac{\mathbb{E}_S\{(A_S - k\bar{\tau}_m)^2 \mid i \in S\}}{k^2\bar{\tau}_m^3} - \frac{\mathbb{E}_S\{(A_S - k\bar{\tau}_m)^3/A_S \mid i \in S\}}{k^2\bar{\tau}_m^3}.$$

We treat the three terms in turn. It is convenient to note once that $z_k = (m-k)/(km)$, so that $z_k \sim 1/k$ whenever $k/m \to 0$.

The linear term. By Lemma 1(3),

$$-\frac{\mathbb{E}_S(A_S - k\bar{\tau}_m \mid i \in S)}{k\bar{\tau}_m^2} = -\frac{(m-k)(\tau_i - \bar{\tau}_m)}{(m-1)k\,\bar{\tau}_m^2} = z_k \left\{ -\frac{m}{m-1} \cdot \frac{\tau_i - \bar{\tau}_m}{\bar{\tau}_m^2} \right\} = z_k \left\{ -\frac{\tau_i - \bar{\tau}_m}{\bar{\tau}_m^2} + O(m^{-1}) \right\},$$

uniformly in i, using $(m-k)/\{(m-1)k\} = z_k\, m/(m-1)$ and $m/(m-1) = 1 + O(m^{-1})$ together with the uniform precision bounds.

The quadratic term. Write $\mathbb{E}_S\{(A_S - k\bar{\tau}_m)^2 \mid i \in S\} = \mathrm{Var}_S(A_S \mid i \in S) + \{\mathbb{E}_S(A_S - k\bar{\tau}_m \mid i \in S)\}^2$ and divide each piece by $k^2\bar{\tau}_m^3$, factoring z_k out of both. Using

Lemma 1(3),

$$\frac{\mathrm{Var}_S(A_S \mid i \in S)}{k^2\bar{\tau}_m^3} = z_k \left\{ \frac{m(k-1)}{k(m-2)} \cdot \frac{\dfrac{m}{m-1} v_{\tau,m} - \dfrac{m}{(m-1)^2}(\tau_i - \bar{\tau}_m)^2}{\bar{\tau}_m^3} \right\},$$

$$\frac{\{\mathbb{E}_S(A_S - k\bar{\tau}_m \mid i \in S)\}^2}{k^2\bar{\tau}_m^3} = z_k \left\{ \frac{m(m-k)}{k(m-1)^2} \cdot \frac{(\tau_i - \bar{\tau}_m)^2}{\bar{\tau}_m^3} \right\}.$$

In the first line $m(k-1)/(k(m-2)) = (1-\frac{1}{k})(1+\frac{2}{m-2}) = 1 + O(k^{-1}+m^{-1})$, and the numerator inside the braces is $v_{\tau,m} + O(m^{-1})$ uniformly in i because $\frac{m}{m-1} v_{\tau,m} = v_{\tau,m} + \frac{v_{\tau,m}}{m-1}$ with $v_{\tau,m} \le (\tau_U - \tau_L)^2$ bounded, and the subtracted term is $O(m^{-1})$ since $\frac{m}{(m-1)^2} = O(m^{-1})$ and $(\tau_i - \bar{\tau}_m)^2 \le (\tau_U - \tau_L)^2$. Combining the prefactor with the numerator yields:

$$\begin{aligned} \frac{\mathrm{Var}_S(A_S \mid i \in S)}{k^2\bar{\tau}_m^3} &= z_k\{1 + O(k^{-1}+m^{-1})\} \frac{v_{\tau,m} + O(m^{-1})}{\bar{\tau}_m^3} \\ &= z_k \left\{ \frac{v_{\tau,m}}{\bar{\tau}_m^3} + O(k^{-1}+m^{-1}) \right\}. \end{aligned}$$

In the second line $m(m-k)/\{k(m-1)^2\} = O(k^{-1})$, so that the whole term $\frac{\{\mathbb{E}_S(A_S - k\bar{\tau}_m \mid i \in S)\}^2}{k^2\bar{\tau}_m^3}$ is absorbed into the $O(k^{-1}+m^{-1})$. Hence

$$\begin{aligned} \frac{\mathbb{E}_S\{(A_S - k\bar{\tau}_m)^2 \mid i \in S\}}{k^2\bar{\tau}_m^3} &= z_k \left\{ \frac{v_{\tau,m}}{\bar{\tau}_m^3} + O(k^{-1}+m^{-1}) \right\} \\ &= z_k \left\{ \frac{v_{\tau,m}}{\bar{\tau}_m^3} + o(1) \right\}, \qquad \text{uniformly in } i. \end{aligned}$$

The cubic remainder. Recall $A_S \ge k\tau_L$ and apply Lemma 1(4):

$$\left| \frac{\mathbb{E}_S\{(A_S - k\bar{\tau}_m)^3 / A_S \mid i \in S\}}{k^2\bar{\tau}_m^3} \right| \le \frac{\mathbb{E}_S(|A_S - k\bar{\tau}_m|^3 \mid i \in S)}{k^3\tau_L\bar{\tau}_m^3} \le \frac{Ck^{3/2}}{k^3\tau_L\bar{\tau}_m^3} = O(k^{-3/2}),$$

uniformly in i. Since $k/m \to 0$, we have $z_k = (1-k/m)/k \ge 1/(2k)$ for all sufficiently large (k,m), so $k^{-3/2} \le 2z_k\, k^{-1/2}$; the cubic term is therefore $z_k \cdot O(k^{-1/2}) = z_k \cdot o(1)$, uniformly in i.

Adding the three terms and using $(\tau_i - \bar{\tau}_m)/\bar{\tau}_m^2 = \bar{\tau}_m(\tau_i - \bar{\tau}_m)/\bar{\tau}_m^3$ gives

$$\begin{aligned} g_i(k) - \frac{1}{\bar{\tau}_m} &= z_k \left\{ -\frac{\tau_i - \bar{\tau}_m}{\bar{\tau}_m^2} + \frac{v_{\tau,m}}{\bar{\tau}_m^3} + o(1) \right\} \\ &= z_k \left\{ \frac{v_{\tau,m} - \bar{\tau}_m(\tau_i - \bar{\tau}_m)}{\bar{\tau}_m^3} + o(1) \right\}, \end{aligned}$$

which is (49). Substitute it into the score form of R_k^{bag} in (24):

$$R_k^{bag} = \frac{z_k^2}{m} \cdot \frac{1}{m} \sum_{i=1}^{m} \tau_i \left\{ \frac{v_{\tau,m} - \bar{\tau}_m(\tau_i - \bar{\tau}_m)}{\bar{\tau}_m^3} + o(1) \right\}^2.$$

The leading expression inside braces is uniformly bounded, and the remainder is uniform in i. After expanding the square and averaging, the cross term and squared remainder are therefore both $o(1)$. Thus

$$R_k^{bag} = \frac{z_k^2}{m}\left[\frac{1}{m}\sum_{i=1}^{m}\frac{\tau_i}{\bar{\tau}_m^6}\{v_{\tau,m} - \bar{\tau}_m(\tau_i - \bar{\tau}_m)\}^2 + o(1)\right] = \frac{z_k^2}{m}\{\mathcal{D}_m + o(1)\},$$

by (4). To justify converting this additive remainder to a multiplicative one, use $\tau_i \geq \tau_L$ and compute the unweighted average:

$$\frac{1}{m}\sum_{i=1}^{m}\big\{v_{\tau,m} - \bar{\tau}_m(\tau_i - \bar{\tau}_m)\big\}^2 = v_{\tau,m}^2 - 2\bar{\tau}_m v_{\tau,m}\cdot\frac{1}{m}\sum_i(\tau_i - \bar{\tau}_m) + \bar{\tau}_m^2\cdot\frac{1}{m}\sum_i(\tau_i - \bar{\tau}_m)^2 = v_{\tau,m}^2 + \bar{\tau}_m^2 v_{\tau,m},$$

since $\sum_i(\tau_i - \bar{\tau}_m) = 0$. Dividing by $\bar{\tau}_m^6$ gives the stated lower bound, which is bounded away from zero because $v_{\tau,m} \to v_\tau > 0$. Consequently,

$$R_k^{bag} = \frac{z_k^2}{m}\{\mathcal{D}_m + o(1)\} = \frac{\mathcal{D}_m}{m}z_k^2\{1 + o(1)\},$$

which is (50). □

B.2 A global lower bound on the approximation error

Proposition 11 describes R_k^{bag} on the interior range $k \to \infty$, $k/m \to 0$. To show that the optimizer lies in that range, we also need a bound that holds for every k.

Lemma 3 (Small-k localization)**.** *Under Assumptions 2 and 4, for all sufficiently large m and every $1 \leq k < m$,*

$$R_k^{bag} \;\geq\; \frac{\tau_L v_\tau^2}{4\tau_U^6\, m}z_k^2. \tag{51}$$

Proof. **Step 1: a lower bound on the excess conditional harmonic mean.** Apply the exact reciprocal identity (47) with $a = A_S$ and $\mu = k\bar{\tau}_m$:

$$\frac{1}{A_S} = \frac{1}{k\bar{\tau}_m} - \frac{A_S - k\bar{\tau}_m}{(k\bar{\tau}_m)^2} + \frac{(A_S - k\bar{\tau}_m)^2}{A_S(k\bar{\tau}_m)^2}.$$

Multiply by k and take $\mathbb{E}_S$. Since $\mathbb{E}_S(A_S) = k\bar{\tau}_m$ by Lemma 1(2), the linear term has expectation zero, leaving the exact identity

$$k\,\mathbb{E}_S(A_S^{-1}) - \frac{1}{\bar{\tau}_m} = \mathbb{E}_S\left\{\frac{(A_S - k\bar{\tau}_m)^2}{A_S\, k\,\bar{\tau}_m^2}\right\}.$$

Using $A_S \leq k\tau_U$ and $\mathrm{Var}_S(A_S) = k(m-k)v_{\tau,m}/(m-1)$ from Lemma 1(2),

$$k\,\mathbb{E}_S(A_S^{-1}) - \frac{1}{\bar{\tau}_m} \;\geq\; \frac{\mathrm{Var}_S(A_S)}{k^2\tau_U\bar{\tau}_m^2} = \frac{m}{m-1}\cdot\frac{v_{\tau,m}}{\tau_U\bar{\tau}_m^2}\,z_k \;\geq\; \frac{v_\tau}{2\tau_U^3}\,z_k,$$

the last step using $m/(m-1) \geq 1$, $\bar{\tau}_m \leq \tau_U$, and $v_{\tau,m} \geq v_\tau/2$ for all sufficiently large m.

Step 2: an aggregation identity for the scores. Additionally, because $\Pr(i \in S) = k/m$,

$$\sum_{i=1}^{m} g_i(k) = k \sum_{i=1}^{m} \mathbb{E}_S(A_S^{-1} \mid i \in S) = m\, \mathbb{E}_S\left(\frac{\sum_i \mathbf{1}\{i \in S\}}{A_S}\right) = mk\, \mathbb{E}_S(A_S^{-1}),$$

so that

$$\sum_{i=1}^{m} \left(g_i(k) - \frac{1}{\bar{\tau}_m}\right) = m\left\{k\, \mathbb{E}_S(A_S^{-1}) - \frac{1}{\bar{\tau}_m}\right\}.$$

Step 3: Cauchy–Schwarz. Now split each summand as $\sqrt{\tau_i}\big(g_i(k) - 1/\bar{\tau}_m\big)/\sqrt{\tau_i}$ and apply Cauchy–Schwarz, then use the score form (24) and $\sum_i \tau_i^{-1} \leq m/\tau_L$:

$$m^2 \left\{k\, \mathbb{E}_S(A_S^{-1}) - \frac{1}{\bar{\tau}_m}\right\}^2 \leq \left\{\sum_i \tau_i \Big(g_i(k) - \frac{1}{\bar{\tau}_m}\Big)^2\right\} \left(\sum_i \frac{1}{\tau_i}\right) \leq m^2\, R_k^{bag} \cdot \frac{m}{\tau_L}.$$

Rearranging and then inserting the lower bound from Step 1 gives

$$R_k^{bag} \geq \frac{\tau_L}{m} \left\{k\mathbb{E}_S(A_S^{-1}) - \frac{1}{\bar{\tau}_m}\right\}^2 \geq \frac{\tau_L}{m} \left(\frac{v_\tau}{2\tau_U^3} z_k\right)^2 = \frac{\tau_L v_\tau^2}{4\tau_U^6 m} z_k^2,$$

which is (51). □

B.3 Exact linearization of the estimated subset weights

We now turn to the feasible estimator. Under Assumptions 2 and 4,

$$e_t = h_t \iota_m + \xi_t, \qquad \Omega_{e,m} = \mathbb{E}(e_t e_t') = b\iota_m \iota_m' + D,$$

and $\Omega_{e,k,S} = P_S' \Omega_{e,m} P_S = b\iota_k \iota_k' + D_S$. The feasible subset weights minimize the uncentered sum of squared realized combination errors, without an intercept and with the zero mean treated as known:

$$\widehat{w}_S = \arg\min_{\iota_k' w = 1} \sum_{t=1}^{T} (w' P_S' e_t)^2 = \frac{\widehat{\Omega}_{e,k,S}^{-1} \iota_k}{\iota_k' \widehat{\Omega}_{e,k,S}^{-1} \iota_k}, \qquad \widehat{\Omega}_{e,k,S} = \frac{1}{T} \sum_{t=1}^{T} P_S' e_t e_t' P_S,$$

and $\widehat{v}_S = P_S \widehat{w}_S$, $\widehat{\bar{v}}_k = \mathbb{E}_S(\widehat{v}_S)$. The population counterpart is $w_S = D_S^{-1}\iota_k / A_S$, because

$$\Omega_{e,k,S} w_S = b\iota_k(\iota_k' w_S) + D_S w_S = \left(b + \frac{1}{A_S}\right) \iota_k.$$

Thus the feasible regression targets the population subset weights, with no bias from the common component. Define

$$K_S = P_S \left[D_S^{-1} - \frac{D_S^{-1} \iota_k \iota_k' D_S^{-1}}{A_S}\right] P_S'. \tag{52}$$

This expression does not depend on b.

Lemma 4 (Properties of K_S)**.** *K_S is symmetric and*

$$K_S\iota_m = 0, \qquad K_S D K_S = K_S, \qquad K_S D v_S = 0, \qquad \operatorname{tr}(DK_S) = k-1. \tag{53}$$

Moreover $D^{1/2}K_S D^{1/2}$ is an orthogonal projection of rank $k-1$.

Proof. The bracket in (52) is symmetric: D_S^{-1} is diagonal, and the second term is the symmetric outer product of $D_S^{-1}\iota_k$ with itself, divided by A_S. Thus

$$K_S' = K_S.$$

Since $\iota_k' D_S^{-1}\iota_k = A_S$,

$$\left[D_S^{-1} - \frac{D_S^{-1}\iota_k\iota_k' D_S^{-1}}{A_S}\right]\iota_k = 0,$$

and hence $K_S\iota_m = 0$, using $P_S'\iota_m = \iota_k$.

Because D is diagonal and P_S selects coordinates, $D^{1/2}P_S = P_S D_S^{1/2}$, so

$$D^{1/2}K_S D^{1/2} = P_S\left[I_k - \frac{(D_S^{-1/2}\iota_k)(D_S^{-1/2}\iota_k)'}{A_S}\right]P_S'.$$

As $\|D_S^{-1/2}\iota_k\|^2 = \iota_k' D_S^{-1}\iota_k = A_S$, the bracket is the orthogonal projection onto the orthogonal complement of $D_S^{-1/2}\iota_k$; it is therefore symmetric, idempotent, and of rank $k-1$. Its embedding inherits all three properties: using $P_S'P_S = I_k$,

$$\operatorname{rank}(D^{1/2}K_S D^{1/2}) = k-1.$$

Idempotence $(D^{1/2}K_S D^{1/2})^2 = D^{1/2}K_S D^{1/2}$, written out as $D^{1/2}K_S D K_S D^{1/2} = D^{1/2}K_S D^{1/2}$, becomes $K_S D K_S = K_S$ after pre- and post-multiplying by the invertible $D^{-1/2}$; and the trace of a projection is its rank, giving $\operatorname{tr}(DK_S) = \operatorname{tr}(D^{1/2}K_S D^{1/2}) = k-1$.

Finally, $(Dv_S)_i = 1/A_S$ for $i \in S$ and 0 otherwise. Therefore $P_S' D v_S = \iota_k/A_S$, and substituting this vector into the bracket in (52) gives $K_S D v_S = 0$. □

Proposition 12 (Exact linearization)**.** *Under Assumptions 2, 3, and 4, with $T \geq k$, let $\widehat{\Delta} = T^{-1}\sum_{t=1}^{T}(e_t e_t' - \Omega_{e,m})$. Then*

$$\widehat{v}_S - v_S = -K_S\widehat{\Delta}\,\widehat{v}_S, \tag{54}$$

and therefore

$$\widehat{v}_S - v_S = \frac{1}{T}\sum_{t=1}^{T}\phi_S(t) + \rho_S, \qquad \phi_S(t) = -K_S\xi_t\{h_t + v_S'\xi_t\}, \tag{55}$$

with $\rho_S = -K_S\widehat{\Delta}(\widehat{v}_S - v_S)$ a product of two estimation errors. Moreover $\mathbb{E}\{\phi_S(t)\} = 0$.

Proof. The two constrained minimisation problems have first-order conditions $\widehat{\Omega}_{e,k,S}\widehat{w}_S = \widehat{\mu}\,\iota_k$ and $\Omega_{e,k,S}w_S = \mu\,\iota_k$ for scalar multipliers $\widehat{\mu}, \mu$. Writing $\widehat{\Omega}_{e,k,S} = \Omega_{e,k,S} + P_S'\widehat{\Delta}P_S$ and subtracting,

$$\Omega_{e,k,S}(\widehat{w}_S - w_S) + P_S'\widehat{\Delta}P_S\widehat{w}_S = (\widehat{\mu} - \mu)\iota_k.$$

Premultiply by $P_S'K_SP_S$. By (52), this matrix annihilates ι_k and, whenever $\iota_k'x = 0$, satisfies

$$P_S'K_SP_S\Omega_{e,k,S}x = \left[D_S^{-1} - \frac{D_S^{-1}\iota_k\iota_k'D_S^{-1}}{A_S}\right]D_Sx = x - \frac{D_S^{-1}\iota_k(\iota_k'x)}{A_S} = x.$$

Both $\widehat{w}_S$ and w_S sum to one, so the displayed identity applies to their difference and gives

$$\widehat{w}_S - w_S + P_S'K_S\widehat{\Delta}P_S\widehat{w}_S = 0.$$

Premultiplying by P_S and using $P_SP_S'K_S = K_S$ together with $P_SP_S'\widehat{v}_S = \widehat{v}_S$ (as $\widehat{v}_S$ is supported on S) gives

$$\widehat{v}_S - v_S + K_S\widehat{\Delta}\widehat{v}_S = 0,$$

which is (54).

Now write $\widehat{v}_S = v_S + (\widehat{v}_S - v_S)$ on the right of (54). The first piece is $-K_S\widehat{\Delta}v_S = T^{-1}\sum_t\{-K_S(e_te_t' - \Omega_{e,m})v_S\}$, and the second is ρ_S. To identify the summand, note that $K_S\iota_m = 0$ gives $K_Se_t = K_S\xi_t$; that $\iota_m'v_S = 1$ gives $e_t'v_S = h_t + v_S'\xi_t$; and that

$$K_S\Omega_{e,m}v_S = bK_S\iota_m(\iota_m'v_S) + K_SDv_S = 0$$

by (53). Hence $-K_S(e_te_t' - \Omega_{e,m})v_S = -K_Se_t(e_t'v_S) = \phi_S(t)$, which is (55).

Finally $\mathbb{E}\{\phi_S(t)\} = -K_S\mathbb{E}(\xi_th_t) - K_S\mathbb{E}(\xi_t\xi_t')v_S = 0 - K_SDv_S = 0$, using $\mathbb{E}(\xi_th_t) = \mathrm{Cov}(h_t, \xi_t) = 0$ from Assumption 2 and (53). □

B.4 The cross-subset coefficient

Infinite bagging averages $\widehat{v}_S$ over *all* subsets, fitted on *one* training sample. Before stating the result, we derive the object it computes, rather than positing it.

Let R be an independent copy of S, and write $\mathbb{E}_R$ for expectation over this second subset draw. Both subset draws are independent of the training sample.

Averaging the exact linearization of Proposition 12 over subsets gives

$$\bar{\widehat{v}}_k - \bar{v}_k = \frac{1}{T}\sum_{t=1}^{T}\mathbb{E}_S\phi_S(t) + \mathbb{E}_S(\rho_S).$$

To identify the first-order risk contribution, temporarily omit the nonlinear remainder and expand the squared D-norm of the linear term. We obtain

$$\begin{aligned}
&\mathbb{E}\left[\left\{\frac{1}{T}\sum_{t=1}^{T}\mathbb{E}_S\phi_S(t)\right\}' D \left\{\frac{1}{T}\sum_{u=1}^{T}\mathbb{E}_R\phi_R(u)\right\}\right] \\
&= \frac{1}{T^2}\sum_{t=1}^{T}\sum_{u=1}^{T}\mathbb{E}\Big[\big\{\mathbb{E}_S\phi_S(t)\big\}' D \big\{\mathbb{E}_R\phi_R(u)\big\}\Big] \\
&= \frac{1}{T^2}\sum_{t=1}^{T}\mathbb{E}\Big[\big\{\mathbb{E}_S\phi_S(t)\big\}' D \big\{\mathbb{E}_R\phi_R(t)\big\}\Big] \\
&= \frac{1}{T}\,\mathbb{E}\Big[\big\{\mathbb{E}_S\phi_S(t)\big\}' D \big\{\mathbb{E}_R\phi_R(t)\big\}\Big] \\
&= \frac{1}{T}\,\mathbb{E}_{S,R}\,\mathbb{E}\big[\phi_S(t)' D \phi_R(t)\big] = \frac{1}{T}\,\mathbb{E}_{S,R}\,\Gamma_{SR}.
\end{aligned}$$

Here

$$\Gamma_{SR} = \mathbb{E}\big\{\phi_S(t)' D\, \phi_R(t)\big\}. \tag{56}$$

The off-time terms vanish by independence across observations and the mean-zero property in Proposition 12; the T surviving terms are identical, and bilinearity moves the independent subset expectations outside. Thus the time average keeps only T of the T^2 time pairs, producing $1/T$, whereas the subset average keeps every pair (S, R). The resulting covariance is nonzero precisely because all subset regressions share the training sample.

Its average over independent subset draws, $\Gamma_{m,k} = \mathbb{E}_{S,R}(\Gamma_{SR})$, is therefore the coefficient multiplying $1/T$ in the loss expansion of Proposition 8.

Proposition 13 (Complete first-order coefficient)**.** *Under Assumptions 2, 3, and 4, let S and R be independent uniform subsets of size k, and let Γ_{SR} be as in (56). Then*

$$\Gamma_{m,k} = \mathbb{E}_{S,R}(\Gamma_{SR}) = \mathbb{E}_{S,R}\Big[\big\{b + v_S' D v_R\big\}\,\mathrm{tr}(DK_S DK_R) + v_R' DK_S DK_R D v_S\Big]. \tag{57}$$

Setting $R = S$ gives, exactly,

$$\Gamma_{SS} = \big\{Q(S)\big\}(k-1). \tag{58}$$

Proof. By Assumptions 2 and 3, h_t and ξ_t are independent centred Gaussians with variances b and D. From (55), $\phi_S(t) = -K_S\xi_t(e_t' v_S)$ with $e_t' v_S = h_t + v_S'\xi_t$, and K_S is symmetric, so

$$\Gamma_{SR} = \mathbb{E}\Big[(h_t + v_S'\xi_t)(h_t + v_R'\xi_t)\,\xi_t' K_S DK_R \xi_t\Big].$$

Expanding the first product gives three kinds of terms.

(a) The h_t^2 term. By independence,

$$\mathbb{E}(h_t^2)\,\mathbb{E}(\xi_t' K_S D K_R \xi_t) = b\,\mathrm{tr}(D K_S D K_R),$$

using $\mathbb{E}(\xi' A \xi) = \mathrm{tr}(AD)$ for centred ξ with variance D.

(b) The two terms with one factor h_t. Each equals $\mathbb{E}(h_t)$ times a moment of ξ_t, hence zero.

(c) The fourth-moment term. Applying Isserlis' theorem directly to the four Gaussian factors gives

$$\mathbb{E}\big[(v_S'\xi_t)(v_R'\xi_t)\xi_t' K_S D K_R \xi_t\big] = (v_S' D v_R)\,\mathrm{tr}(D K_S D K_R) + v_S' D K_S D K_R D v_R + v_R' D K_S D K_R D v_S.$$

The middle term vanishes, because K_S is symmetric and $K_S D v_S = 0$ by (53):

$$v_S' D K_S D K_R D v_R = (K_S D v_S)' D K_R D v_R = 0.$$

Adding (a) and (c) and averaging over the independent draws gives

$$\Gamma_{m,k} = \mathbb{E}_{S,R}\Big[\{b + v_S' D v_R\}\,\mathrm{tr}(D K_S D K_R) + v_R' D K_S D K_R D v_S\Big],$$

which is (57).

For (58), put $R = S$. By (53), $K_S D K_S = K_S$ and $\mathrm{tr}(D K_S) = k-1$, so $\mathrm{tr}(D K_S D K_S) = k-1$. The surviving pairing vanishes because

$$v_S' D K_S D K_S D v_S = (K_S D v_S)' D (K_S D v_S) = 0.$$

Finally,

$$v_S' D v_S = \sum_{i \in S} \left(\frac{\tau_i}{A_S}\right)^2 \tau_i^{-1} = \frac{\sum_{i\in S}\tau_i}{A_S^2} = \frac{1}{A_S},$$

and hence, by Proposition 3,

$$\Gamma_{SS} = \left(b + \frac{1}{A_S}\right)(k-1) = Q(S)(k-1),$$

which proves (58). □

B.5 The order of the coefficient

Lemma 5 (Two subset-weight pairings)**.** *Under Assumptions 2 and 4, let $J = |S \cap R|$. There is a constant C depending only on τ_L, τ_U such that, pointwise in (S, R),*

$$\big|(v_S' D v_R)\,\mathrm{tr}(D K_S D K_R)\big| \le \frac{CJ}{k}, \qquad \big|v_R' D K_S D K_R D v_S\big| \le \frac{CJ}{k^2},$$

and

$$\mathbb{E}_{S,R}\big|(v_S' D v_R)\,\mathrm{tr}(DK_S DK_R)\big| \le \frac{Ck}{m}, \qquad \mathbb{E}_{S,R}\big|v_R' DK_S DK_R D v_S\big| \le \frac{C}{m}. \tag{59}$$

Both are $o(k^2/m)$ as $k \to \infty$.

Proof. Throughout, $k\tau_L \le A_S, A_R \le k\tau_U$ by Lemma 1(1).

First pairing. Only shared indices contribute to $v_S' D v_R$, and

$$v_S' D v_R = \sum_{i \in S\cap R} \frac{\tau_i}{A_S A_R}, \qquad \text{so} \qquad |v_S' D v_R| \le \frac{J\tau_U}{k^2\tau_L^2}.$$

For the trace, $D^{1/2}K_S D^{1/2}$ and $D^{1/2}K_R D^{1/2}$ are orthogonal projections of rank $k-1$ (Lemma 4), so each has Frobenius norm $\sqrt{k-1}$ and Cauchy–Schwarz for the Frobenius inner product gives $|\,\mathrm{tr}(DK_S DK_R)| \le k-1$. Multiplying the two bounds gives $\frac{CJ}{k}$.

Second pairing. Since $(Dv_S)_i = \mathbf{1}\{i \in S\}/A_S$ and $(Dv_R)_i = \mathbf{1}\{i \in R\}/A_R$, direct substitution in (52) gives

$$v_R' DK_S DK_R D v_S = \frac{1}{A_S A_R} \sum_{a \in S\cap R} \tau_a \left\{1 - \frac{A_{S\cap R}}{A_S}\right\}\left\{1 - \frac{A_{S\cap R}}{A_R}\right\}, \qquad A_{S\cap R} = \sum_{i\in S\cap R} \tau_i.$$

Both braces lie in $[0,1]$ because $0 \le A_{S\cap R} \le \min(A_S, A_R)$, so the absolute value is at most $J\tau_U/(k^2\tau_L^2) = \frac{CJ}{k^2}$.

Finally $J = \sum_i \mathbf{1}\{i \in S\}\mathbf{1}\{i \in R\}$ and S, R are independent with $\Pr(i \in S) = k/m$, so $\mathbb{E}J = m(k/m)^2 = k^2/m$. Hence the two pointwise bounds give explicitly

$$\mathbb{E}_{S,R}\big|(v_S' D v_R)\,\mathrm{tr}(DK_S DK_R)\big| \le \frac{C}{k}\mathbb{E}J = \frac{Ck}{m}, \qquad \mathbb{E}_{S,R}\big|v_R' DK_S DK_R D v_S\big| \le \frac{C}{k^2}\mathbb{E}J = \frac{C}{m},$$

which is (59). Relative to k^2/m, the two bounds have ratios $(Ck/m)/(k^2/m) = C/k$ and $(C/m)/(k^2/m) = C/k^2$, respectively. Both ratios converge to zero as $k \to \infty$. □

Proposition 14 (Order of $\Gamma_{m,k}$)**.** *Under Assumptions 2, 3, and 4, let $\bar{K} = \mathbb{E}_S(K_S)$. As $m \ge k \to \infty$,*

$$\mathrm{tr}(D\bar{K}D\bar{K}) = \frac{k^2}{m}\big\{1 + O(k^{-1} + m^{-1})\big\}, \tag{60}$$

and consequently

$$\Gamma_{m,k} = b\frac{k^2}{m}\{1 + o(1)\}. \tag{61}$$

Moreover, for every $1 \le k \le m$ with $m \ge 2$, without asymptotics,

$$\mathrm{tr}(D\bar{K}D\bar{K}) \;\ge\; \frac{(k-1)^2}{m-1}, \tag{62}$$

with equality when all precisions are equal.

A single subset fit has effective dimension $k-1$. Averaging over subsets does *not* average $k-1$ independent errors: the same m coordinates are reused, and $\bar{K}$ has entries of order k/m rather than order one. Squaring and summing m diagonal entries turns $(k/m)^2 \cdot m$ into k^2/m. In the orthonormal complete-subset calculation of Elliott et al. (2013), the shrinkage factor is k/K, so the associated variance term has the same k^2/K order. Boot and Nibbering (2019) give a broader random-subspace bias–variance comparison. Proposition 14 shows that the complete-subset order continues to hold under the adding-up restriction and heterogeneous precisions.

Proof. **Step 1: entries of $\bar{K}$.** From (52), on the selected coordinates K_S has diagonal entry $\tau_i - \tau_i^2/A_S$ and off-diagonal entry $-\tau_i\tau_j/A_S$. Since $\Pr(i \in S) = k/m$ and $\Pr(i, j \in S) = k(k-1)/\{m(m-1)\}$ for $i \neq j$, taking expectations gives exactly

$$\bar{K}_{ii} = \frac{k}{m}\tau_i\Big\{1-\tau_i\mathbb{E}_S(A_S^{-1} \mid i \in S)\Big\}, \qquad \bar{K}_{ij} = -\frac{k(k-1)}{m(m-1)}\tau_i\tau_j\,\mathbb{E}_S(A_S^{-1} \mid i, j \in S) \;\; (i \neq j).$$

By Lemma 1(1), $A_S^{-1} \leq 1/(k\tau_L)$ holds pointwise on every subset S, hence under any conditioning event; both conditional expectations are therefore $O(k^{-1})$ uniformly in the conditioning indices.

Step 2: the trace. Since $\bar{K}$ is symmetric and $D = \mathrm{diag}(\tau_i^{-1})$,

$$\mathrm{tr}(D\bar{K}D\bar{K}) = \sum_{i,j}\frac{\bar{K}_{ij}^2}{\tau_i\tau_j} = \sum_i \frac{\bar{K}_{ii}^2}{\tau_i^2} + \sum_{i\neq j}\frac{\bar{K}_{ij}^2}{\tau_i\tau_j}.$$

For the diagonal part, $\tau_i\mathbb{E}_S(A_S^{-1} \mid i \in S) = O(k^{-1})$ uniformly, so

$$\sum_i \frac{\bar{K}_{ii}^2}{\tau_i^2} = \frac{k^2}{m^2}\sum_{i=1}^m\big\{1+O(k^{-1})\big\}^2 = \frac{k^2}{m}\big\{1+O(k^{-1})\big\}.$$

For the off-diagonal part, $\bar{K}_{ij} = O(k/m^2)$ uniformly. Hence each summand $\bar{K}_{ij}^2/(\tau_i\tau_j)$ is $O(k^2/m^4)$, because the precisions are bounded above and away from zero. Summing at most $m(m-1)$ such terms gives $O(k^2/m^2)$, which is $O(m^{-1})$ relative to k^2/m. Therefore

$$\mathrm{tr}(D\bar{K}D\bar{K}) = \frac{k^2}{m}\{1+O(k^{-1})\} + O\left(\frac{k^2}{m^2}\right) = \frac{k^2}{m}\{1+O(k^{-1}+m^{-1})\},$$

which is (60).

Step 3: from the trace to $\Gamma_{m,k}$. In (57), the term generated by the common component is $b\mathbb{E}_{S,R}\,\mathrm{tr}(DK_SDK_R)$; because S and R are independent, the expectation

passes inside the trace and $\mathbb{E}_{S,R}(K_S D K_R) = \bar{K} D \bar{K}$, giving $b\,\mathrm{tr}(D\bar{K}D\bar{K})$. The two remaining terms are $o(k^2/m)$ by Lemma 5. Combining with (60) gives explicitly

$$\Gamma_{m,k} = b\frac{k^2}{m}\{1 + O(k^{-1} + m^{-1})\} + o(k^2/m) = b\frac{k^2}{m}\{1 + o(1)\},$$

which is (61).

Step 4: the non-asymptotic lower bound. The matrix $D^{1/2}\bar{K}D^{1/2} = \mathbb{E}_S(D^{1/2}K_S D^{1/2})$ is an average of orthogonal projections (Lemma 4), hence positive semidefinite. If $k = 1$, then $K_S = 0$ by (52), and both sides of (62) are zero. For $k \geq 2$, its rank is positive and at most $m - 1$: since $\bar{K}\iota_m = \mathbb{E}_S(K_S\iota_m) = 0$, the nonzero vector $D^{-1/2}\iota_m$ satisfies

$$D^{1/2}\bar{K}D^{1/2}(D^{-1/2}\iota_m) = D^{1/2}\bar{K}\iota_m = 0.$$

Its trace is $\mathbb{E}_S\,\mathrm{tr}(DK_S) = k - 1$ by (53). Moreover, by cyclicity of the trace,

$$\mathrm{tr}(D\bar{K}D\bar{K}) = \mathrm{tr}\big\{(D^{1/2}\bar{K}D^{1/2})^2\big\} = \sum_{j=1}^{\ell}\lambda_j^2,$$

where $\lambda_1, \ldots, \lambda_\ell$ are the positive eigenvalues of $D^{1/2}\bar{K}D^{1/2}$ and $\ell \leq m - 1$. Cauchy–Schwarz gives $(\sum_{j=1}^{\ell}\lambda_j)^2 \leq \ell\sum_{j=1}^{\ell}\lambda_j^2$, and therefore

$$\mathrm{tr}(D\bar{K}D\bar{K}) = \sum_{j=1}^{\ell}\lambda_j^2 \geq \frac{(\sum_{j=1}^{\ell}\lambda_j)^2}{\ell} \geq \frac{(k-1)^2}{m-1},$$

which is (62). Under homogeneity ($\tau_i \equiv \tau$, so $A_S = k\tau$ and the conditional expectations in Step 1 equal $1/(k\tau)$), the entrywise formulas give $\bar{K}_{ii} = \tau(k-1)/m$ and $\bar{K}_{ij} = -\tau(k-1)/\{m(m-1)\}$. Summing the m diagonal and the $m(m-1)$ off-diagonal squared entries,

$$\mathrm{tr}(D\bar{K}D\bar{K}) = m\cdot\frac{(k-1)^2}{m^2} + m(m-1)\cdot\frac{(k-1)^2}{m^2(m-1)^2} = \frac{(k-1)^2}{m} + \frac{(k-1)^2}{m(m-1)} = \frac{(k-1)^2}{m-1},$$

so (62) holds with equality. □

B.6 One Gaussian regression, two consequences

The adding-up restriction implies that every deviation from w_S lies in $\iota_k^{\perp}$. Expressing these deviations in coordinates normalized by D_S converts the constrained problem into an ordinary Gaussian regression. The same calculation yields the exact single-subset factor $(T-1)/(T-k)$ and the bound on the nonlinear remainder ρ_S. The exact random-design OLS risk calculation is known (Rosset and Tibshirani, 2020). The lemma below

verifies that the restricted forecast-combination problem satisfies its conditions and also supplies the remainder representation required for the shared-sample result.

Lemma 6 (The subset regression in sum-to-one coordinates)**.** *Under Assumptions 2 and 3, let $T \geq k$, fix S, and choose a $k \times (k-1)$ matrix U_S whose columns span $\{a \in \mathbb{R}^k : \iota_k' a = 0\}$ and satisfy $U_S' D_S U_S = I_{k-1}$. Put*

$$\eta_t = v_S' e_t, \qquad x_t = -U_S' P_S' e_t \in \mathbb{R}^{k-1},$$

and let X be the $T \times (k-1)$ matrix with rows x_t'. Then:

(i) $P_S' K_S P_S = U_S U_S'$;

(ii) the restricted least squares problem defining $\widehat{w}_S$ is exactly the unrestricted regression of η on X: writing $\widehat{\beta} = (X'X)^{-1}X'\eta$, we have $\widehat{w}_S = w_S + U_S\widehat{\beta}$, and any sum-to-one $w = w_S + U_S\beta$ satisfies $(w - w_S)' D_S (w - w_S) = \|\beta\|^2$;

(iii) $x_t \sim N(0, I_{k-1})$ i.i.d. and $X'X \sim W_{k-1}(I_{k-1}, T)$. Moreover, η_t is independent of x_t with variance $Q(S)$;

(iv) the influence-function term of (55) corresponds to $\widetilde{\beta} = T^{-1}X'\eta$, i.e. $T^{-1}\sum_t \phi_S(t) = P_S U_S \widetilde{\beta}$ and $\rho_S = P_S U_S(\widehat{\beta} - \widetilde{\beta})$, so that $\rho_S' D \rho_S = \|\widehat{\beta} - \widetilde{\beta}\|^2$.

Proof. Such a matrix exists: if $U_{S,0}$ is any full-column-rank basis of $\iota_k^{\perp}$, then $U_{S,0}' D_S U_{S,0}$ is positive definite and $U_S = U_{S,0}(U_{S,0}' D_S U_{S,0})^{-1/2}$ has the required two properties.

(i) The columns of $D_S^{1/2} U_S$ are orthonormal and orthogonal to $D_S^{-1/2}\iota_k$. Since they span its $(k-1)$-dimensional orthogonal complement,

$$D_S^{1/2} U_S U_S' D_S^{1/2} = I_k - \frac{D_S^{-1/2}\iota_k \iota_k' D_S^{-1/2}}{A_S}.$$

Premultiplying and postmultiplying by $D_S^{-1/2}$ gives

$$U_S U_S' = D_S^{-1} - \frac{D_S^{-1}\iota_k \iota_k' D_S^{-1}}{A_S} = P_S' K_S P_S,$$

which proves (i).

(ii) Every sum-to-one w is $w_S + U_S\beta$ for a unique β, and $(U_S\beta)' D_S (U_S\beta) = \|\beta\|^2$ by the normalisation. Substituting into the training criterion,

$$\sum_t (w' P_S' e_t)^2 = \sum_t \big(\eta_t - \beta' x_t\big)^2,$$

which is minimised at $\widehat{\beta} = (X'X)^{-1}X'\eta$.

(iii) e_t is Gaussian, so $(\eta_t, x_t')'$ is Gaussian. Because $\iota_k' U_S = 0$ kills the rank-one part of $\Omega_{e,k,S}$,

$$\mathbb{E}(x_t x_t') = U_S' \Omega_{e,k,S} U_S = U_S' D_S U_S = I_{k-1}.$$

Since $w_S = D_S^{-1}\iota_k / A_S$, $\Omega_{e,k,S} w_S = (b + A_S^{-1})\iota_k = Q(S)\iota_k$, so

$$\mathbb{E}(\eta_t x_t) = -U_S' \Omega_{e,k,S} w_S = -Q(S) U_S' \iota_k = 0, \qquad \mathrm{Var}(\eta_t) = w_S' \Omega_{e,k,S} w_S = Q(S).$$

Jointly Gaussian and uncorrelated means independent, and independence over t gives $X'X \sim W_{k-1}(I_{k-1}, T)$.

(iv) By Proposition 12 and (i),

$$P_S' \phi_S(t) = -U_S U_S' P_S' e_t \, \eta_t = U_S x_t \eta_t,$$

because $P_S' K_S P_S \Omega_{e,k,S} w_S = Q(S) P_S' K_S P_S \iota_k = 0$ by (53). Since $\phi_S(t)$ is supported on S, averaging and embedding gives $T^{-1}\sum_t \phi_S(t) = P_S U_S (T^{-1} X'\eta) = P_S U_S \widetilde{\beta}$. Since $\widehat{v}_S - v_S = P_S U_S \widehat{\beta}$, the remainder is $P_S U_S (\widehat{\beta} - \widetilde{\beta})$, and its D-quadratic form is $\|\widehat{\beta} - \widetilde{\beta}\|^2$ by the normalisation in (ii). □

Proposition 15 (Exact single-subset factor)**.** *Under Assumptions 2 and 3, write $L(v) = v'\Omega_{e,m} v$ for the independent-test loss of a sum-to-one weight vector v. For $T > k$, the expected independent-test loss of one fitted subset is*

$$\mathbb{E}\, L(\widehat{v}_S) = \{Q(S)\} \frac{T-1}{T-k}.$$

For $T > m$, the same argument applied to the full set $S = \{1, \ldots, m\}$ (with $P_S = I_m$) gives

$$\mathbb{E} L(\widehat{v}_S)|_{S=\{1,\ldots,m\}} = \left(b + \frac{1}{\sum_{i=1}^m \tau_i}\right) \frac{T-1}{T-m}.$$

Lemma 6 gives the precise correspondence with the cited random-design regression: $n = T$, $p = k-1$, regressor covariance I_{k-1}, zero conditional-mean coefficient, disturbance variance $Q(S)$, and an independent evaluation draw. The condition $T > k$ is exactly $p < n - 1$. Assumption 3 is therefore at least as strong as the conditions needed for this exact OLS calculation. The proof is included because the contrast-coordinate mapping and its remainder representation are used again below.

Proof. Since $\iota_m'\widehat{v}_S = 1$, the test loss is $L(\widehat{v}_S) = \widehat{v}_S'\Omega_{e,m}\widehat{v}_S$. Writing $\widehat{w}_S = w_S + U_S\widehat{\beta}$ and expanding,

$$\widehat{v}_S'\Omega_{e,m}\widehat{v}_S = w_S'\Omega_{e,k,S}w_S + 2\widehat{\beta}'U_S'\Omega_{e,k,S}w_S + \widehat{\beta}'U_S'\Omega_{e,k,S}U_S\widehat{\beta} = \{Q(S)\} + 0 + \|\widehat{\beta}\|^2,$$

using the facts established in Lemma 6(iii). Because $T > k$, the required inverse moment exists. Now $\mathbb{E}(\|\widehat{\beta}\|^2 \mid X) = \{Q(S)\}\,\mathrm{tr}\{(X'X)^{-1}\}$, and, because $X'X \sim W_{k-1}(I_{k-1}, T)$, $\mathbb{E}\{(X'X)^{-1}\} = I_{k-1}/(T-k)$. Hence

$$\mathbb{E}\|\widehat{\beta}\|^2 = \{Q(S)\}\frac{k-1}{T-k},$$

and therefore

$$\mathbb{E}L(\widehat{v}_S) = Q(S)\left\{1 + \frac{k-1}{T-k}\right\} = Q(S)\frac{T-1}{T-k}.$$

For $S = \{1, \ldots, m\}$, we have $k = m$ and $Q(S) = b + 1/\sum_{i=1}^m \tau_i$, so the same equality becomes

$$\mathbb{E}L(\widehat{v}_S)|_{S=\{1,\ldots,m\}} = \left(b + \frac{1}{\sum_{i=1}^m \tau_i}\right)\frac{T-1}{T-m},$$

as stated. □

Remark 2 (Consistency check)**.** *When $k/T \to 0$, the exact single-subset factor satisfies*

$$\frac{T-1}{T-k} = 1 + \frac{k-1}{T} + O\left(\frac{k^2}{T^2}\right).$$

Its first-order term is Γ_{SS}/T divided by $Q(S)$, so (58) reproduces the exact single-subset result. This $1 + (k-1)/T$ first-order form is the estimation-error factor used by Elliott and Liao (2026) for restricted OLS forecast-combination weights.

Corollary 1 (Bound for the nonlinear remainder)**.** *Under Assumptions 2 and 3, for $T > k$,*

$$\mathbb{E}\big(\rho_S'D\rho_S\big) = \{Q(S)\}\frac{k(k-1)}{T(T-k)} = O\left(\frac{k^2}{T^2}\right) \quad \textit{when } k \le m = o(T).$$

Proof. By Lemma 6(iv), $\rho_S'D\rho_S = \|\widehat{\beta} - \widetilde{\beta}\|^2$ with $\widehat{\beta} - \widetilde{\beta} = \{(X'X)^{-1} - T^{-1}I\}X'\eta$. Conditioning on X and using $\mathbb{E}(\eta\eta' \mid X) = \{Q(S)\}I_T$ gives directly

$$\mathbb{E}\big(\|\widehat{\beta} - \widetilde{\beta}\|^2 \mid X\big) = Q(S)\,\mathrm{tr}\left\{(X'X)^{-1} - \frac{2}{T}I_{k-1} + \frac{X'X}{T^2}\right\}.$$

Taking expectations with $\mathbb{E}(X'X/T) = I_{k-1}$ and $\mathbb{E}\{(X'X/T)^{-1}\} = TI_{k-1}/(T-k)$ leaves

$$\frac{Q(S)}{T}(k-1)\left\{\frac{T}{T-k} - 1\right\} = \{Q(S)\}\frac{k(k-1)}{T(T-k)}.$$

When $m = o(T)$ and $k \le m$ we have $T - k \asymp T$, giving $O(k^2/T^2)$. □

B.7 The shared-sample inflation

We now combine the coefficient calculation and the nonlinear-remainder bound to prove Proposition 8.

Proof of Proposition 8. **Step 1: the estimation error is measured in D and is unbiased.** Both $\widehat{\bar{v}}_k$ and $\bar{v}_k$ sum to one. Therefore, $\iota_m'(\widehat{\bar{v}}_k - \bar{v}_k) = 0$ and

$$(\widehat{\bar{v}}_k - \bar{v}_k)'\Omega_{e,m}(\widehat{\bar{v}}_k - \bar{v}_k) = b\{\iota_m'(\widehat{\bar{v}}_k - \bar{v}_k)\}^2 + (\widehat{\bar{v}}_k - \bar{v}_k)'D(\widehat{\bar{v}}_k - \bar{v}_k) = (\widehat{\bar{v}}_k - \bar{v}_k)'D(\widehat{\bar{v}}_k - \bar{v}_k).$$

For unbiasedness, fix S and work in the coordinates of Lemma 6. By part (iii), $(\eta_t, x_t')'$ is Gaussian with η_t independent of x_t for each t, and the pairs are i.i.d. across t; the joint law therefore factorizes, so the whole vector η is independent of the whole matrix X. Hence $\mathbb{E}(\widehat{\beta} \mid X) = (X'X)^{-1}X'\mathbb{E}(\eta \mid X) = 0$; the conditional expectation is integrable because $\mathbb{E}\|\widehat{\beta}\|^2 = Q(S)(k-1)/(T-k) < \infty$ under $k < T$ (Proposition 15). Thus $\mathbb{E}(\widehat{v}_S) = v_S + P_S U_S \mathbb{E}(\widehat{\beta}) = v_S$ for every S, and averaging over the finitely many subsets gives $\mathbb{E}(\widehat{\bar{v}}_k) = \bar{v}_k$. Expanding,

$$L_{T,m}^{\mathrm{bag},\infty}(k) = \mathbb{E}\big(\widehat{\bar{v}}_k'\Omega_{e,m}\widehat{\bar{v}}_k\big) = \bar{v}_k'\Omega_{e,m}\bar{v}_k + \mathbb{E}\big[(\widehat{\bar{v}}_k - \bar{v}_k)'D(\widehat{\bar{v}}_k - \bar{v}_k)\big],$$

the cross term vanishing by unbiasedness. The first term is $Q_k^{\mathrm{bag},\infty}$.

Step 2: use the decomposition in (55). By Proposition 12, the linear part of $\widehat{\bar{v}}_k - \bar{v}_k$ is $T^{-1}\sum_{t=1}^{T}\mathbb{E}_S\phi_S(t)$ and its remainder is $\mathbb{E}_S(\rho_S)$. The leading term is exactly the one computed at the start of Section B.4, where Γ_{SR} was derived:

$$\mathbb{E}\left[\left\{\frac{1}{T}\sum_{t=1}^{T}\mathbb{E}_S\phi_S(t)\right\}' D \left\{\frac{1}{T}\sum_{t=1}^{T}\mathbb{E}_S\phi_S(t)\right\}\right] = \frac{\Gamma_{m,k}}{T},$$

which is $\asymp k^2/(mT)$ by Proposition 14.

Step 3: the remainder is negligible. By convexity of the D-quadratic form and Corollary 1,

$$\mathbb{E}\left[\{\mathbb{E}_S(\rho_S)\}'D\{\mathbb{E}_S(\rho_S)\}\right] \le \mathbb{E}\big[\mathbb{E}_S(\rho_S'D\rho_S)\big] = O\left(\frac{k^2}{T^2}\right),$$

whose ratio to $k^2/(mT)$ is $O(m/T) = o(1)$ by Assumption 3. For the cross term, Cauchy–Schwarz in the D-inner product gives

$$\left|\mathbb{E}\left[\left\{\frac{1}{T}\sum_{t=1}^{T}\mathbb{E}_S\phi_S(t)\right\}' D\,\mathbb{E}_S(\rho_S)\right]\right| \le \left(\frac{\Gamma_{m,k}}{T}\right)^{1/2}\left(\mathbb{E}\left[\{\mathbb{E}_S(\rho_S)\}'D\{\mathbb{E}_S(\rho_S)\}\right]\right)^{1/2}$$
$$= O\left(\frac{k}{\sqrt{mT}}\cdot\frac{k}{T}\right).$$

whose ratio to $k^2/(mT)$ is $O(\sqrt{m/T}) = o(1)$. Thus Steps 2 and 3 give

$$\begin{aligned}&\mathbb{E}\big[(\widehat{\bar{v}}_k - \bar{v}_k)' D (\widehat{\bar{v}}_k - \bar{v}_k)\big] \\ &= \frac{\Gamma_{m,k}}{T} + o\left(\frac{k^2}{mT}\right).\end{aligned}$$

Using Step 1 therefore gives

$$L_{T,m}^{\mathrm{bag},\infty}(k) = Q_k^{\mathrm{bag},\infty} + \frac{\Gamma_{m,k}}{T} + o\left(\frac{k^2}{mT}\right),$$

which is the expansion in (27).

Step 4: multiplicative form. By (23), (24), and Lemma 1(1), $g_i(k)$ is bounded above and below, so $\bar{v}_{i,k} = O(m^{-1})$ uniformly and $Q_k^{\mathrm{bag},\infty} = b + \sum_i \bar{v}_{i,k}^2/\tau_i = b + O(m^{-1})$. Hence $Q_k^{\mathrm{bag},\infty} = b + O(m^{-1}) \to b > 0$; in particular, it is bounded away from zero. Therefore, dividing the preceding expansion by $Q_k^{\mathrm{bag},\infty}$ and using $\Gamma_{m,k} = bk^2 m^{-1}\{1+o(1)\}$ from (61) gives explicitly

$$L_{T,m}^{\mathrm{bag},\infty}(k) = Q_k^{\mathrm{bag},\infty}\left[1 + \frac{bk^2}{mTQ_k^{\mathrm{bag},\infty}}\{1+o(1)\}\right] = Q_k^{\mathrm{bag},\infty}\left\{1 + \frac{k^2}{mT}\{1+o(1)\}\right\},$$

because $b/Q_k^{\mathrm{bag},\infty} \to 1$. □

B.8 The fourth-root rule

We now prove Proposition 10. The main text states the economic intuition and its distinction from the square-root criterion; the appendix focuses on localization and argmin transfer.

Proof of Proposition 10. **Step 1: solve the leading minimisation problem directly.** Since $Q_k^{\mathrm{bag},\infty} = Q^{\mathrm{opt}} + R_k^{bag}$ by Proposition 5, the part of the leading excess-loss criterion that depends on k is

$$\mathcal{J}_{T,m}(k) = \frac{\mathcal{D}_m}{k^2} + b\frac{k^2}{T}, \qquad k > 0.$$

Treating k temporarily as a positive real number,

$$\mathcal{J}'_{T,m}(k) = -\frac{2\mathcal{D}_m}{k^3} + \frac{2bk}{T}, \qquad \mathcal{J}''_{T,m}(k) = \frac{6\mathcal{D}_m}{k^4} + \frac{2b}{T} > 0.$$

Thus $\mathcal{J}_{T,m}$ is strictly convex and its unique minimiser is

$$\widetilde{k}_{T,m} = \left(\frac{\mathcal{D}_m T}{b}\right)^{1/4}, \tag{63}$$

obtained from $\mathcal{J}'_{T,m}(k) = 0$, or equivalently $\widetilde{k}^4_{T,m} = \mathcal{D}_m T/b$. Because $\mathcal{D}_m \to \mathcal{D} > 0$, Assumption 3 implies $\widetilde{k}_{T,m} \to \infty$ and $\widetilde{k}_{T,m}/m \to 0$. Let $k^0_{T,m}$ be a feasible integer nearest to $\widetilde{k}_{T,m}$; feasibility, that is $1 \le k^0_{T,m} \le \min\{m, T-1\}$, holds for all large (T, m) because $\widetilde{k}_{T,m} \asymp T^{1/4} = o(\min\{m, T\})$. By Taylor's theorem around $\widetilde{k}_{T,m}$, the first-order condition $\mathcal{J}'_{T,m}(\widetilde{k}_{T,m}) = 0$, $|k^0_{T,m} - \widetilde{k}_{T,m}| \le 1$, and the bound $\mathcal{J}''_{T,m}(\xi) = 6\mathcal{D}_m/\xi^4 + 2b/T = O(T^{-1})$ for $\xi \in [\widetilde{k}_{T,m} - 1, \widetilde{k}_{T,m} + 1]$ (as $\xi \asymp T^{1/4}$ implies $\xi^4 \asymp T$),

$$\mathcal{J}_{T,m}(k^0_{T,m}) - \mathcal{J}_{T,m}(\widetilde{k}_{T,m}) = O(T^{-1}) = o(T^{-1/2}). \tag{64}$$

Moreover, Propositions 11 and 8 apply at $k^0_{T,m}$, since $k^0_{T,m} \asymp T^{1/4} \to \infty$ and $k^0_{T,m}/m \to 0$ by $T^{1/4} = o(m)$. They give

$$mR^{bag}_{k^0_{T,m}} = \mathcal{D}_m\, z^2_{k^0_{T,m}} \{1 + o(1)\} = \frac{\mathcal{D}_m}{(k^0_{T,m})^2} \left(1 - \frac{k^0_{T,m}}{m}\right)^2 \{1 + o(1)\} = \frac{\mathcal{D}_m}{(k^0_{T,m})^2} + o(T^{-1/2}),$$

because $(k^0_{T,m})^{-2} \asymp T^{-1/2}$ while $(1 - k^0_{T,m}/m)^2 = 1 + o(1)$, and

$$\begin{aligned} & m\{L^{\mathrm{bag},\infty}_{T,m}(k^0_{T,m}) - Q^{\mathrm{bag},\infty}_{k^0_{T,m}}\} \\ &= \frac{m\Gamma_{m,k^0_{T,m}}}{T} + m \cdot o\left(\frac{(k^0_{T,m})^2}{mT}\right) \\ &= \frac{b(k^0_{T,m})^2}{T}\{1 + o(1)\} + o(T^{-1/2}) \\ &= \frac{b(k^0_{T,m})^2}{T} + o(T^{-1/2}). \end{aligned}$$

using (61) and $(k^0_{T,m})^2/T \asymp T^{-1/2}$. Since $Q^{\mathrm{bag},\infty}_{k^0_{T,m}} - Q^{\mathrm{opt}} = R^{bag}_{k^0_{T,m}}$ (Proposition 5), adding the two displays yields

$$m\{L^{\mathrm{bag},\infty}_{T,m}(k^0_{T,m}) - Q^{\mathrm{opt}}\} = \mathcal{J}_{T,m}(k^0_{T,m}) + o(T^{-1/2}) = O(T^{-1/2}). \tag{65}$$

Step 2: localise the exact global minimiser. This step verifies that the direct solution in (63) lies in the region where the leading expansion describes the exact criterion. The non-asymptotic approximation bound in Lemma 3, valid for every $1 \le k < m$, gives

$$mR^{bag}_k \ge \frac{\tau_L v^2_\tau}{4\tau^6_U} z^2_k, \tag{66}$$

and (66) holds trivially at $k = m$, where both sides vanish because $z_m = 0$ and $R^{bag}_m = 0$. For the estimation component, use the linear and remainder terms in (55). The averaged influence function satisfies

$$\mathbb{E}_S \phi_S(t) = -\bar{K}\xi_t h_t - \mathbb{E}_S\big[K_S \xi_t (v'_S \xi_t)\big].$$

Both terms have mean zero; for the second, this follows from $\mathbb{E}_S(K_S D v_S) = 0$ by (53). Independence across t therefore leaves only same-t terms, and the linear contribution is

$$\frac{1}{T}\Big\{\mathbb{E}\big[h_t^2\,\xi_t'\bar{K}D\bar{K}\xi_t\big]+2\,\mathbb{E}\big[h_t\,\xi_t'\bar{K}D\,\mathbb{E}_S\{K_S\xi_t(v_S'\xi_t)\}\big]+\mathbb{E}\big[\mathbb{E}_S\{K_S\xi_t(v_S'\xi_t)\}'D\mathbb{E}_S\{K_S\xi_t(v_S'\xi_t)\}\big]\Big\}.$$

The middle term vanishes because h_t is independent of ξ_t with $\mathbb{E}(h_t) = 0$; the first term equals $(b/T)\,\mathrm{tr}(D\bar{K}D\bar{K})$ by $\mathbb{E}(\xi_t'A\xi_t) = \mathrm{tr}(AD)$; and the third term is nonnegative. Hence, using (62),

$$\mathbb{E}\left[\left\{\frac{1}{T}\sum_{t=1}^{T}\mathbb{E}_S\phi_S(t)\right\}'D\left\{\frac{1}{T}\sum_{t=1}^{T}\mathbb{E}_S\phi_S(t)\right\}\right] \geq \frac{b}{T}\,\mathrm{tr}(D\bar{K}D\bar{K}) \geq \frac{b(k-1)^2}{T(m-1)}.$$

Corollary 1 and $m/T \to 0$ imply, uniformly over $2 \leq k \leq m$ (using $Q(S) \leq b + 1/(2\tau_L)$, $T - k \geq T - m$, and $k^2/(k-1)^2 \leq 4$ for $k \geq 2$),

$$\frac{\mathbb{E}\left[\{\mathbb{E}_S(\rho_S)\}'D\{\mathbb{E}_S(\rho_S)\}\right]}{b(k-1)^2/\{T(m-1)\}} = O\Big(\frac{m}{T}\Big) = o(1).$$

The reverse triangle inequality therefore gives, uniformly over $2 \leq k \leq m$,

$$\big\{\mathbb{E}\big[(\widehat{\bar{v}}_k - \bar{v}_k)'D(\widehat{\bar{v}}_k - \bar{v}_k)\big]\big\}^{1/2} \geq \left\{\frac{b(k-1)^2}{T(m-1)}\right\}^{1/2} - \left\{\mathbb{E}\left[\{\mathbb{E}_S(\rho_S)\}'D\{\mathbb{E}_S(\rho_S)\}\right]\right\}^{1/2}.$$

The preceding ratio is $o(1)$, so squaring this inequality yields

$$\mathbb{E}\big[(\widehat{\bar{v}}_k - \bar{v}_k)'D(\widehat{\bar{v}}_k - \bar{v}_k)\big] \geq \frac{b(k-1)^2}{T(m-1)}\{1 - o(1)\}. \tag{67}$$

At $k = 1$ the bound (67) is vacuous, but $k = 1$ is excluded from the optimum directly by (66): $z_1^2 = (1 - 1/m)^2 \geq 1/4$ is not $O(T^{-1/2})$.

Optimality of $k_{T,m}^*$ and the benchmark bound (65), together with the two nonnegative components (66) and (67), imply

$$z_{k_{T,m}^*}^2 = O(T^{-1/2}), \qquad \frac{(k_{T,m}^* - 1)^2}{T} = O(T^{-1/2}).$$

The second relation gives $k_{T,m}^* = O(T^{1/4}) = o(m)$. Consequently, $z_{k_{T,m}^*} = (k_{T,m}^*)^{-1}\{1 - k_{T,m}^*/m\} \sim (k_{T,m}^*)^{-1}$, and the first relation then gives $(k_{T,m}^*)^{-1} = O(T^{-1/4})$. Hence there are fixed constants $0 < c < C < \infty$ such that, eventually,

$$cT^{1/4} \leq k_{T,m}^* \leq CT^{1/4}. \tag{68}$$

Step 3: compare with the exact criterion. Because $\widetilde{k}_{T,m}/T^{1/4} \to (\mathcal{D}/b)^{1/4}$, the constants c and C in (68) may be enlarged, if necessary, so that $\widetilde{k}_{T,m}$ and $k_{T,m}^0$ also belong

to the band. Uniformly over this band, $k \to \infty$, $k/m \to 0$, and $z_k \sim k^{-1}$. Equations (50), (61), and (27) therefore give

$$m\{L_{T,m}^{\mathrm{bag},\infty}(k) - Q^{\mathrm{opt}}\} = \mathcal{J}_{T,m}(k) + \varepsilon_{T,m}(k), \qquad \sup_{cT^{1/4} \le k \le CT^{1/4}} |\varepsilon_{T,m}(k)| = o(T^{-1/2}). \quad (69)$$

The uniform statement follows from the sequential criterion. For each (T, m) the band contains finitely many integers, so there exists $k_{T,m}$ in the band attaining $\sup_{cT^{1/4} \le k \le CT^{1/4}} |\varepsilon_{T,m}(k)|$. Any such extremizing sequence satisfies $k_{T,m} \asymp T^{1/4} \to \infty$ and $k_{T,m}/m \to 0$ (by $T^{1/4} = o(m)$), so Propositions 11 and 8 apply along it and force $T^{1/2}|\varepsilon_{T,m}(k_{T,m})| \to 0$, which is exactly the stated uniform bound.

By optimality of $k_{T,m}^*$, equations (64) and (69) give

$$\mathcal{J}_{T,m}(k_{T,m}^*) - \mathcal{J}_{T,m}(\widetilde{k}_{T,m}) = o(T^{-1/2}). \quad (70)$$

This gap can be evaluated exactly. Using $\mathcal{D}_m/\widetilde{k}_{T,m}^2 = b\widetilde{k}_{T,m}^2/T = \sqrt{\mathcal{D}_m b/T}$ (both equalities from $\widetilde{k}_{T,m}^4 = \mathcal{D}_m T/b$), we obtain

$$\begin{aligned}\mathcal{J}_{T,m}(k_{T,m}^*) - \mathcal{J}_{T,m}(\widetilde{k}_{T,m}) &= \sqrt{\frac{\mathcal{D}_m b}{T}} \left\{ \left(\frac{k_{T,m}^*}{\widetilde{k}_{T,m}}\right)^{-2} + \left(\frac{k_{T,m}^*}{\widetilde{k}_{T,m}}\right)^{2} - 2 \right\} \\ &= \sqrt{\frac{\mathcal{D}_m b}{T}} \left\{ \frac{k_{T,m}^*}{\widetilde{k}_{T,m}} - \frac{\widetilde{k}_{T,m}}{k_{T,m}^*} \right\}^2 .\end{aligned}$$

Combining this identity with (70) and $\mathcal{D}_m \to \mathcal{D} > 0$ gives

$$\left\{ \frac{k_{T,m}^*}{\widetilde{k}_{T,m}} - \frac{\widetilde{k}_{T,m}}{k_{T,m}^*} \right\}^2 \longrightarrow 0.$$

On the band, $k_{T,m}^*/\widetilde{k}_{T,m}$ is bounded above and below by positive constants. Multiplying the expression in braces by $k_{T,m}^*/\widetilde{k}_{T,m}$ therefore shows explicitly that $(k_{T,m}^*/\widetilde{k}_{T,m})^2 - 1 \to 0$, and hence $k_{T,m}^*/\widetilde{k}_{T,m} \to 1$. Finally, (63) yields

$$\frac{k_{T,m}^*}{T^{1/4}} = \frac{k_{T,m}^*}{\widetilde{k}_{T,m}} \left(\frac{\mathcal{D}_m}{b}\right)^{1/4} \longrightarrow \left(\frac{\mathcal{D}}{b}\right)^{1/4},$$

which is (33). □

B.9 The boundary-corrected finite-m criterion

Proposition 16 (Local risk justification of the boundary correction)**.** *Under Assumptions 2, 3, and 4, for every fixed $0 < c < C < \infty$,*

$$\sup_{\substack{1 \le k \le \min\{m, T-1\}: \\ cT^{1/4} \le k \le CT^{1/4}}} \left| m\{L_{T,m}^{\mathrm{bag},\infty}(k) - Q^{\mathrm{opt}}\} - \left\{ \mathcal{D}_m \left(\frac{1 - k/m}{k}\right)^2 + b\frac{k^2}{T} \right\} \right| = o(T^{-1/2}). \quad (71)$$

If

$$k^{\circ}_{T,m} \in \arg \min_{1\le k\le \min\{m,T-1\}} \left\{ \mathcal{D}_m \left(\frac{1-k/m}{k} \right)^2 + b\frac{k^2}{T} \right\},$$

then

$$\frac{k^{\circ}_{T,m}}{T^{1/4}} \longrightarrow \left(\frac{\mathcal{D}}{b} \right)^{1/4}, \qquad \frac{k^{\circ}_{T,m}}{k^{*}_{T,m}} \longrightarrow 1.$$

Thus the true-parameter boundary-corrected criterion is first-order equivalent to infinite-bag risk on fourth-root bands; it is not an exact fixed-m risk identity.

Proof. Proposition 5 gives

$$L^{\mathrm{bag},\infty}_{T,m}(k) - Q^{\mathrm{opt}} = R^{\mathrm{bag}}_k + \{L^{\mathrm{bag},\infty}_{T,m}(k) - Q^{\mathrm{bag},\infty}_k\}.$$

On every band in (71), Assumption 3 implies $k \to \infty$, $k/m \to 0$, and $k^2/T \to 0$. Proposition 11, with $z_k = (1-k/m)/k$, and Proposition 8 with (29) therefore give, uniformly on the band,

$$mR^{\mathrm{bag}}_k = \mathcal{D}_m \left(\frac{1-k/m}{k} \right)^2 + o(T^{-1/2}), \qquad m\{L^{\mathrm{bag},\infty}_{T,m}(k) - Q^{\mathrm{bag},\infty}_k\} = b\frac{k^2}{T} + o(T^{-1/2}).$$

Uniformity follows by the extremizing-sequence argument used for (69). Adding the two expansions gives

$$m\{L^{\mathrm{bag},\infty}_{T,m}(k) - Q^{\mathrm{opt}}\} = \mathcal{D}_m \left(\frac{1-k/m}{k} \right)^2 + b\frac{k^2}{T} + o(T^{-1/2}),$$

uniformly on the band, which is (71).

For the argmin statement, treat k temporarily as continuous. The derivative of the displayed boundary-corrected objective has the sign of

$$\frac{bk^4}{T} - \mathcal{D}_m \left(1 - \frac{k}{m} \right).$$

The first term is strictly increasing and the second is strictly decreasing on $(0, m]$, so there is a unique minimizer, characterized by

$$\frac{bk^4}{T} = \mathcal{D}_m \left(1 - \frac{k}{m} \right).$$

It is bounded above by $(\mathcal{D}_m T/b)^{1/4} = O(T^{1/4})$; hence $k/m \to 0$, and the first-order condition gives

$$\frac{k}{(\mathcal{D}_m T/b)^{1/4}} = \left(1 - \frac{k}{m} \right)^{1/4} \longrightarrow 1.$$

The integer minimizer is one of the two adjacent integers, so rounding does not affect the ratio. Since $\mathcal{D}_m \to \mathcal{D}$,

$$\frac{k^\circ_{T,m}}{T^{1/4}} \longrightarrow \left(\frac{\mathcal{D}}{b}\right)^{1/4}.$$

Together with Proposition 10, this yields

$$\frac{k^\circ_{T,m}}{k^*_{T,m}} = \frac{k^\circ_{T,m}/T^{1/4}}{k^*_{T,m}/T^{1/4}} \longrightarrow 1.$$

□

Proposition 16 explains the boundary-corrected shape of (35). Replacing b and $\mathcal{D}_m$ by estimated, numerically safeguarded quantities is an additional plug-in step; the result does not assert exact fixed-m optimality or equality of the integer argmins.

The fixed-pool boundary case is stated separately in Remark 1; it is outside the triangular-array argument used here.

B.10 Finite-G transfer

This subsection separates two effects of using only G subset draws: the additional diagonal term in the estimation coefficient and the population-level subset-draw error. The finite-bag loss approaches the infinite-bag loss as G grows, and the corollary gives the rate sufficient to preserve the fourth-root optimizer.

Proposition 17 (Second moment of the finite-G averaged influence function)**.** *Under Assumptions 2, 3, and 4, let $S_1, \dots, S_G$ be independent subset draws and define the averaged influence function*

$$\ell_G = \frac{1}{T}\sum_{t=1}^{T}\frac{1}{G}\sum_{g=1}^{G}\phi_{S_g}(t).$$

Then

$$\mathbb{E}(\ell_G' D \ell_G) = \frac{\Gamma^G_{m,k}}{T}, \qquad \Gamma^G_{m,k} = \frac{1}{G}\,\mathbb{E}_S\Gamma_{SS} + \Big(1-\frac{1}{G}\Big)\Gamma_{m,k}, \qquad \Gamma_{SS} = \big\{Q(S)\big\}(k-1). \quad (72)$$

Thus $\Gamma^G_{m,k}$ plays for the finite-G estimator the role that $\Gamma_{m,k}$ plays for the infinite bag in Proposition 8: it is the second moment of the linear term of the estimation error. (A complete finite-G loss expansion is not needed in what follows; Proposition 18 compares the complete feasible losses exactly.)

Proof. Set $\ell_{G,t} = G^{-1}\sum_{g=1}^{G}\phi_{S_g}(t)$. Conditional on $S_1,\dots,S_G$, the vectors $\ell_{G,1},\dots,\ell_{G,T}$ are i.i.d. across t and Proposition 12 gives $\mathbb{E}(\ell_{G,t} \mid S_1,\dots,S_G) = G^{-1}\sum_g \mathbb{E}\{\phi_{S_g}(t) \mid S_g\} = 0$. Hence every off-time term satisfies

$$\mathbb{E}(\ell'_{G,t}D\ell_{G,u} \mid S_1,\dots,S_G) = \mathbb{E}(\ell_{G,t} \mid S_1,\dots,S_G)'D\mathbb{E}(\ell_{G,u} \mid S_1,\dots,S_G) = 0, \qquad t \neq u.$$

For a surviving same-time term, the double sum over (g,ℓ) produces G diagonal pairs, each with expectation $\mathbb{E}_S\Gamma_{SS}$, and $G(G-1)$ off-diagonal pairs, each with expectation $\Gamma_{m,k}$ by independence of S_g and S_ℓ. Dividing by G^2 and by the factor T from the time average gives

$$\mathbb{E}(\ell'_G D\ell_G) = \frac{1}{T}\left\{\frac{1}{G}\mathbb{E}_S\Gamma_{SS} + \left(1-\frac{1}{G}\right)\Gamma_{m,k}\right\},$$

which is (72); (58) gives the stated expression for Γ_{SS}. □

The coefficient calculation does not by itself compare the complete finite-G and infinite-bag losses, because finite subset sampling also creates a population-level Monte Carlo term. The following exact identity accounts for both effects.

Proposition 18 (Exact feasible finite-bag bridge)**.** *Under Assumptions 2, 3, and 4, let*

$$\widehat{\bar{v}}_{k,G} = \frac{1}{G}\sum_{g=1}^{G}\widehat{v}_{S_g}, \qquad L_{T,m}^{\mathrm{bag},G}(k) = \mathbb{E}\big(\widehat{\bar{v}}'_{k,G}\Omega_{e,m}\widehat{\bar{v}}_{k,G}\big).$$

For every finite G and $T > k$,

$$L_{T,m}^{\mathrm{bag},G}(k) = L_{T,m}^{\mathrm{bag},\infty}(k) + \frac{1}{G}\left\{L_{T,m}^{\mathrm{sub}}(k) - L_{T,m}^{\mathrm{bag},\infty}(k)\right\}. \tag{73}$$

If $k\to\infty$, $k/m\to 0$, and $k^2/T\to 0$, then

$$L_{T,m}^{\mathrm{bag},G}(k) - L_{T,m}^{\mathrm{bag},\infty}(k) = \frac{1}{Gk\bar{\tau}_m}\{1+o(1)\}. \tag{74}$$

Proof. Condition on the training sample and abbreviate $\widehat{\bar{v}}_k = \mathbb{E}_S(\widehat{v}_S)$. Expanding the quadratic form produces G diagonal and $G(G-1)$ ordered off-diagonal pairs. Conditional independence of two distinct subset draws allows their expectations to factor, so the pair count gives

$$\begin{aligned}\mathbb{E}_{S_1,\dots,S_G}\big(\widehat{\bar{v}}'_{k,G}\Omega_{e,m}\widehat{\bar{v}}_{k,G} \mid \text{training sample}\big) = \frac{1}{G}\mathbb{E}_S\big(\widehat{v}'_S\Omega_{e,m}\widehat{v}_S \mid \text{training sample}\big)\\ + \left(1-\frac{1}{G}\right)\widehat{\bar{v}}'_k\Omega_{e,m}\widehat{\bar{v}}_k.\end{aligned}$$

Taking expectation over the training sample gives

$$L_{T,m}^{\mathrm{bag},G}(k) = \frac{1}{G} L_{T,m}^{\mathrm{sub}}(k) + \left(1 - \frac{1}{G}\right) L_{T,m}^{\mathrm{bag},\infty}(k),$$

and therefore

$$L_{T,m}^{\mathrm{bag},G}(k) = L_{T,m}^{\mathrm{bag},\infty}(k) + \frac{1}{G}\left\{L_{T,m}^{\mathrm{sub}}(k) - L_{T,m}^{\mathrm{bag},\infty}(k)\right\},$$

which is (73).

For the order calculation, Proposition 15 and Proposition 4 give

$$L_{T,m}^{\mathrm{sub}}(k) = Q_k^{\mathrm{sub}} \frac{T-1}{T-k} = Q_k^{\mathrm{sub}} + O\left(\frac{k}{T}\right),$$

because $Q_k^{\mathrm{sub}} = b + O(k^{-1}) = O(1)$. Also, Propositions 5 and 11 imply

$$Q_k^{\mathrm{sub}} - Q_k^{\mathrm{bag},\infty} = \frac{1}{k\bar{\tau}_m} - \frac{1}{m\bar{\tau}_m} + O(k^{-2}) + O\left(\frac{1}{mk^2}\right) = \frac{1}{k\bar{\tau}_m}\{1 + o(1)\}.$$

Finally, Proposition 8 gives $L_{T,m}^{\mathrm{bag},\infty}(k) = Q_k^{\mathrm{bag},\infty} + O(k^2/(mT))$. The conditions $k^2/T \to 0$ and $k/m \to 0$ ensure that both k/T and $k^2/(mT)$ are $o(k^{-1})$. Consequently,

$$L_{T,m}^{\mathrm{sub}}(k) - L_{T,m}^{\mathrm{bag},\infty}(k) = \frac{1}{k\bar{\tau}_m}\{1 + o(1)\},$$

and substitution into (73) gives

$$L_{T,m}^{\mathrm{bag},G}(k) - L_{T,m}^{\mathrm{bag},\infty}(k) = \frac{1}{G}\frac{1}{k\bar{\tau}_m}\{1 + o(1)\} = \frac{1}{Gk\bar{\tau}_m}\{1 + o(1)\},$$

which is (74). □

Corollary 2 (Finite-bag fourth-root rule)**.** *Under Assumptions 2, 3, and 4, let $G = G_{T,m}$ satisfy*

$$\frac{mT^{1/4}}{G} \longrightarrow 0, \tag{75}$$

which is Assumption 5. $k \asymp T^{1/4}$. Then every minimizer

$$k_{T,m}^{*,G} \in \arg\min_{1 \le k \le \min\{m,T-1\}} L_{T,m}^{\mathrm{bag},G}(k)$$

satisfies

$$\frac{k_{T,m}^{*,G}}{T^{1/4}} \longrightarrow \left(\frac{\mathcal{D}}{b}\right)^{1/4}.$$

Proof. Let $k_{T,m}^0$ be the integer benchmark defined in the proof of Proposition 10. The gap in the exact bridge (73) can be bounded non-asymptotically. First, $0 \le Q_k^{\mathrm{sub}} - Q_k^{\mathrm{bag},\infty} \le Q_k^{\mathrm{sub}} - Q^{\mathrm{opt}} \le \mathbb{E}_S(A_S^{-1}) \le 1/(k\tau_L)$ by (19) and Lemma 1(1). Second, $0 \le$

$L^{\mathrm{sub}}_{T,m}(k) - Q^{\mathrm{sub}}_k = Q^{\mathrm{sub}}_k(k-1)/(T-k)$ by Proposition 7, and $0 \le L^{\mathrm{bag},\infty}_{T,m}(k) - Q^{\mathrm{bag},\infty}_k \le \mathbb{E}_S\{Q(S)\}(k-1)/(T-k)$ by conditional Jensen's inequality and Proposition 15. Since Q^{sub}_k and $\mathbb{E}_S\{Q(S)\}$ are bounded and $T-k \asymp T$ on every band $cT^{1/4} \le k \le CT^{1/4}$, combining the three bounds inside (73) gives, uniformly over the band,

$$0 \le L^{\mathrm{bag},G}_{T,m}(k) - L^{\mathrm{bag},\infty}_{T,m}(k) = \frac{1}{G}\left\{L^{\mathrm{sub}}_{T,m}(k) - L^{\mathrm{bag},\infty}_{T,m}(k)\right\} = O\left(\frac{1}{Gk} + \frac{k}{GT}\right).$$

After multiplication by the fourth-root normalization $m\sqrt{T}$, the two terms are respectively $O(mT^{1/4}/G)$ and $O\{m/(GT^{1/4})\}$, both $o(1)$ under (75). Equivalently, uniformly on such a band,

$$m\{L^{\mathrm{bag},G}_{T,m}(k) - L^{\mathrm{bag},\infty}_{T,m}(k)\} = o(T^{-1/2}). \tag{76}$$

Moreover, (73) and conditional Jensen's inequality show that the finite-G loss is never below the infinite-bag loss. At the benchmark $k^0_{T,m}$, (65) and (76) give

$$m\{L^{\mathrm{bag},G}_{T,m}(k^0_{T,m}) - Q^{\mathrm{opt}}\} = O(T^{-1/2}).$$

Because the finite-G loss dominates the infinite-bag loss pointwise in k, optimality of $k^{*,G}_{T,m}$ gives the chain

$$m\{L^{\mathrm{bag},\infty}_{T,m}(k^{*,G}_{T,m}) - Q^{\mathrm{opt}}\} \le m\{L^{\mathrm{bag},G}_{T,m}(k^{*,G}_{T,m}) - Q^{\mathrm{opt}}\} \le m\{L^{\mathrm{bag},G}_{T,m}(k^0_{T,m}) - Q^{\mathrm{opt}}\} = O(T^{-1/2}).$$

The global lower bounds (66) and (67), applied to $L^{\mathrm{bag},\infty}_{T,m}$ at $k^{*,G}_{T,m}$, therefore force $z^2_{k^{*,G}_{T,m}} = O(T^{-1/2})$ and $(k^{*,G}_{T,m}-1)^2/T = O(T^{-1/2})$ — exactly the two inequalities used in Step 2 of Proposition 10 — which localise every finite-G minimiser in a band $cT^{1/4} \le k^{*,G}_{T,m} \le CT^{1/4}$.

On that band, combining (69) and (76) yields

$$m\{L^{\mathrm{bag},G}_{T,m}(k) - Q^{\mathrm{opt}}\} = \mathcal{J}_{T,m}(k) + o(T^{-1/2})$$

uniformly in k. Optimality of $k^{*,G}_{T,m}$ relative to $k^0_{T,m}$ and (64) consequently imply

$$\mathcal{J}_{T,m}(k^{*,G}_{T,m}) - \mathcal{J}_{T,m}(\widetilde{k}_{T,m}) = o(T^{-1/2}).$$

Applying the exact squared-gap identity from Step 3 of Proposition 10 with $k^{*,G}_{T,m}/\widetilde{k}_{T,m}$ in place of $k^*_{T,m}/\widetilde{k}_{T,m}$ gives $k^{*,G}_{T,m}/\widetilde{k}_{T,m} \to 1$. Combining this with (63) gives

$$\frac{k^{*,G}_{T,m}}{T^{1/4}} = \frac{k^{*,G}_{T,m}}{\widetilde{k}_{T,m}}\left(\frac{\mathcal{D}_m}{b}\right)^{1/4} \longrightarrow \left(\frac{\mathcal{D}}{b}\right)^{1/4},$$

which proves the stated limit. □

Proposition 19 (Feasible selector and oracle size)**.** *Under Assumptions 2, 3, and 4, estimate $\widehat{b}$ and $\widehat{\mathcal{D}}_m$ from the T training errors as in (35). Complete the rule with its stated fallback on samples where the estimates are invalid. Then*

$$\widehat{b} - b = O_p(T^{-1/2}), \qquad \widehat{\mathcal{D}}_m - \mathcal{D}_m = O_p(T^{-1/2}). \tag{77}$$

For every fixed $0 < c < C < \infty$, the feasible criterion satisfies

$$\sup_{\substack{1\le k\le \min\{m,T-2\}:\\ cT^{1/4}\le k\le CT^{1/4}}} \left| \widehat{\mathcal{D}}_m \left(\frac{1-k/m}{k}\right)^2 + \widehat{b}\frac{k^2}{T} - m\{L_{T,m}^{\mathrm{bag},\infty}(k) - Q^{\mathrm{opt}}\} \right| = o_p(T^{-1/2}). \tag{78}$$

If $k_{T,m}^*$ minimizes the fixed-size expected risk, then

$$\frac{\widehat{k}}{T^{1/4}} \xrightarrow{p} \left(\frac{\mathcal{D}}{b}\right)^{1/4}, \qquad \frac{\widehat{k}}{k_{T,m}^*} \xrightarrow{p} 1, \tag{79}$$

and

$$m\sqrt{T}\{L_{T,m}^{\mathrm{bag},\infty}(\widehat{k}) - L_{T,m}^{\mathrm{bag},\infty}(k_{T,m}^*)\} \xrightarrow{p} 0. \tag{80}$$

Here $L_{T,m}^{\mathrm{bag},\infty}(\widehat{k})$ means the fixed-size expected-risk curve evaluated at the random index $\widehat{k}$.

Proof. The Gaussian model and the uniform bounds on b and the precisions give uniformly bounded sub-exponential norms for e_{it}^2 and $e_{it}e_{jt}$. Bernstein's inequality across the independent training origins and a union bound imply, for every sufficiently small fixed $\delta > 0$,

$$\Pr\left(\max_i \left|\widehat{C}_{ii} - \widehat{b} - \tau_i^{-1}\right| > \delta \text{ or } |\widehat{b} - b| > \delta\right) \le C_\delta m e^{-c_\delta T}.$$

The off-diagonal average defining $\widehat{b}$ is itself an average of uniformly sub-exponential one-origin statistics; its second moment about b is $O(T^{-1})$. Similarly,

$$\frac{1}{m}\sum_{i=1}^m E\left(\widehat{C}_{ii} - \widehat{b} - \tau_i^{-1}\right)^2 = O(T^{-1}).$$

Since $m = o(T)$, the displayed exceptional probability tends to zero. On its complement, all implied idiosyncratic variances lie in a fixed positive compact interval, $\widehat{b} > 0$, and the variance floor is inactive. Reciprocation and the functional in (4) are uniformly Lipschitz there. In particular,

$$|\widehat{\mathcal{D}}_m - \mathcal{D}_m| \le \frac{C}{m}\sum_i \left|\widehat{C}_{ii} - \widehat{b} - \tau_i^{-1}\right|.$$

The second-moment bounds prove (77); $\mathcal{D}_m \to \mathcal{D} > 0$ also makes $\widehat{\mathcal{D}}_m > 0$ with probability tending to one. On a fourth-root band, both $((1-k/m)/k)^2$ and k^2/T are $O(T^{-1/2})$.

Thus coefficient estimation changes the criterion by $o_p(T^{-1/2})$ uniformly. Proposition 16 gives (78).

With positive estimated coefficients, the continuous feasible objective has one minimizer in $(0, m)$. Its first-order condition is

$$\frac{\widehat{b}k^4}{T} = \widehat{\mathcal{D}}_m \left(1 - \frac{k}{m}\right).$$

It is at most $(T\widehat{\mathcal{D}}_m/\widehat{b})^{1/4} = O_p(T^{1/4})$, so $k/m = o_p(1)$. The first-order condition and (77) imply that the continuous minimizer divided by $T^{1/4}$ converges in probability to $(\mathcal{D}/b)^{1/4}$. It is eventually interior to the admissible integer grid with probability tending to one. The integer minimizer is one of its two adjacent integers, so rounding changes its normalized value by at most $T^{-1/4}$. Proposition 10 establishes (79).

Both $\widehat{k}$ and $k^*_{T,m}$ belong with probability tending to one to a common fixed fourth-root band. By (71), their normalized fixed-size risks converge to the same minimum $2\sqrt{b\mathcal{D}}$ of $\mathcal{D}/a^2 + ba^2$, where $a = k/T^{1/4}$. Their difference therefore obeys (80). □

B.11 Numerical example of the RSM selector

Suppose that the forecast pool contains $m = 4$ forecasters and that the estimation window is $T = 20$ observations. The estimated second-moment matrix of the realized forecast errors $\widehat{C}$ is as follows:

$$\widehat{C} = \frac{1}{T}\sum_{t=1}^{T} e_t e_t' = \begin{pmatrix} 1.200 & 0.160 & 0.250 & 0.190 \\ 0.160 & 0.700 & 0.220 & 0.180 \\ 0.250 & 0.220 & 0.450 & 0.200 \\ 0.190 & 0.180 & 0.200 & 0.325 \end{pmatrix}.$$

First, we estimate the common component $\widehat{b}$ as the mean of the non-diagonal elements of $\widehat{C}$:

$$\widehat{b} = \frac{1}{m(m-1)}\sum_{i\neq j}\widehat{C}_{ij} = \frac{2}{4(4-1)}\left(0.160 + 0.250 + 0.190 + 0.220 + 0.180 + 0.200\right) = 0.2$$

The corresponding estimated precisions are estimated as $\widehat{\tau}_i = \frac{1}{\widehat{C}_{ii} - \widehat{b}}$. Hence:

$$(\widehat{\tau}_1, \widehat{\tau}_2, \widehat{\tau}_3, \widehat{\tau}_4) = \left(\frac{1}{1.200 - 0.2}, \frac{1}{0.700 - 0.2}, \frac{1}{0.450 - 0.2}, \frac{1}{0.325 - 0.2} \right) = (1, 2, 4, 8).$$

Then, compute the cross-sectional mean $\widehat{\bar{\tau}}_m$:

$$\widehat{\bar{\tau}}_m = \frac{1}{m} \sum_{i=1}^{m} \widehat{\tau}_i = \frac{1 + 2 + 4 + 8}{4} = 3.75,$$

and the corresponding cross-sectional variance:

$$\widehat{v}_{\tau,m} = \frac{1}{m} \sum_{i=1}^{m} \left(\widehat{\tau}_i - \widehat{\bar{\tau}}_m \right)^2 = \frac{1}{4} \left[(1 - 3.75)^2 + (2 - 3.75)^2 + (4 - 3.75)^2 + (8 - 3.75)^2 \right] = 7.1875.$$

Plug the afore-estimated quantities into $\widehat{\mathcal{D}}_m$:

$$\widehat{\mathcal{D}}_m = \frac{1}{3.75^6} \frac{1}{4} \Big[1(17.50)^2 + 2(13.75)^2 + 4(6.25)^2 + 8(-8.75)^2 \Big] \approx 0.1306.$$

For the boundary-corrected RSM, each $k \in \{1, 2, 3, 4\}$ is then evaluated to find k^* which minimizes the criterion. In this case:

$$\widehat{Q}(1) = 0.1306 \left(\frac{1 - 1/4}{1} \right)^2 + 0.2 \frac{1^2}{20} = 0.0735 + 0.0100 = 0.0835.$$

$$\widehat{Q}(2) = 0.1306 \left(\frac{1 - 2/4}{2} \right)^2 + 0.2 \frac{2^2}{20} = 0.0082 + 0.0400 = 0.0482.$$

$$\widehat{Q}(3) = 0.1306 \left(\frac{1 - 3/4}{3} \right)^2 + 0.2 \frac{3^2}{20} = 0.0009 + 0.0900 = 0.0909.$$

$$\widehat{Q}(4) = 0.1306 \left(\frac{1 - 4/4}{4} \right)^2 + 0.2 \frac{4^2}{20} = 0 + 0.1600 = 0.1600.$$

Thus, the RSM selector therefore yields:

$$\boxed{\widehat{k}_{\text{RSM}} = \underset{k \in \{1,2,3,4\}}{\arg\min}\ \widehat{Q}(k) = 2.}$$

We can also use the direct fourth-root rule using the same estimates:

$$\widehat{k}_{\text{fourth-root}} = \left(\frac{0.1306 \times 20}{0.200} \right)^{1/4} \approx 1.90.$$

Rounding to the nearest admissible integer yields

$$\boxed{\widehat{k}_{\text{fourth-root}} = 2.}$$

Thus, in this example, the boundary-corrected RSM selector and the direct fourth-root plug-in rule select the same subset size of 2:

$$\boxed{\widehat{k}_{\mathrm{RSM}} = \widehat{k}_{\text{fourth-root}} = 2.}$$

C Additional Monte Carlo Results

C.1 General precision profiles

Profile	T	m	H	$\widehat{k}^*$	$\widehat{k}^*_{\text{smooth}}$	k^*_Γ	$(\mathcal{D}_m T/b)^{1/4}$
Two-point	256	80	2.78	8	8.66	9	8.76
Two-point	512	114	2.78	10	10.63	10	10.42
Two-point	1024	160	2.78	13	12.27	12	12.39
Two-point	2048	226	2.78	15	15.06	15	14.74
Two-point	4096	320	2.78	18	18.43	17	17.53
Smooth	256	80	1.25	7	7.33	7	8.76
Smooth	1024	160	1.25	11	10.81	11	12.39
Smooth	4096	320	1.25	17	17.82	18	17.53
Skewed	256	80	1.14	7	7.20	7	8.76
Skewed	1024	160	1.13	11	10.50	10	12.39
Skewed	4096	320	1.13	17	16.54	17	17.53
Log-log slope on $\log T$, two-point				0.29	0.27	0.24	0.25
smooth / skewed				0.32	0.32 / 0.30	0.34 / 0.32	0.25

Notes: All designs use $b = 0.2$ and $\bar{\tau}_m = 0.05$, and every profile is calibrated so that $\mathcal{D}_m = 4.608$ exactly; the profiles therefore share the two constants that enter the leading rule and differ in the dispersion index $H = \bar{\tau}_m\, m^{-1} \sum_i \tau_i^{-1}$, which does not. $\widehat{k}^*$ is the empirical control-variate optimizer and $\widehat{k}^*_{\text{smooth}}$ its five-point local-quadratic counterpart. k^*_Γ minimizes the complete criterion $Q_k^{\text{bag},\infty} + \Gamma_{m,k}/T$, with $\Gamma_{m,k}$ integrated conditional on the exact hypergeometric overlap distribution. The last column is the leading expression of Proposition 10. Empirical risk is the $G = 800$ conditional-bridge estimate of expected infinite-bag risk. The bottom rows report descriptive log-log slopes. They are given for the theoretical columns as well, so that the empirical slope is compared against the criterion the theory implies on this finite grid rather than against its asymptote; these five-point and three-point slopes are descriptive summaries, not formal inference.

Table 8: Optimal subset size for shared-sample infinite bagging under alternative precision profiles

C.2 Comparative statics in b, $\mathcal{D}$, and $\bar{\tau}_m$

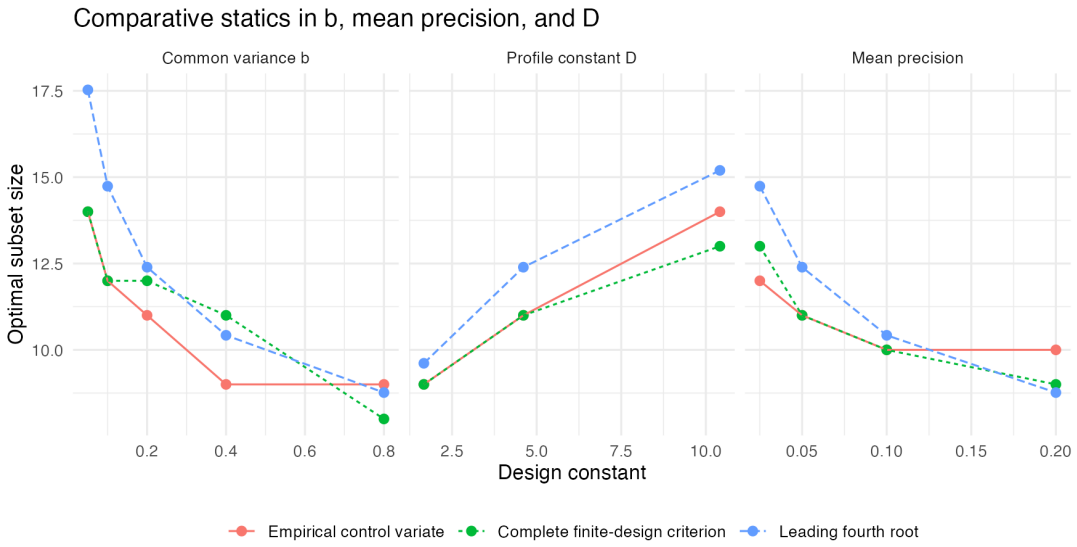


Figure 8: Comparative statics of the shared-sample optimizer

Experiment	Setting	$\mathcal{D}_m$	b	$\widehat{k}^*$	k^*_Γ	$(\mathcal{D}_m T/b)^{1/4}$
Common variance b	0.05	4.61	0.05	14	14	17.53
	0.10	4.61	0.10	12	12	14.74
	0.20	4.61	0.20	11	12	12.39
	0.40	4.61	0.40	9	11	10.42
	0.80	4.61	0.80	9	8	8.76
Dispersion $\mathcal{D}_m$	1.67	1.67	0.20	9	9	9.61
	4.61	4.61	0.20	11	11	12.39
	10.42	10.42	0.20	14	13	15.20
Precision scale $\bar{\tau}_m$	0.025	9.22	0.20	12	13	14.74
	0.050	4.61	0.20	11	11	12.39
	0.100	2.30	0.20	10	10	10.42
	0.200	1.15	0.20	10	9	8.76
Fitted elasticity of $\widehat{k}^*$				b: −0.17	$\mathcal{D}_m$: 0.24	$\bar{\tau}_m$: −0.09
complete criterion / leading rule				−0.17 / −0.25	0.20 / 0.25	−0.17 / −0.25

Notes: Each row is a separate experiment with 120 training replications and $G = 800$.

Table 9: Comparative statics of the shared-sample optimizer

C.3 Pool size

Figure 9 supports the absence of a leading m effect on the optimal subset size k^*.

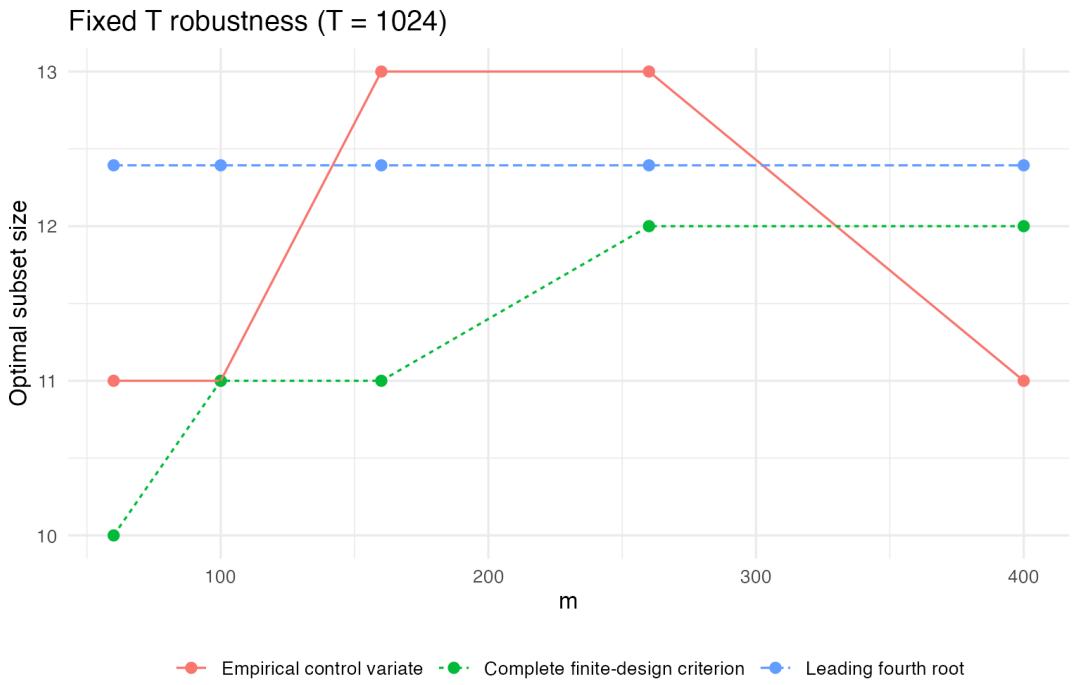


Figure 9: Pool-size diagnostic

C.4 The average-single-subset criterion

This final exercise evaluates each fitted subset separately and then averages losses, which corresponds to square-root rule of selecting k^*. We set $b = 1$, $m = 180$, $T \in \{128, 256, 512, 1024\}$, $G = 300$ and repeat the exercise 150 times. The balanced two-point profile is defined as before, with the following values $\tau_i \in \{0.5, 1.5\}$. Again, the smooth profile is matched on mean precision and $\mathcal{D}_m$.

Figure 10 illustrates the optimal subset size k^* over T.

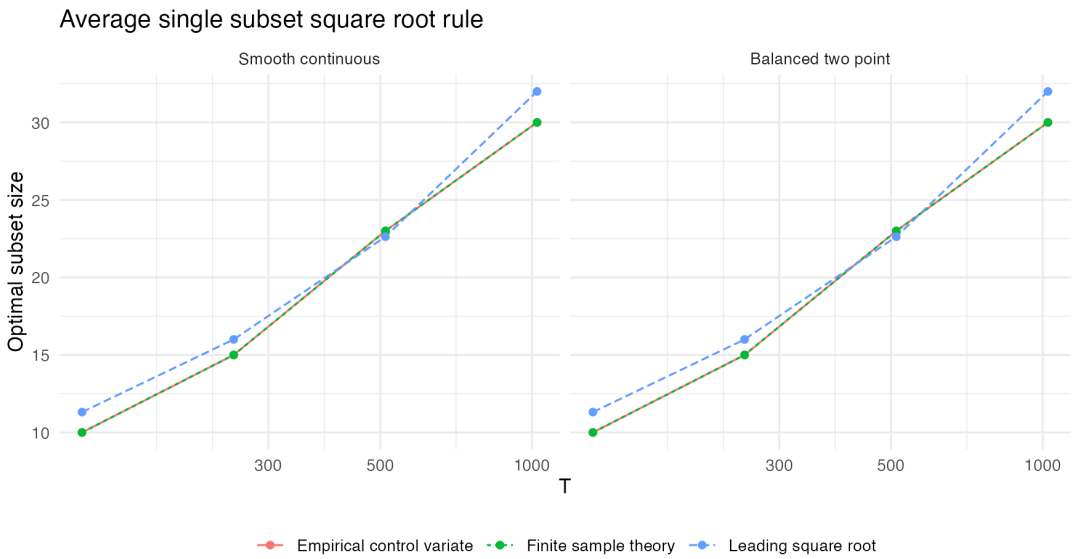


Figure 10: Average-single-subset criterion: empirical and exact-factor risk curves and their optimizers.

D U.S. SPF Designs and Robustness Checks

D.1 Additional Empirical Results

All models use the same dataset at each origin to compute the model-specific forecast combination weights.

The non-negative sum-to-one weights are computed by solving the following minimization problem at each origin:

$$\widehat{w}^{\text{non-neg}} = \arg\min_{w} \; w'\widehat{C}w \quad \text{subject to} \quad \iota_m' w = 1, \;\; w \geq 0.$$

The unrestricted sum-to-one weights are computed as before:

$$\widehat{w}^{\text{unr}} = \frac{\widehat{C}^{-1}\iota_m}{\iota_m'\widehat{C}^{-1}\iota_m},$$

Additionally, we estimate partially-egalitarian Lasso (peLASSO) (Diebold and Shin, 2019) and the shrinkage-toward-equality rule as used in Stock and Watson (2004).

Following Stock and Watson (2004), first, we solve the unconstrained least-squares combination problem:

$$\widehat{w}^{\text{SW}} = \arg\min_{w} \|y_t - w f_t\|^2, \qquad \widehat{y}_t^{\text{ols}} = \widehat{w}^{SW} f_t,$$

where the y_t is the observed outcome and f_t is the vector of forecasts at time t.

The final prediction is obtained by shrinking as follows:

$$\widehat{y}_t^{\text{SW}}(\kappa) = \big(1 - \lambda_t(\kappa)\big)\, \bar{f}_t + \lambda_t(\kappa)\, \widehat{y}_t^{\text{ols}}, \qquad \lambda_t(\kappa) = \max\left\{0,\; 1 - \frac{\kappa\, m}{T - 1 - m}\right\},$$

where $\bar{f}_r^{\mathcal{S}}$ is the screened equal-weight combination and κ scales the imposed penalty: larger κ shrinks weights toward equality.

For more detail on the peLASSO estimation; see Diebold and Shin (2019). It is estimated in two stages, with the Lasso used for selection only and the selected forecasts then equally weighted. In our case, feasible peLASSO uses no intercept or standardization, 100 log-spaced penalty ratios from 0.95 to 0.01, and ordered validation rows 25–48 within each 48-observation window. Selected forecasts receive equal weights, as stated above. The ex-post Lasso diagnostic searches 200 penalties on $exp\{seq(15, -15)\}$ using completed evaluation losses and is therefore infeasible in real time.

	Full	Calm
Panel A: $h = 1$		
peLASSO, ex-post common $\lambda^{\dagger}$	1.007 (.300)	1.014 (.265)
Stock–Watson, $\kappa = 1$	1.012 (.217)	1.018 (.221)
Stock–Watson, $\kappa = .25$	1.315^{**} (.011)	1.196^{**} (.014)
Stock–Watson, $\kappa = .50$	1.040^{*} (.088)	1.011 (.319)
Panel B: $h = 4$		
peLASSO, ex-post common $\lambda^{\dagger}$	1.000 (.475)	1.003 (.209)
Stock–Watson, $\kappa = 1$	1.005 (.181)	1.010^{*} (.070)
Stock–Watson, $\kappa = .25$	1.503^{***} (< .001)	1.597^{***} (.001)
Stock–Watson, $\kappa = .50$	1.080 (.104)	1.097 (.124)

Entries are each method's RMSFE divided by the RSM RMSFE. Parentheses contain the one-sided HAC(4) Diebold–Mariano p-values. *, **, and *** denote significance at the 10, 5, and 1 percent levels. † The common Lasso penalty is selected from evaluation losses separately by horizon and is infeasible.

Table 10: Additional Regularization Benchmarks

Table 10 reports relative to RSM forecasting performance of these methods. The Stock-Watson method crucially depends on κ. Recall that in our setup, $T = 48$, $m \approx 30$ for $h = 1$ and 29 for $h = 4$. If we apply a strong regularization, $\kappa = 1$, the penalty $\lambda = 0$ is observed in 96% of the origins at $h = 1$ and 89% at $h = 4$, respectively. Thus, for the case when $\kappa = 1$, this method yields similar results to equal-weighting.

D.2 CPI and Selector-Form Robustness

The CPI exercise applies the same 48-observation window, 12-of-48 eligibility rule, current-response requirement, own-history imputation, 1990Q2 cutoff, and 2004Q2–2025Q2 evaluation period. The 2020Q2–2020Q3 targets enter subsequent training windows once released. CPI training outcomes use the mid-quarter real-time vintage available at each origin, and target quarter q is scored using the first subsequent mid-quarter vintage containing all three months of that quarter.

The quarterly CPI level is constructed, within each vintage, as the arithmetic average of the three monthly index levels in the quarter. Realized inflation applies the same annualized quarter-on-quarter transformation as the GDP price-inflation applica-

tion, $100[(\bar{X}_q/\bar{X}_{q-1})^4 - 1]$, with both quarterly averages taken from the same vintage. CPI forecasts are collected directly as annualized quarterly percentage rates.

As a selector-form diagnostic, we compare the RSM Selector with the direct fourth-root rule obtained by rounding $\{\widehat{\mathcal{D}}_m T/\widehat{b}\}^{1/4}$ to the nearest admissible integer. This direct rule drops the boundary correction and is therefore an asymptotic diagnostic. For CPI, the median RSM and direct-rule selected sizes are $(1, 1)$ in Full and $(2, 1)$ in Calm at $h = 1$, and $(1, 1)$ in both samples at $h = 4$; origin-level agreement ranges from 91.8 to 98.8 percent, and their RMSFEs differ by at most 0.17 percent. For GDP-price inflation, the median sizes coincide and the RMSFE difference is at most 0.26 percent.

Target	Horizon	Median m	RSM Selector k	Direct fourth-root k	Ex-post best k
GDP-price inflation	1	30	3	3	2
GDP-price inflation	4	29	2	2	4
CPI inflation	1	31	1	1	1
CPI inflation	4	29	1	1	1

Notes: The table reports results for the full evaluation sample. Median m is the median number of eligible forecasts. For the RSM Selector and direct fourth-root rule, the table reports the median selected k across forecast origins. The ex-post best k minimizes RMSFE over the corresponding evaluation sample and is infeasible in real time.

Table 11: Selected Subset Sizes for GDP-Price and CPI Inflation: Full Sample

Target	Horizon	RSM Selector	Direct fourth-root	Ex-post best fixed k
GDP-price inflation	1	100.00	100.01	100.53
GDP-price inflation	4	100.00	100.19	98.46
CPI inflation	1	100.00	99.91	100.23
CPI inflation	4	100.00	100.00	99.66

Notes: The table reports relative RMSFE for the full evaluation sample. For each target and horizon, the RMSFE of the RSM Selector is normalized to 100.00. Values below 100 indicate lower RMSFE than the RSM Selector, while values above 100 indicate higher RMSFE. The ex-post best fixed k is selected using the corresponding evaluation sample and is infeasible.

Table 12: Relative Forecasting Performance of the Subset-Size Rules: Full Sample

Target	Horizon	Median m	RSM Selector k	Direct fourth-root k	Ex-post best k
GDP-price inflation	1	30	3	3	2
GDP-price inflation	4	29	2	2	4
CPI inflation	1	31	2	1	1
CPI inflation	4	29	1	1	1

Notes: The table reports results for the calm evaluation sample. Median m is the median number of eligible forecasts. For the RSM Selector and direct fourth-root rule, the table reports the median selected k across forecast origins. The ex-post best k minimizes RMSFE over the corresponding evaluation sample and is infeasible in real time. The Calm sample excludes the GFC and COVID-19 evaluation periods.

Table 13: Selected Subset Sizes for GDP-Price and CPI Inflation: Calm Sample

Target	Horizon	RSM Selector	Direct fourth-root	Ex-post best fixed k
GDP-price inflation	1	100.00	100.01	101.04
GDP-price inflation	4	100.00	100.26	98.05
CPI inflation	1	100.00	99.83	100.28
CPI inflation	4	100.00	100.00	99.94

Notes: The table reports relative RMSFE for the calm evaluation sample. For each target and horizon, the RMSFE of the RSM Selector is normalized to 100.00. Values below 100 indicate lower RMSFE than the RSM Selector, while values above 100 indicate higher RMSFE. The ex-post best fixed k is selected using the corresponding evaluation sample and is infeasible. The Calm sample excludes the GFC and COVID-19 evaluation periods.

Table 14: Relative Forecasting Performance of the Subset-Size Rules: Calm Sample

D.3 Robustness of the Safeguards

For the *RSM Selector*, we specify safeguards to make the estimation feasible. Let us restate the proposed procedure.

RSM Selector

1. Construct the $T \times m$ matrix of training forecast errors and compute $\widehat{C} = T^{-1} \sum_{t=1}^{T} e_t e_t'$.
2. Estimate $\widehat{b}$ by the mean off-diagonal elements of $\widehat{C}$. Namely, $\widehat{b} = \frac{1}{m(m-1)} \sum_{i \neq j} \widehat{C}_{ij}$.
3. Compute $\widehat{\tau}_i = \frac{1}{\widehat{C}_{ii} - \widehat{b}}$ and $\widehat{\mathcal{D}}_m = \frac{1}{\widehat{\bar{\tau}}_m^6} \frac{1}{m} \sum_{i=1}^{m} \widehat{\tau}_i \{\widehat{v}_{\tau,m} - \widehat{\bar{\tau}}_m(\widehat{\tau}_i - \widehat{\bar{\tau}}_m)\}^2$.
4. Evaluate the criterion in Equation (35) on the admissible integer grid and choose its smallest minimizer $\widehat{k}$.
5. At $\widehat{k}$, estimate subset-optimal weights and average the resulting subset forecasts. Whenever computationally feasible, use exact enumeration of the available subsets.

Sometimes, sampling error can make some implied variances $\widehat{C}_{ii} - \widehat{b}$ and, correspondingly, $\widehat{\tau}_i = \frac{1}{\widehat{C}_{ii} - \widehat{b}}$ nonpositive. In that case, we replace only those values by $0.05 \times \text{median}\{\widehat{C}_{ii} - \widehat{b}\}$, where the median is taken over all positive differences. We then compute $\widehat{\tau}_i$ and $\widehat{\mathcal{D}}_m$ as proposed. If there is no positive $\widehat{C}_{ii} - \widehat{b}$, $\widehat{b}$ is nonpositive or nonfinite, or $\widehat{\mathcal{D}}_m$ is nonfinite, we set $\widehat{k} = 1$.

Table 15 summarizes the panel and covariance diagnostics for the current sample.

h	Origins	Median m	Mean imputed (%)	Off-diag. dispersion	Floored (%)	$\widehat{b} > 0$ (%)
1	85	30	30.9	0.191	8.1	100.0
4	85	29	32.4	0.203	17.3	100.0

Notes: Mean imputed is the origin-average share of the retained training panel filled from a forecaster's earlier observed history. Off-diagonal dispersion is the origin-average standard deviation of off-diagonal training covariances divided by the mean diagonal variance. Floored is the origin-average share of implied idiosyncratic variances replaced by the numerical floor. The last column is the share of origins with a positive estimated common component.

Table 15: Current-Sample Panel and Structural Diagnostics

As Table 15 indicates, the empirical SPF panel exhibits off-diagonal dispersion and requires variance flooring, indicating that the true data-generating process likely features heterogeneous factor loadings or residual cross-sectional dependence.

We recompute the selector after replacing nonpositive implied variances by 1%, 5% (baseline), or 10% of the median positive implied variance. Table 16 reports changes in selected subset sizes and forecast RMSFE; all other training and evaluation rules are held fixed. The alternative floors change the selected size at 20–51% of Full-sample origins. At the 1% floor, the median selected size at $h = 4$ falls from two to one. Even so, every Full- and Calm-sample RMSFE differs from the 5% baseline by less than 1%.

h	Floor	Median $\widehat{k}$	Changed (%)	RMSFE Full	RMSFE Calm
1	1% of median	3	41.2	100.75	100.90
1	5% (baseline)	3	0.0	100.00	100.00
1	10% of median	3	41.2	100.03	99.72
4	1% of median	1	50.6	100.19	100.47
4	5% (baseline)	2	0.0	100.00	100.00
4	10% of median	2	20.0	100.01	100.08

Notes: Only nonpositive $d_i = \widehat{C}_{ii} - \widehat{b}$ are replaced; the replacement is the stated fraction of the median positive d_i. Targets 2020Q2 and 2020Q3 are eligible for training once available. All specifications use the same 85 Full and 73 Calm evaluation targets. Changed is the percentage of Full-sample origins with a different selected size from the 5% baseline. RMSFE columns are percentages of the baseline RMSFE within each sample and horizon; values below 100 indicate lower loss. Baseline RMSFEs are 1.430289 and 1.286424 at $h = 1$ (Full and Calm), and 1.645514 and 1.538784 at $h = 4$. Subset weights use the original sample second-moment matrix and exact enumeration.

Table 16: Floor Sensitivity: GDP-Price Inflation

D.4 Panel-Construction Robustness

The panel-completion procedure is motivated by Conflitti et al. (2015), but it is adapted to the real-time, origin-specific design used here. Their procedure first constructs a fixed panel by excluding forecasters with more than 25 missing observations over 48 survey rounds. An internal missing value is replaced by that forecaster's most recent reported forecast, while a missing value in the first survey round is replaced by the contemporaneous cross-sectional average. Their combination weights are then estimated

over the most recent five years of data. Our objective differs: at every forecast origin, we require a complete recent training matrix while allowing the set of active forecasters to change as respondents enter and leave the survey. The result is therefore a panel balanced conditionally on each origin, not one globally fixed balanced panel.

We use a slightly modified approach. An internal gap (missing values inside) is filled with the mean of that respondent's earlier genuine forecasts, whereas a leading gap (first unassigned forecasts) is backfilled with the respondent's first genuine forecast in the window. Also, we exclude forecasters who reported fewer than 12 rather than 23 forecasts.

These modifications serve three purposes. Requiring a genuine current report prevents a stale forecast from being treated as a respondent's current belief. Applying the participation rule separately at each origin accommodates survey entry and exit without using future participation information. Finally, averaging only earlier genuine reports reduces sensitivity to a single last observation and prevents imputation errors from propagating through the training window. All values used for this completion are available by origin t.

This choice involves a transparent trade-off. Relative to last observation carried forward, averaging earlier reports is more stable but may smooth genuine changes in an individual's forecast level. The 12-of-48 requirement limits that extrapolation while retaining substantially more current respondents than a globally balanced-panel restriction.

D.5 Forecasting Performance of the SPF Variables

Specification	h	N	Median m	Missing (%)	$\widehat{k}_{\mathrm{RSM}}$	$\widehat{k}_4$	k_{best}	RSM vs. EW (%)	Fourth-root rRMSFE	Best-fixed rRMSFE
Panel A: Full sample										
12/48, own mean	1	85	30	30.9	3	3	2	1.05	100.01	100.53
12/48, own mean	4	85	29	32.4	2	2	4	0.03	100.19	98.46
24/48, own mean	1	85	24	21.2	3	3	2	0.21	100.45	99.94
24/48, own mean	4	85	22	22.3	2	2	1	0.72	100.13	100.72
36/48, own mean	1	84	16	12.7	2	3	1	-0.03	99.98	99.97
36/48, own mean	4	82	15	13.2	2	1	1	-0.11	99.77	99.89
12/48, LOCF	1	85	30	30.9	3	3	2	0.13	100.00	99.78
12/48, LOCF	4	85	29	32.4	2	2	2	-0.17	100.17	99.36
12/48, consensus	1	85	30	30.9	2	2	2	0.53	99.96	100.44
12/48, consensus	4	85	29	32.4	1	1	2	0.46	100.00	99.69
Panel B: Calm sample										
12/48, own mean	1	73	30	30.7	3	3	2	1.63	100.01	101.04
12/48, own mean	4	73	29	32.4	2	2	4	0.30	100.26	98.05
24/48, own mean	1	73	24	21.0	3	3	2	1.16	100.68	100.86
24/48, own mean	4	73	22	22.1	2	1	2	1.09	100.18	100.96
36/48, own mean	1	72	16	12.6	3	3	1	0.20	99.97	100.20
36/48, own mean	4	71	15	13.3	2	2	1	-0.07	99.69	99.93
12/48, LOCF	1	73	30	30.7	3	3	2	0.28	99.99	99.76
12/48, LOCF	4	73	29	32.4	2	2	2	-0.16	100.19	99.04
12/48, consensus	1	73	30	30.7	2	2	2	0.94	100.51	100.33
12/48, consensus	4	73	29	32.4	1	1	3	0.51	100.00	98.83

Notes: The baseline requires at least 12 genuine responses among the 48 eligible training rows. An internal missing forecast is replaced by the mean of that forecaster's earlier genuine reports in those rows; a leading gap is filled with the forecaster's first genuine report. Imputed values do not enter later means. LOCF denotes last observation carried forward; consensus imputation uses the corresponding cross-sectional mean. Missing is the pre-imputation missing share. $\widehat{k}_{\mathrm{RSM}}$ and $\widehat{k}_4$ are the median sizes selected by the RSM Selector and direct fourth-root rule. The ex-post best fixed k is infeasible. RSM vs. EW is the percentage RMSFE reduction relative to screened equal weights. Relative RMSFE is indexed to the RSM Selector, normalized to 100.00. The Calm sample excludes the GFC and COVID-19 evaluation periods.

Table 17: Robustness to Alternative Forecast-Panel Construction

		RMSFE relative to the RSM Selector					
Variable	Origins	Equal	All-resp.	Inv.-MSE	Plug-in	Non-neg.	Unrestr.
Payroll employment growth	37	1.003	1.005	1.000	1.021	1.156	3.217**
Unemployment rate	85	1.009	1.022	0.987	0.967	0.972	1.491
Housing starts	85	1.017*	1.008	1.007	1.020	0.975	1.313**
Industrial production growth	85	1.002	1.010	1.007	1.037	1.345	2.892*
Nominal GDP growth	85	0.997	0.996	0.997	1.041	1.084	2.693*
Real consumption growth	85	1.005	1.004	1.004	1.061	1.121	2.435**
Real GDP growth	85	1.001	1.001	1.002**	0.993	1.050*	3.813*
Real nonresidential investment growth	85	0.990	0.984	0.993	1.026*	1.013	2.332**
Real residential investment growth	85	0.999	0.997	0.996	0.988	1.037*	2.013***
Real state & local government growth	85	1.000	1.001	0.988	0.979	1.073	3.135***
Core CPI inflation	24	1.010	1.014	1.004	0.966	0.996	1.847**
Core PCE inflation	24	0.995	0.991	0.992	1.007	0.998	2.454***
CPI inflation	85	1.002	1.003	1.001	1.023***	1.082**	2.424***
PCE inflation	24	0.999	1.001	0.988	0.985	1.050	2.657***
GDP price inflation	85	1.011	1.012	1.005	0.980	1.043**	1.902***
10-year Treasury yield	83	1.004**	1.001	1.002	1.002	1.019	1.086
3-month T-bill rate	83	1.003	1.024*	0.999	1.010	0.995	1.524**

Notes:

Table 18: Forecast Comparisons: Seventeen SPF Variables, $h = 1$